Bharat Sanchar Nigam Limited
(A Govt. of IndiaEnterprise)
O/o the Chief General Manager,BSNL,
Odisha Circle,
3rd Floor, MM Cell,
Bhubaneswar-751009
email: agmmmor@gmail.com

BID DOCUMENT

E-Tender for PLB Pipe/Duct Laying by HDD/Open Trench Method, Optical Fiber Cable Pulling/Blowing, Splicing, earthing and associated works for **BPR02 (Chatrapur) Zone in** Berhampur BA of Odisha Telecom Circle

NIT No. GEM/2026/B/7453892

Particulars of the payment of Price of Tender Document

Transaction ID/ Receipt No.

……dated……..

Signature of the Bidder with Seal

Particulars of EMD

Transaction ID/ Receipt No.

…… dated……..

Signature of the Bidder with Seal

Note: Refer Clause-2, 12& 13 under Section-I reg. Price of Tender Document and EMD.

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SECTION–I

Introduction about the service

Notice Inviting Tender

NIT No. **GEM/2026/B/7453892**

E-tenders (Digitally Signed) are invited for and on behalf of CGM BSNL Odisha Circle from the experienced contractors, for the following works:

1. **Name of work:**Optical Fiber Cable Construction work.
2. **Scope and Jurisdiction of Work:** The scope and particulars of work put to tender are given in the Table below.

Sl No	NIT No. (Suffixing Zone no.)	Zone Number/ Name	Total (KMs)	Estimated cost with GST (in Rs.)	Price of Bid Document (in Rs.)	EMD @2% of Est cost (in Rs.)
1	GEM/2026/B/7453892	BPR02 (Chatrapur)	43.9	13079745.40	2360	261595

Note-1: The quantities stipulated in the tender are estimated and the BSNL reserves the right to vary the value of works to the extent of -25% to + 25% of contract value during contract period at the same rates and terms & conditions with prior approval of the tender approving authority.

However, **in exceptional circumstances only**, due to change in scope of work or otherwise, any variation in the value of work may be permitted upto 50% of contract value with prior approval of the next higher authority after recording adequate reasons and justification for the same.

Note-2: There is restriction on the number of tenders that a bidder can participate. Each bidder is allowed to participate in a maximum of three Bids/Zones out of the following 13 nos of Bids/Zones floated separately for Berhampur BA of Odisha Telecom Circle.

Sl No	NIT No. (Suffixing Zone no.)	Zone Number/ Name	Total (KMs)	Estimated cost with GST (in Rs.)	Period of Completion (in Days)
1	GEM/2026/B/7451710	**BPR01 (Berhampur)**	**48.3**	**14459071**	**150**
2	GEM/2026/B/7453892	**BPR02 (Chatrapur)**	**43.9**	**13079745.4**	**150**
3	GEM/2026/B/7454231	**BPR03 (Aska)**	**55.3**	**16835337.3**	**180**
4	GEM/2026/B/7454352	**BPR04 (Mohana)**	**68.928**	**20375927.3**	**210**
5	GEM/2026/B/7454458	**BPR05 (Paralakhemundi)**	**51.17**	**14898632.8**	**180**
6	GEM/2026/B/7453964	**BPR06 (Bharatnet-1)**	**30**	**8912658**	**150**
7	GEM/2026/B/7454009	**BPR07 (Bharatnet-2)**	**28**	**8530102**	**150**
8	GEM/2026/B/7454054	**BPR08 (Bharatnet-3)**	**33.15**	**10103632**	**150**
9	GEM/2026/B/7454110	**PHU01 (Boudh)**	**40**	**12011880.8**	**120**
10	GEM/2026/B/7454304	**PHU02 (Harbhanga)**	**45.5**	**13593097.32**	**150**
11	GEM/2026/B/7454345	**PHU03**	**30.5**	**9128527.2**	**90**

		(Tumudibandha)			
12	GEM/2026/B/7454369	**PHU04 (Baliguda)**	**63**	**18804810.4**	**210**
13	GEM/2026/B/7454411	**PHU05 (Phulbani)**	**77**	**23113415.2**	**240**

Therefore, bidders are advised to carefully select a maximum of three Bids/Zones out of the 13 nos of Bids/Zones floated separately for Berhampur BA of Odisha Telecom Circle when submitting their bids. The tendering authority reserves the right to reject the 4th or subsequent bids in this tender during the technical evaluation stage, without providing any reason thereof. The tendering authority reserves right to reject all the bids of concerned bidder or consider three nos of zones in ascending order (Starting from Sl No.1 to Sl.No.13).Bidders who have been rescinded /debarred by Tendering authority/Higher BSNL Unit as the case may be including for reasons for failing to sign necessary agreement with the concerned Tendering authority in stipulated time on earlier occasions, will not be eligible to participate in the tender for the same work.

Note-4: Bidders Whose contract has been rescinded or debarred by any tendering authority in BSNL/MTNL/TCIL/ITI/Rites/HCL/ PGCIL/ RailTel/any other Govt. (Central/ State) undertaking during preceding 12 Months (from the date of NIT) or the debarment is in currency, will not be eligible to participate in the tender.

Note-5: For participating in e-tendering process, bidder shall access e-bid document uploaded on the e-tender portal and pay requisite tender processing fee as defined for the tender.

Note-6: It is the responsibility of Successful Bidder to obtain the RoW permission from the concerned authorities viz., NH, NHAI, SHAI, Municipality, Corporation, Panchayat, IOCL, Water Department, Electricity Department, Forest Department and etc. if required**,** within the stipulated time**.** The successful bidder will be required to prepare all the necessary documents as asked by RoW authority for obtaining ROW permission like chainage route diagram in required format, valuation details of the Land (for PWD, NH, NHAI etc.), NOC from the different authorities related with ROW permission etc. In this regard, BSNL will not pay any expenditure for the said processing. However, BSNL will pay only for the amount of demandnoteissuedtowardsRoWchargesdirectlytotheconcernedauthorities.

Note-7: **WORK COMPLETION TIME =As per Note-2 Table (Counted from 7th Day from the date of issue of Work Order)** excluding the time allowed for obtaining ROW permission **if required**

Note-8: **Time for obtaining of ROW Permission=30 Days (Counted from 3rd Day from the date of issue of Work Order)** and that may be extended for another 15 days (max.) subject to valid justification and approval of the competent authority in this regard.

Note-8: **Details of the OFC Route work is attached in Section – III.**

Eligibility Criteria:

3.Work Experience

3.1 The tenderers who possess experience of having successfully completed the **Optical Fiber Cable Construction works** in BSNL/MTNL/TCIL/ITI/Rites/HCL/ PGCIL/ RailTel/any other Govt. (Central/ State) undertaking having mandate to carry out OFC construction works and also have received the payment for such completed works to the extent as mentioned below, are eligible to participate in the tender:

a) For an amount equal to at least 90% of estimated cost of work put to tender during last 5 Financial years plus the subsequent period ending the month preceding the one in which NIT is issued.

OR

b) For an amount equal to at least 120% of estimated cost of work put to tender during last 7 Financial years plus the subsequent period ending the month preceding the one in which NIT is issued.

The 'Estimated Cost of Work' for this purpose shall be the same as noted in Clause 2 of this Section.

3.2 Other mandatory requirement:

3.2.1 The tenderer whose near relative(s) is/are employed in BSNL, is not eligible to Participate in the tender, as per limitations mentioned in Annexure-C

3.2.2 Contractor should have valid EPF registration certificate.

3.2.3 Contractor should have valid ESI registration certificate.

3.2.4 Contractor should have valid GSTIN registration certificate.

3.2.5 Contractor should have valid PAN No.

3.2.6 A self- declaration along with the evidence that the bidder is not black listed by GST authorities and BSNL/MTNL/TCIL/ITI/Rites/HCL/ PGCIL/ RailTel/any other Govt. (Central/ State) undertaking as the case may be.

3.2.7 **Financial Turn over**: The total financial turnover for the last three years, ending on 31st March of the previous financial years (i.e., 2022-23, 2023-24, and 2024-25), must be at least 150% of the estimated tender cost (inclusive of GST). Self-attested copies of the Profit & Loss statements, duly audited by a Chartered Accountant (CA), along with a Financial Turnover Certificate issued by the CA, must be submitted. Additionally, the certificates issued by the Chartered Accountants should include a verified UDIN for the corresponding years.

3.2.8 Scanned copy of Solvency certificate for 25% of the Tender cost or Rs. 5 Lac whichever is higher. ***The solvency certificate shall not be older than the date of issue of NIT***.

Note1: The Experience should be of Optical Fiber Cable construction work only and does not include any other work like Pole less/Upgradation/Maintenance works of OFC/External Plant/ UG Cable (Copper) etc.

Note 2: The experience certificate for carrying out the works and amounts received should be in the prescribed pro-forma and the certificate should be under the signature of officer responsible for maintaining the contractor's ledger e.g. DGM/ AO (Cash)/DDO or equivalent officer of the concerned PSU i.e. BSNL/ MTNL/ TCIL/ ITI/ Rites/ HCL/ PGCIL/ RailTel /any other Govt. (Central/ State) undertaking as the case may be. BSNL may verify and ask the bank statement of the certified amount which should be reflected in the bank statement of the bidder.

Note 3: **The experience criteria as mentioned above (ref. Section IX, Clause 1.1) shall supersede any other experience criteria mentioned in GeM portal by default.**

4. Period of contract: The period of this contract will be one year from the date of agreement. However, BSNL reserves the right to extend the contract further, in terms of six months, up to one year on the same rates, terms & conditions, which shall be binding on the contractor.

5. Date of sale of tender document, Last date of submission of bid, Date of opening of Bids:

Availability of tender Document Online on the GeM portal	Deadline for Submission of e-Bids Online on the GeM Portal	Date of Opening of Bids Online	
		Qualifying Bid	Financial Bid
As mentioned in the GeM Portal	Date and time as mentioned in the GeM Portal	Date, time and venue as mentioned in the GeM Portal	To be notified later

6. If the date of opening of the bids happens to be a holiday, the tenders will be opened on the next working day at the same time and the same venue.

7. The tender is invited through **e-tendering process** through the portal (https://gem.gov.in). Prospective bidders should get their registration done well in time with the Digital Signature Certificate (DSC) issuing authority, which is essential for participating in the tendering process.

8. Tender will not be accepted/received on the e-tendering portal after due date and time. The DGM (MM), Odisha Circle, Bhubaneswar reserves the right to reject any or all tenders without assigning any reason whatsoever.

9. As the tender is invited through e-tendering process, physical copy of the tender document would not be available for sale. The bidder needs to download the tender document using the link as mentioned above.

10. All the documents in Qualifying Bid (Part-A) are to be uploaded in the e-tendering portal including proof towards e-payment (RTGS/NEFT/IMPS) of EMD and Cost of Tender document.

11. No off-line document will be received. Hence, there shall not be any tender dropping box.

12. Cost of Bid Document =Rs 2360/- (Rs 2000 +18% GST)

The cost of Tender Document and EMD has to be paid through e-payment i.e. NEFT/RTGS/IMPS.

Details of e-payment Account of BSNL:

Accounts Officer (A&P), O/o CGM, Odisha Circle Account No. 380801010035282,

IFS Code: UBIN0538086, Name of the Bank: Union Bank of India,

Branch Address: Main Branch,38-Ashok Nagar, Rajmahal Square, Bhubaneswar, 751009.

'Accounts officer (A & P) O/o CGM BSNL Odisha Circle, Bhubaneswar'

Exempted in case of MSE with UAM (Udyog Aadhar memorandum) Certificate vendors on production of valid MSME Certificate

13. The Units registered under single point registration scheme of NSIC/MSE units for OF Cable laying works are exempted from payment of cost of the tender document and earnest money deposit (EMD). Such bidders shall have to upload the document in proof of registration of valid NSIC/MSE for carrying out OFC work along with bid document.

14. The successful bidder including NSIC/MSE bidders shall have to submit a Performance Bank Guarantee (PBG) for 10%* of tender approved value and additional Bank Guarantee for Material Security as per Letter of Intent (LoI)/APO.

**** PBG amount mentioned in GeM bid (i.e default value of 5 % of estimated value) is taken for bid creation purpose only. Actual PBG amount is 10 % of contract value including GST valid for 36 months as per ATC.***

15. All the documents in Qualifying Bid (Part-A) are to be uploaded in the e-tendering portal as per clause 7 of Section – II for establishing the bidder's eligibility which are mandatory. In case of non-submission of any document as mentioned in clause 7 of Section – II, the bid shall be outrightly rejected.

16. If a vendor registered with body specified by Ministry of MSME claiming concessional benefits is awarded the work by BSNL and subsequently fails to discharge any of the contractual obligations, he may be debarred from participating in future tenders in Odisha Circle for one year from the date of issue of such order and MSME authority issuing the certificate shall also be informed for taking suitable action.

17. Procedure for Acceptance and Testing will be followed as per Clause No.17 of Section - IV.

DGM (MM)
O/o the CGM, Odisha Circle, Bhubaneswar

----- END OF SECTION-I ----

SECTION II

Instructions to Bidders

A. **INTRODUCTION:**

1. DEFINITIONS

1.1 **President of India:** The President of India means the President of India and his successors.

1.2 **Government of India:** The Government or Government of India shall mean the President of India.

1.3 **BSNL** means Bharat Sanchar Nigam Limited, the company with Chairman& Managing director and Board of directors, with Head Quarter at New Delhi.

All references of:

Department
Chief General Manager/CGM
Principal General Manager/PGM
Sr. General Manager/ General Manager/Sr.GM/GM
Addl. General Manager/ Joint General Manager
Deputy General Manager/DGM
Divisional Engineer/AGM (DE)/AGM
Sub Divisional Engineer/AGM/SDE
Junior Telecom Officer/JTO
Chief Accounts Officer/CAO
Accounts Officer/AO
Assistant Accounts Officer/AAO
Junior Accounts Officer/JAO

Including other officers in BSNL, whatever designations assigned to them from time to time, who may be the in-charge of direction, supervision, testing, acceptance and maintenance including their successor(s) in the office appearing in various clauses shall be taken to mean BSNL, an enterprise, under the Ministry of Communications, Government of India

1.4 The jurisdiction of GM/Addl.GM/ DGM/DET shall mean the territorial area under their control for the purpose of administration and management of the projects and other associated functions

1.5 Representative of GM/Addl.GM/ DGM / DET mean Officer and staff under their control for the time being deputed for supervising the work or testing etc.

1.6 Engineer-in-charge: The Engineer-in-charge means the Engineering Officer nominated by the GM/Addl.GM/ DGM to supervise the work, under the contract. (Minimum Divisional Engineer/AGM level officer to be nominated while approving the work order).

1.7 Site Engineer: Site Engineer shall mean a JTO/SDE of BSNL who may be placed by the work order issuing authority for supervising the work.

1.8 A/T Unit: A/T unit shall mean Acceptance and testing unit of BSNL.

1.9 A/T Officer: An officer authorized by BSNL to conduct A/T.

1.10 The words bidder and tenderer have been used interchangeably

1.11 Contract: The term contract means, the documents forming the tender and acceptance thereof and the formal agreement executed between the competent authority on behalf of BSNL and the contractor, together with the documents referred to therein including these conditions, the specifications, designs, drawings and instructions issued from time to time, by the engineer in-charge and all these documents taken together shall be deemed to form one contract and shall be complementary to one another.

1.12 Contractor: The contractor shall mean the individual, firm or company, approved to undertake the works and shall include the legal personal representative of such individual or the persons composing such firm or company, or the successors of such firm or company and the permitted assignees of such individual, firm or company.

1.13 Work: The expression "works" shall unless there be something either in the subject or context repugnant to such construction be construed and taken to mean the works by or by virtue of the contract contracted to be executed whether temporary or permanent and whether original, altered, substituted or additional.

1.14 Schedule(s): Schedule(s) referred to in these conditions shall mean the relevant schedule(s) or the standard schedule of rates mentioned in the document.

1.15 Site: The site shall mean the land / or other places on, into or through which work is to be executed under the contract or any adjacent land, path or street through which, the work is to be executed under the contract or any adjacent land, path or street which may be allotted or used for the purpose of carrying out the contract.

1.16 Normal time or stipulated time: Normal time or stipulated time means time specified in the work order to complete the work.

1.17 Extension of Time: Extension of Time means the time granted by BSNL to complete the work beyond the normal time or stipulated time. The EOT may be with/without Liquidated damages as may be decided by the competent authority.

1.18 Date of Commencement of work: Date of commencement of work means the date of actual commencement of work or 7th day from the date of issue of work order, whichever is earlier.

1.19 Due date of completion: Due date of completion shall be the date by which the work shall be completed at site including clearance of site.

1.20 Duration of completion of work: The duration of completion of work or completion time shall be time specified in the work order plus extension of time granted, if any.

1.21 Excepted risk: Excepted risk are risks due to war (whether declared or not), invasion, act of foreign enemies, hostilities, civil war, rebellion, revolution, insurrection, military or usurped power, any acts of Government damages from aircraft, acts of God, such as earthquake, lightening and unprecedented floods and other causes over which, the contractor has no control and the same having been accepted as such, by the Accepting Authority or causes solely due to use or occupation by the government of the part of the work, in respect of which a certificate of completion has been issued.

1.22 Extra work: as used herein any work or compliance with any requirements, other than a change, which is not expressly or impliedly contemplated by the contract documents, and which is necessary to be performed for the proper completion of the contracted work. For the purpose of clarifications, it is declared that, any work or operation which shall be necessarily incidental to the proper performance of any item of work or part there of shall be deemed to have been by implication provided for in the relevant item of work or part thereof and shall not constitute extra work.

1.23 Competent authority: For the work related to the tendering process, competent authority means DGM(MM)/ PGM (CFA)/ CGM, Odisha Circle, Bhubaneswar as per delegation of financial power. For the work related to the execution process, competent authority means the concerned BA head.

2. <u>ELIGIBILITY OF BIDDERS</u>:

2.1 The invitation of bids is open to all bidders as per the eligibility mentioned in NIT of this tender document.

2.2 The experience certificate for carrying out the works and amounts received should be in the prescribed proforma and the certificate should be under the signature of officer responsible for

maintaining the contractor's ledger e.g. DGM/ AO (Cash)/DDO or equivalent officer of the concerned PSU i.e. BSNL/MTNL/TCIL/ITI/Rites/HCL/ PGCIL/ RailTel/any other Govt. (Central/ State) undertaking as the case may be.

Of late big Govt. contracts are being carried out by the Public Sector Undertakings mentioned herein through a chain of contractors. In such case, the experience certificate, if issued, in favor of a contractor, in the chain, by such PSUs in the prescribed format shall be acceptable.

B. THE BID DOCUMENTS

3. <u>BID DOCUMENTS</u>:

3.1 The construction work to be carried out, goods required, bidding procedures and contract terms are prescribed in the Bid Documents. The Bid Documents include:

3.1.1**. Part-A(Qualifying Bid)**:

3.1.1.1 Introduction about the Service
3.1.1.2 Instruction to Bidders
3.1.1.3 Scope of the Work and jurisdiction of contract
3.1.1.4 Special conditions of the contract & General (Commercial) conditions of the contract
3.1.1.5 Service level Agreement Proforma.
3.1.1.6 Payment Terms
3.1.1.7 Penalties
3.1.1.8 Schedule of work / Bill of Quantity (BOQ)
3.1.1.9 Qualifying criteria and Experience
3.1.1.10 Bid Form & Tenderer's Profile

3.1.2 **Part-B (Financial Bid)**

3.1.2.1 Schedule of Quoting Rates.

3.2 The Bidder is expected to examine all instructions, forms, terms &conditions and the specifications in the bid documents. Failure to furnish all information required as per the Bid Documents or submission of bids not substantially responsive to the Bid Documents in every respect will be at the bidder's risk and shall result in rejection of the bid.

4. <u>QUERIES ON BID DOCUMENTS:</u>

A prospective bidder, requiring any clarification of the Bid Documents shall notify the tender issuing authority in writing through representation in GeM Portal. It may be appreciated that such communications are preferably done through GeM Portal only as any other mode of communications may result in delay for which BSNL shall not be responsible. The tender issuing authority shall respond in writing, in GeM Portal, to any request for clarification of the Bid Documents, which it receives not later than 5 working days prior to the date of opening of the bids. Copies of such queries (without identifying the source) and the clarifications, if any, by the tender issuing authority shall be placed on e-tendering website (https://gem.gov.in) in form of Addendum/Corrigendum and all such clarifications issued by the tender issuing authority will form part of the bid document.

5. <u>AMENDMENT OF BID DOCUMENTS:</u>

At any time, prior to the date for submission of bids, the tender issuing authority may, for any reason whether Suo motu or in response to a clarification requested by a prospective Bidder, modify the bid documents by amendments.

The amendments shall be notified in writing to such prospective bidders who seek clarifications on their mail id, besides uploading the same on e-tendering portal.
In order to afford prospective bidders reasonable time to take the amendments into account in preparing their bids, the tender issuing authority may, at its discretion, extend the deadline for the submission/opening of bids suitably.

All the amendments/corrigendum/addenda/extension of date(s) shall be published only one-tendering website i.e**. https://gem.gov.in**. Therefore, the prospective bidders are advised to visit the website regularly.

PREPARATION OF BIDS

6. **Cost of Bidding:**

The bidder shall bear all costs associated with the preparation and submission of the bid through e-tendering. BSNL, will in no case, be responsible or liable for these costs, regardless of the conduct or outcome of the bidding process.

7. **DOCUMENTS ESTABLISHING BIDDER'S ELIGIBILITY AND QUALIFICATION**:

The bidder shall submit the following documents online on the e-tender portal (along with the Qualifying Bid) for establishing the bidder's eligibility which are mandatory:

List of Documents in the 'Technical Bid Folder' to be uploaded on e-tendering portal	
1.	Scanned copy of the duly filled and signed Tenderer's Profile &Bid Form, Section-X of the tender document.
2.	Scanned Copy of receipt of e-payment (RTGS/NEFT/IMPS) of requisite value towards the price of tender document as per NIT or MSME Certificate for availing exemption thereof.
3.	Scanned Copy of receipt of e-payment (RTGS/NEFT/IMPS) of requisite value for the Bid Security/EMD as per NIT or MSME Certificate for availing exemption thereof.
4.	Scanned copy of Latest and valid MSE Certificate for OF Cable laying work (if any).
5.	Scanned copy of PAN card along with assessed ITR for the last two assessment years.
6.	Scanned copy of the valid EPF& ESI Registration Certificates along with EPF challan cum return for the last two years.
7.	Scanned copy of Annual turnover details as per eligibility criteria in the Bid document, Audited balance sheet, P&L A/C, 3CB and Turnover certificate by CA with valid UDIN.
8.	Scanned copy of valid labour license under Contract Labour (R&A) Act 1970 and Contract Labour (Regulation and Abolition) Central Rules 1971 or undertaking to submit the same before execution of work.
9.	Scanned copy of certificate under "The Building and other Construction Workers Welfare Cess Act 1996" or undertaking to submit the certificate before execution of work.
10.	Scanned copy of the valid GSTIN Registration Certificate along with GSTR-1 and GSTR-3B for last 2 years.
11.	Scanned copy of Solvency certificate for 25% of the Tender cost or Rs. 5 Lac whichever is higher. ***The solvency certificate shall not be older than the date of issue of NIT***.
12.	Scanned copy of the Experience Certificate(s) as per NIT in the prescribed Proforma (Annexure-D).
13.	Scanned copy of the documents, as applicable, in support of registration of the Company/Firm/Proprietorship: ➢ The registration of the firm, authenticated copy of partnership deed in case of partnership firm. ➢ Scanned copy of Articles of Association duly registered with Registrar of Company affairs, in case of Limited/Pvt. Limited Company. ➢ Duly executed Affidavit, stating that tenderer is the sole proprietor of the firm or any document issued by the taxation authorities stating that tenderer is the sole proprietor of the firm, in case of proprietorship firm. ***The Affidavit shall not be older than the date of publication of NIT.***
14.	Scanned copy of Power of Attorney (duly registered with sub-registrar)/Resolution of Board of Directors, authorizing an individual with whose DSC the tender is submitted on e-tendering portal (if and as applicable). ***The Power of Attorney shall not be older than the date of publication of NIT.***
15.	Scanned copy of No Near relative certificate(s) in the prescribed proforma as per Annexure-C.
16.	Duly completed Declaration about Genuineness of Documents/Certificates as per Annexure-E.

17.	Undertaking for completion of the work as per Annexure-G (To be submitted on non-judicial stamp paper of Rs. 100/-)
18.	Duly completed Declaration about Non-Banned/Non-Blacklist/Debarred as per Annexure-H.
19.	Declaration w.r.t. Rule 144 (xi) to GFR 2017 as per Annexure-I in Section-XI
20.	Make in India Certificate as per Annexure-II in Section-XI

8. BID SECURITY(EMD):

8.1 The bidder shall furnish, as part of his bid, a bid security (EMD) for an amount as mentioned in NIT. No interest shall be paid by BSNL on the bid security for any period, whatsoever.

8.2 The bid security is required to protect BSNL against the risk of bidder's conduct, which would warrant the security's forfeiture.

8.3 The bid security shall be paid through e-payment(RTGS/NEFT/IMPS) in favour of

Accounts Officer (A&P), O/o CGM, Odisha Circle Account No. 380801010035282,

IFS Code: UBIN0538086, Name of the Bank: Union Bank of India,

Branch Address: Main Branch,38-Ashok Nagar, Rajmahal Square, Bhubaneswar, 751009.

8.4 A bid not secured in accordance with **para 8.1 & 8.3** shall be rejected by BSNL as non-responsive.

8.5 The bid security of the unsuccessful bidder will be refunded as promptly as possible after finalization of the contract but not later than 30 days after the expiry of the period of bid validity prescribed by BSNL.

8.6 The MSE units shall be exempted from submission of Bid Security deposit on production of requisite proof in respect of valid certification from MSME or NSIC for the tendered item/work. If a vendor registered with body specified by Ministry of Micro, Small & Medium Enterprise claiming concessional benefits is awarded work by BSNL and subsequently fails to obey any of the contractual obligations; he may be debarred from any further work/ contract by BSNL, Odisha Circle for one year from the date of issue of such order.

8.7 The bid security shall be forfeited:

8.7.1 If a bidder withdraws his bid during the period of bid validity specified in the bid document or

8.7.2 If the bidder makes any modifications in the terms and conditions of the tender before acceptance of the tender, which are not acceptable to BSNL or

8.7.3 In case of a successful bidder, if the bidder fails:

8.7.3.1 to sign the agreement in accordance with **Section V** or

8.7.3.2 to furnish Material Security in accordance with clause 11.5 of Section IV.

8.7.3.3 to observe material terms and conditions stipulated in the tender document including mandatory legal requirements e.g. Labour, taxation laws.

9 BID PRICES:

9.1 The bidder shall give the total composite price inclusive of all levies and taxes, packing, forwarding, freight and insurance in case of materials to be supplied and inclusive of all taxes and levies including GST in case of works to be executed. The contractor shall be responsible for loading/unloading and transporting the materials, to be supplied by BSNL from any BSNL store / Telecom factory to the work site. The charges on this account shall be regulated as per the rates in Standard Schedule of Rates (SOR). The offer shall be firm in Indian Rupees.

9.2 Prices shall be quoted by the bidder as percentage below / above / at par the standard schedule of rates in the schedule of quoting rates (Financial Bid). Prices quoted at any other place shall not be considered.

9.3 The price quoted by the bidder shall remain fixed during the entire period of contract and shall not be subject to variation on any account. A bid submitted with an adjustable price quotation will be treated as non-responsive and rejected.

9.4 Discount, if any, offered by the bidders shall not be considered unless they are specifically indicated in the schedule of quoting rates (Financial Bid). Bidders desiring to other discount shall therefore modify their offers suitably while quoting and shall quote clearly net price taking all such factors like Discount, free supply, etc. into account.

9.5 The estimated "cost of work per meter" is computed for all the works put to tender based on surveyed quantities and prevailing SOR as mentioned in **Section–VIII**. The bidders are supposed to ascertain the soil strata existing in the area of the contract and quote their rates keeping in view their own ascertainment. The bidder should note that the approved percentage shall be loaded on to the estimated cost of work per meter as mentioned above to indicate the approved rate per meter. The final rate arrived post execution against this contract by and large will be within this approved rate per meter.

10. PERIOD OF VALIDITY OF BIDS:

Bid shall remain valid for **150 days** from date of opening of the bid (Qualifying Bid). **A Bid valid for a shorter period shall be rejected by BSNL as non-responsive.**
In exceptional circumstances, the tendering authority may request the consent of the bidder for an extension of bid validity. The request and response thereto, shall be made in writing. The bid security provided under **clause-8 of section-II** shall also be suitably extended. The bidder may refuse the request without forfeiting his bid security. A bidder accepting the bid and granting extension will not be permitted to modify its bid.

11. SIGNING OF BID:

11.1 The bidder shall prepare and submit digitally signed bid document, establishing the conformity of his bid to the bid documents of all the works to be executed by the bidder under the contract.

11.2 The bid shall contain no interlineations, erasures or overwriting except as necessary to correct errors made by the bidder in which case such corrections shall be signed with date by the person or persons signing the bid.

11.3 Power of Attorney:

11.3.1 The power of attorney **<u>in original</u>**, in case, person other than the sole proprietor/ authorized signatory has signed the tender document, shall be on a non-judicial stamp paper of Rs. 50/- in case same has been executed in between blood relations and in other cases, it should be on a non–judicial stamp paper of Rs.100/- and should be attested by the Notary Public or same should be registered before the sub registrar. The power of attorney in original should be submitted.

11.3.2 The power of Attorney in original is executed by a person who has been authorized by the Board of Directors of the bidder in this regard, on behalf of the company/ institution/ Body corporate.

11.3.3 In case of the bidder being a partnership firm, the said Power of Attorney should be executed by all the partner(s) in favour of the said Attorney.

11.3.4 Attestation of the specimen signature of such authorized signatory of the bid by the Company's/Firm's bankers shall be furnished. Name, designation, phone number, mobile number, email address and postal address of the authorized signatory shall be provided.

11.3.5 In case the representative of bidder company who uploads the document on e-tender portal using his Digital Signature Certificate (DSC) is different from the authorized signatory for the bid (Power of Attorney holder) then the representative who uploads the document on e- tender portal using DSC issued in his name shall also be made as one of the power of attorney holder by the bidder company, in addition to the authorized signatory for the bid.

11.3.6 The power of Attorney should be executed after the date of publication of NIT/Bid.

D. SUBMISSION OF BIDS

12.Method of Preparation & Submission of Bids in e-Tendering System:

12.1 Bid should be submitted in the following manner:

12.1.1 Documents to be submitted:
All the documents in **Qualifying Bid Part-A** as mentioned in **Clause 7** of this section, are to be uploaded in the e-tender portal including proof of e-payment (RTGS/NEFT/IMPS) towards cost of Tender document and EMD or MSME Certificate as the case may be.

12.1.2 Submission of Bids on e-tendering website-
The bidder shall upload his bid for this tender in two separate folders marked as 'Qualifying Bid', and 'Financial Bid'.

The bid folders shall contain the documents as given below-

12.1.2.1 'Qualifying Bid' folder –The scanned documents in electronic form as per **clause 7** of **Section II** must be uploaded in this folder.

12.1.2.2 'Financial Bid' folder-The Rates duly quoted in the prescribed format as given in BOQ for Quoting the Rates.

12.2 The tenders which are not submitted in above mentioned manner shall be summarily rejected.

12.3 The tenderer will be bound by all terms, conditions & specifications as detailed in the tender documents.

12.4 Any tender with conditions other than those specified in the tender document is liable to be summarily rejected. No modification by the contractor in any of the conditions will be permitted.

13 SUBMISSION OF e-BIDS:

13.1 E-Tenders should be submitted strictly as per the procedure mentioned in clause 12 above. The tenderer is to ensure submission of the Qualifying Bid and Financial Bid online as mentioned above. BSNL shall not be held responsible for wrong submission/non-submission of the bids online.

13.2 If tender opening day is declared as holiday by the government then the bids technical or financial will be opened on the next working day. The Time & Venue will remain unchanged.

13.3 LATE BIDS:

Tenders shall not be accepted online after the specified date and time of closing of the tender. It is the sole responsibility of the tenderer that he should ensure timely online submission of tender and in the manner prescribed there, by the e-tendering website. The time settings fixed in the server side & displayed at the top of the e-tendering site, will be valid for all actions of requesting, bid submission, bid opening etc., in the e-tender system. The bidders should follow this time during bid submission.

14. MODIFICATIONS AND WITHDRAWAL OF BIDS:

14.1 The bidder may modify or withdraw his bid before the due date and time of submission in the manner prescribed there for by the e-tendering website.

14.2 No bid can be modified subsequent to the deadline for submission of bids.

E. BID OPENING AND EVALUATION:

15. OPENING OF BIDS BY BSNL:

The designated Tender Opening Committee (TOC) of BSNL the bid opening committee shall count the number of bids downloaded from the e-tendering portal and assign serial numbers to the bids. For example, if 10 tenders have been received the bids shall be numbered as 1 of 10, 2 of 10 etc. All the members of TOC shall initial on all the pages of all the bids downloaded with date.

After opening the qualifying bid, all the documents contained therein shall be serially numbered and signed by the bid opening committee members and TOC should prepare and submit TOC report.

The Qualifying Bids downloaded shall be evaluated by the designated TEC and the result of evaluation after approval by the competent authority shall be uploaded on the E –tender portal for information of all concerned clearly mentioning the qualified bidders and not qualified bidders (with reasons for non-qualification).

The **FINANCIAL BID** shall be opened in the following manner:

15.1 The folder marked **"Financial Bid",** will be opened only for those bidders who have been found eligible after evaluation of Qualifying Bids.

15.2 The date and time of opening of Financial Bid shall be notified on e-tendering portal for information of all the bidders.

15.3 After opening the "Financial Bid" the TOC shall prepare the report containing the bidder's name, bid prices, modifications, etc.

15.4 In case there is a discrepancy in figures and words in the quote, the same shall be announced in the bid opening, but the quoted rates in words shall prevail.

15.5 TOC will hand over the Financial Bid opening report to CET for Financial evaluation.

16. CLARIFICATION OF BIDS BY BSNL:

To assist in examination, evaluation and comparison of bids, BSNL may, at its discretion ask the bidder for clarification of its bid. The request for its clarification and its response shall be in writing. However, no post bid clarification at the initiative of the bidder shall be entertained.

17. VERIFICATION OF BIDS:

If any of the documents required to be submitted along with technical bid is found wanting, the offer is liable to be rejected. However, BSNL at its discretion may call for any clarification regarding the document within a stipulated time period. In case of non-compliance to such queries, the bid shall be outrightly rejected without entertaining further correspondence in this regard.

18. PRELIMINARY EVALUATION:

BSNL shall evaluate the bids to determine whether they are complete, whether any computational errors have been made, whether required sureties have been furnished, whether the documents have been properly signed and whether the bids are generally in order.

Prior to the detailed evaluation, BSNL will determine the substantial responsiveness of each bid to the bid document. For purpose of these clauses a substantially responsive bid is one which conforms to all the terms and conditions of the bid documents without deviations.

BSNL may waive any minor infirmity or non-conformity or irregularity in a bid which does not constitute a material deviation, provided such waiver does not prejudice or affect the relative ranking of the bidder.

19. EVALUATION AND COMPARISON OF SUBSTANTIALLY RESPONSIVE BIDS:

BSNL shall evaluate in detail and compare the bids previously determined to be substantially responsive.

The evaluation and comparison of responsive bids shall be on the percentage deviation (above/below/at par) offered and indicated in schedule of quoting rates of the bid documents. Bidder quoting the lowest rates will be L-1 for this tender.

20. CONTACTING BSNL:

20.1 No bidder shall try to influence BSNL on any matter relating to its bid, from the time of bid opening till the time the contract is awarded.

20.2 Any effort by the bidder to modify his bid or influence BSNL in the BSNL's bid evaluation, bid comparison or the contract award decisions shall result in the rejection of the bid.

21. AWARD OF CONTRACT & DISTRIBUTION OF WORK:

21.1 BSNL shall consider award of contract only to those eligible bidders whose offers have been found technically, commercially and financially acceptable. The L-1 bidder is one who quotes the lowest rate.

21.2 If there are more than two tenderers at the same rate, tenderer having highest experience in category shall precede to the one having lower experience while deciding the ranking for considering the award of work. The final ranking of L1, L2, L3 etc. shall be worked out using the experience criteria of 3 years period category.

21.3. If L1 bidder refuses to accept the offer, L2 bidder may be given counter offer at L1 rate if deemed necessary.

22. BSNL'S RIGHT TO VARY QUANTUM OF WORK:

The quantities stipulated in the tender are estimated and the BSNL reserves the right to vary the value of works to the extent of -25% to + 25% of contract value during contract period at the same rates and terms & conditions with prior approval of the tender approving authority.

However, **in exceptional circumstances only**, due to change in scope of work or otherwise, any variation in the value of work may be permitted up to 50% of contract value with prior approval of the next higher authority after recording adequate reasons and justification for the same.

However, BSNL will not be liable to the contractor for any losses or damages, costs, charges the contractor may in any way sustain/suffer due to non-issue of work order/delay in making store available or delay in receipt of permission from road authorities / local bodies/forest department.

23. BSNL'S RIGHT TO ACCEPT ANY BID AND TO REJECT ANY OR ALL BIDS;

BSNL reserves the right to accept or reject any bid and to annul the bidding process and reject all bids, at any time prior to award of contract without assigning any reason whatsoever and without thereby incurring any liability to the affected bidder or bidders on the grounds for the BSNL's action.

24. <u>ISSUE OF LETTER OF INTENT:</u>

24.1 The issue of letter of intent shall constitute the intention of BSNL to enter into the contract with the bidder. Letter of intent will be issued as offer to the successful bidder.

24.2 The bidder shall within 15 days of issue of letter of intent, give his acceptance along with E-Performance Bank Guarantee& material security in conformity with **clause 11.5 of Section IV,** provided with the bid documents. However, Tender Issuing Authority reserves the right to extend the time limit if the reasons on which extension is sought deemed fit.

25. SIGNING OF AGREEMENT:

25.1 The signing of agreement shall constitute the award of contract on the bidder. Once tender is approved by the competent authority and LoI/APO is issued, the contractor shall deposit the Performance Bank Guarantee and material security in the form of E-Bank Guarantee(E-BG) /Bank Guarantee/DD within 15 working days. The bid security paid along with tender shall be refunded to the bidder after signing of the agreement.

25.2 In case, the successful bidder doesn't submit the Bank Guarantee(s) within 15 working days, and within the time permitted by the Tender approving authority, the bid will stand terminated. The EMD will be forfeited and the bidder will be debarred from participating in the same tender.

25.3 In such an event, the work can be awarded to L2 bidder at L1 price with approval from the tendering authority.

26. ANNULMENT OF AWARD:

Failure of the successful bidder to comply with the requirement of **clause 25 of section II** shall constitute sufficient ground for the annulment of the award and forfeiture of the bid security, in which event; BSNL may award the work to L2 bidder at L1 rate at the discretion of BSNL or call for new bids.

27. IMPORTANT INSTRUCTIONS

27.1 After the work commences the contractor or his authorized representative(s) shall be present at the site. The representative shall be authorized by the Divisional Engineer/AGM in charge based on the contractor's request.

27.2 Work shall be recorded in the site register by the site Engineer. The contractor or his authorized representative shall sign in the site register held by the site Engineer.

27.3 Since the time provided for the work is only a few days/weeks, all the communication shall be done at the site. If the contractor or his authorized representatives shall not sign the register, then it shall be considered that nobody is present at the site. Then only written communication shall be sent through registered post.

27.4 If the contractor does not provide the barricades, warning cones, warning tapes and work notice board, 5% additional penalty shall be levied. The same shall be provided by BSNL and 150% of the cost of the items shall be recovered from the contractor.

27.5 Due to underground utilities, if the manhole cannot be constructed at the site shown, the decision of the site engineer shall be final and binding on the contractor. If it is decided to shift the manhole within the neighborhood, the contractor shall construct at the place shown by the site engineer.

27.6 The Right of Way (RoW) shall be obtained by the successful bidder on behalf of BSNL along with the conditions imposed by the RoW authority or Traffic Police. RoW charges shall be paid by BSNL directly to the RoW authority based on demand note issued by the concerned authority.

27.7 In general, the permission is available for working at night only. The probable time for allowing trenchless/HDD/Open trench is from 10 PM to next day 6 AM. Delay in the work and non-commencement of the work due to these reasons shall not be entertained. The work may be accordingly restricted or terminated.

27.8 If the work extends beyond the time given in the work order, then the permissions from concerned authorities shall be obtained by the contractor. If the contractor finds the time allowed for completion of work to be less, he may accordingly deploy more men/machines to complete the work in time.

27.9 If the work is not progressing as per the schedule, the work may be either restricted or terminated.

27.10 If the manhole work is abandoned after commencing, the work may be either restricted or terminated.

27.11 In both the cases where the work is either restricted or terminated, apart from invoking necessary tender clauses, the contractor shall not be allowed to participate in any tender for a similar work under Tender Issuing Authority, for a period of three years.

27.12 After the ducts are laid by either Open trenching method or by HDD, the duct integrity test shall be carried out by the contractor on the next day. This is to facilitate the Manhole contractor to commence the manhole work.

27.13 If any one or more number of ducts is found to fail the duct integrity test, additional laying/ HDD of pipe/s shall be carried out by the contractor at no extra cost. The cost of the damaged pipe(s) shall be recovered from the contractor with additional 17.5% towards Overhead charges.

27.14 After the re-laying of additional pipe(s), duct integrity test shall be performed in the failed segment, on all the ducts once again.

27.15 After the ducts are laid by either Open trenching method or by HDD, if duct integrity test is not carried out by the contractor, the test shall be carried out by BSNL and only 50% of the actual payment shall be made to the contractor for PLB pipe laying.

27.16 For any unfinished work BSNL shall not effect any payment.

27.17 The material abandoned shall be the property of BSNL. No claim in this regard from the contractor shall be entertained.

27.18 BSNL reserves the right to construct the manhole, from the stage at which the construction is abandoned by the contractor. This is because the pipe is already terminated at the said location.

27.19 Rate should be quoted in figures as well as in words and corrections should be attested, otherwise tender shall be rejected.

27.20 It is the responsibility of the contractor to keep the pipes inside the existing cement concrete duct, where the HDD/OT commences. Duct poking shall be arranged by BSNL.

27.21 The contractor shall restore the cut portions of BT/Concrete roads to their original conditions immediately on completion of the pipe laying works. If the restoration work is not satisfactory with reference to the standards prescribed by the Corporation / Highways, the amount equal to the charges as claimed by the Corporation/Highways will be recovered from the contractor.

27.22 The contractor shall deploy as many machines as required for the timely completion of the work.

27.23 The fact that the contractor is working simultaneously in one or more sections, the same Division/other Divisions will not absolve him from the responsibility of completing the work in time.

27.24 Wastage of PLB pipes will be permitted to an extent of 5% of the completed length as measured on the road surface.

27.25 In case of wastage of PLB pipes beyond 5% the cost of PLB pipe will be recovered from the contractor with additional 17.5%.

28. RATE OF EMPTY CABLE DRUM

These are the rate of the empty cable drums which have to be deducted from contractor bills as per tender document if cable pulling is to be carried out by the same contractor.

S/No	Particulars of drum	Rate of disposal
1	Optical Fibre cable drum 24F/12F/48F	Rs.100/- per empty cable drum
2	Optical Fibre cable drum 96F/144F/288F	Rs.200/- per empty cable drum

Note: -These are fixed rates and no variations shall be acceptable from these rates.

29. PROFORMAS

Annexure-A

MATERIAL SECURITY BOND FORM

Where…… (hereinafter called "the Contractor") has been awarded the contract of cable construction work, as per tender Number_____

KNOW ALL MEN by these Present that WE ……OF…….having our registered office at _____(hereinafter called "the Contractor") are bound unto…….....(hereinafter called the "BSNL) in the sum of ……. for which payment will and truly to be made of the said BSNL, the bank binds itself, its successors and assigns by these presents.

THE CONDITIONS of the obligations are:

1. If the Contractor is unable to keep stores issued to him, property, i.e. the store provided to the contractor, by BSNL are damaged or

2. The store issued to the contractor by BSNL are stolen or

3. The Contractor is not able to provide proper account of the stores issued to him/ her / them by BSNL. We undertake to pay to BSNL up to the above upon receipt of its first written demand without BSNL having to substantiate its demand, provided that in its demand, BSNL will note that the amount claimed by it is due to it owing to the occurrence of one or two or all of three conditions specifying the occurred condition or conditions.

   This guarantee will remain in force up to and including One hundred and eighty (180) days after the period of Contract validity, and any demand in respect thereof should reach the Bank not later than the above date.

Signature of the bank

Signature of the Witness

Name of Witness

Address of Witness

Annexure-B

Proforma of Performance Bank Guarantee (PBG)

(To be typed on non-judicial stamp paper of appropriate value)

Dated:…….

Performance Bank Guarantee (PBG)

Whereas ……<Contract awarding authority>(hereinafter referred to as BSNL)has issued an Award Letter no...…… Dated……/……/20…….. for awarding the work of Optical Fibre Cable Laying work to M/s ……..…… (hereinafter referred to as "Contractor") and BSNL has asked them to submit a e-performance bank guarantee in favour of …… O/o …… of Rs. ……..……/- (hereinafter referred to as "PBG. Amount") valid up to ……/……/20……(hereinafter referred to as "Validity Date"). Now at the request of the Contractor, We …… Bank ……Branch having ……. (Address) and Regd. office address as ……...…… …… …… …… (hereinafter called "the Bank") agreed to give this guarantee as hereinafter contained:

1. We, "the Bank" do hereby undertake and assure to BSNL that if in the opinion of BSNL, the Contractor has in any way failed to observe or perform the terms and conditions of the said agreement or has committed any breach of its obligations there-under, the Bank shall on demand and without any objection or demur pay to BSNL the said sum limited to PBG Amount or such lesser amount as BSNL may demand without requiring BSNL to have recourse to any legal remedy that may be available to it to compel the Bank to pay the same.

2. Any such demand from BSNL shall be conclusive as regards the liability of Contractor to pay to BSNL or as regards the amount payable by the Bank under this guarantee. The Bank shall not be entitled to withhold payment on the ground that the Contractor had disputed its liability to pay or has disputed the quantum of the amount or that any arbitration proceeding or legal proceeding is pending between Contractor and BSNL regarding the claim. Liability of the bank under this present being absolute and unequivocal.

3. We, the Bank further agree that the guarantee shall come into force from the date of its issue and shall remain in full force and effect up to its Validity date.

4. The Bank further agrees that BSNL shall have the fullest liberty without the consent of the Bank and without affecting in any way the obligations hereunder to vary any of the terms and conditions of the said agreement or to extend the time for performance of the said agreement from any of the powers exercisable by BSNL against the Contractor and to forebear to enforce any of the terms and conditions relating to the said agreement and the Bank shall not be relieved from its liability by reason of such failure or extension being granted to Contractor or through any forbearance, act or omission on the part of BSNL or any indulgence by BSNL to Contractor or any other matter or thing whatsoever which under the law relating to sureties would but for this provision have the effect of relieving or discharging the guarantor.

5. Notwithstanding anything herein contained; (a) The liability of the Bank under this guarantee is restricted to the PBG Amount and it will remain in force up to its Validity date. (b) The guarantee Page shall stand completely discharged and all rights of BSNL under this Guarantee shall be extinguished if no claim or demand is made on us in writing on or before its validity date.

6. In case BSNL demands for any money under this bank guarantee, the same shall be paid through DD/Banker's Cheque/RTGS …….

7. The Bank guarantees that the below mentioned officer who have signed it on behalf of the Bank have authority to give this guarantee under its delegated power.

Place: ……

Date: …….

(Signature of the Bank Officer)
Rubber stamp of the bank
Authorized Power of Attorney Number
……
Name of the Bank Officer:
……
Designation: ……
Complete Postal address of Bank:
…….
Telephone Numbers …….
Fax Numbers …….

Annexure-C

NEAR-RELATIONSHIP CERTIFICATE

1. The bidder should give a certificate that none of his/ her near relative, as defined below, is working in the BSNL units*

In case of bidder being proprietorship firm, this certificate will be given by the proprietor. For partnership firm, the certificate will be given by all the partners.

In case of bidder being Company/Limited company, the certificate will be given by all the Directors of the company, but excluding following:

a) Government of India/ Financial institution nominees and independent non-Official part time Directors, appointed by Govt. of India or the Governor of the state and
b) Full time Directors of PSUs, both Central and State

Due to any breach of these conditions or incorrect declarations by the bidder [Company or firm or any other person], the tender /bid submitted will be cancelled and Bid Security will be forfeited at any stage, whenever it is noticed and BSNL will not pay any damage to the company or firm or the concerned person.

*Unit, shall be taken as following:

(a) In case of any near relative of the bidder being non-executive employees, the "BSNL unit" is defined as "Business Area (BA)".
(b) In case of any near relative of the bidder being executive (up to AGM/STS level), the "BSNL unit" is defined as "BSNL Circle".
(c) In case of any near relative of the bidder being higher executive (DGM/JAG or higher), "BSNL unit" is defined as "BSNL as a whole".

2. The near relatives for this purpose are defined as: -

(a) Members of a Hindu undivided family.
(b) They are husband and wife.
(c) The one is related to the other in the manner as father, mother, son(s) & Son's wife (daughter in law), Daughter(s) and daughter's husband (son in law), brother(s) and brother's wife, sister(s) and sister's husband (brother-in-law).

NEAR-RELATIONSHIP CERTIFICATE

(Format of the Certificate to be given as per the clause 34.3 of Section-4 Part-A by the bidder in respect of status of employment of his/ her near relation in BSNL)

I……..……S/o……..…R/o……hereby certify that none of my relative(s) as defined in the tender document is/are employed in any BSNL unit as defined in the clause in the Tender enquiry, on Near relationship.

OR

* Following are the details of near relatives working with the BSNL.

S. No.	Name of the Relative	Designation	Name of the Unit (Office & section of BSNL) where working

* Strike off whichever is not applicable.

In case at any stage, it is found that the information given by me is false/incorrect, BSNL shall have the absolute right to take any action as deemed fit/without any prior intimation to me.

DATE_____ SIGNATURE OF TENDERER WITH SEAL

Annexure-D

Proforma for Experience Certificate

No:_____ Dated _____

Subject: Experience certificate for optical fiber cable construction work.

It is certified that M/s _____ having office at _____ whose Proprietor(s)/Partner(s)/Director(s)are

_____

_____

_____have successfully carried out **optical fiber cable construction work** for the amount and Route KM mentioned hereunder. It is certified that these amounts have been paid to the above-mentioned contractor.

S.N.	**Period during which payments Made**		**Contract No(s).**	**Route km**	**Amount paid (In Rs.)**
	From	**To**		(km)	
1.	01-04-YYYY	31-03-YYYY+1			

DGM/AO/DDO/ Officer responsible for
maintaining contractors' ledger
Signature with office seal

Note:

The Above Proforma is indicative of the intent that the experience certificate should clearly mention that the route km of OFC construction done, and the amount has been paid as the experience is to be reckoned only in terms of amount paid and not in any other physical values like km of work done etc. Therefore, the available certificates with the bidder in any other format will be taken as valid provided it mentions the requisite wordings that the amount has been paid and period during which payments have been made.

Annexure-E

Declaration about Genuineness of Documents/Certificates
(to be duly filled, signed, stamped, scanned and submitted
with other documents as per **clause-7 of Section-II**)

I/We……. hereby declare that the information furnished in the bid in response to the **NIT No. ----------**is true and correct. I/we verify the genuineness and correctness of all documents, including experience certificates attached with the bid submitted electronically or as a hard copy. Further I also declare that I have submitted the tender document digitally signed, without any additions/deletions/modifications, as a token of having read, understood and accepted the terms and conditions therein. I/we am/are aware that I/we shall be held responsible in case any document attached is found false/forged/fabricated/ tempered/manipulated at any stage and BSNL is fully competent to take any action against me/my firm as deemed fit in accordance with the terms and conditions of the contract and law of the land.

Place:Signature of tenderer / Authorized signatory…….

Date:Name of the tenderer…….

Seal of the tenderer

Annexure-F

SAMPLE APPLICATION FOR DEPTH RELAXATION IN CASE OF ACHIEVING LOWER DEPTH

To

The Engineer-in-Charge,
……..

Sub: Approval for lower depth than 165/135/90 cm (depending on the terrain as applicable)

Ref: Work Order No……

Sir,

In the following sections the normal depth of 165 cm could not be achieved due to reasons stated against each section vide above reference work order.

Your approval is requested.

Sl. No.	Section No.	Location From	Location To	Length in metres	Actual Depth Achieved in metres	Reason
1						
2						
3						

Yours faithfully,

Name of the Bidder with address & date.

Recommendation of the Site-in-Charge.

Annexure-G

UNDERTAKING FOR COMPLETION OF WORK

(To be submitted on non-judicial stamp paper of Rs. 100/-)

I_____S/o_____ R/O_____ have submitted Bid for NIT No. _____ Dated _____ I hereby undertake that, in case of successful bidder as L1 against the said NIT and the said work allotted to me, I will complete the said job within stipulated time as mentioned in Work Order issued by the authority. In case of failure of the said job within the stipulated time as prescribed in Work Order, the Tender Issuing Authority having the full right to forfeit the PBG submitted by me against the aforesaid NIT as well as I may be rescinded from the contract by barring participation in future tenders under Odisha Circle, BSNL for next financial years with effect from the date of rescindment.

DATE_____ SIGNATURE OF TENDERER WITH SEAL

Annexure-H

DECLARATION FOR NON-BANNED/NON-BLACKLIST/ DEBARRED

(On Stamp Paper, Rs. 10/-)

(In case of E-stamp only Article 4 Affidavit must be used)

I/We……..……..……..…. Proprietor / Partner(s)/ Director(s) of M/s……..... hereby declare that the firm/company namely M/s……has not been blacklisted and debarred by BSNL/MTNL/TCIL/ITI/Rites/HCL/ PGCIL/ RailTel/any other Govt. (Central/ State) undertaking in the past & at the time of submission of bid.

OR

Was blacklisted or debarred by ……..from taking part in tender of BSNL/MTNL/TCIL/ITI/Rites/HCL/ PGCIL/ RailTel/any other Govt. (Central/ State) undertaking for a period of .....years w.e.f. ..... The period is over on ……. or going to over on .....

In case the above information is found false. I/We are fully aware that the tender / contract will be rejected / cancelled by BSNL and EMD / SD shall be forfeited.

In addition to the above BSNL will not be responsible to pay the bills for any completed / partially completed works.

Place:Signature of bidder

Date:Or Officer authorized to sign the bid document on behalf of the bidder

Name & address of the firm……

30. OF CABLE CONSTRUCTION SPECIFICATION MANUAL

The guidelines in the form of Engineering Instructions (E.Is.) on Construction Practices of Optical Fibre Cables are issued by T & D wing of the Department, from time to time. However, the present practices are summarized here under, from the point of view of describing scope of work under various items of work. This document will need revision as and when practices undergo any revision.

General

The Optical Fibre Cable is planned considering the following objectives of the scheme:

1.1 Minimum possible route length vis-a-vis route having maximum number of towns with potential telecom growth.

1.2 Linking of small exchanges off main road by leading in O.F.C. vis-a-vis routing the main cable itself via such exchanges.

After deciding above mentioned issues, a detailed measurement of lengths of cable route along with details of rail/road crossings, culverts, causeways, etc. may be recorded in the detailed survey register. The probable locations of joints, terminations and re-generators may also be decided and marked on the route map.

On the basis of surveys, general permission from road and rail authorities for laying the Optical Fibre Cable along the decided routes and permission for rail /road crossings will have to be obtained. Generally, O.F.C. is laid straight as far as possible along the road near the boundaries, away from the burrow pits. The O.F.C. is laid along the roads at a minimum distance of 15 meters from the center line of the road or in accordance with the permission from the concerned road authorities in view of their road widening plan. As the O.F.C. carries high-capacity traffic and is planned for 40 years of life, it is imperative that the cable is laid after obtaining due permission from all the concerned authorities to avoid any damage/shifting at a later stage and also disruption of services/ revenue loss.

In special cases, where it may be necessary to avoid burrow pits or low-lying areas, the cable may be run underneath the shoulders at a distance of 0.6meter from the outer edge of the road embankment provided the same is located at least 4.5 meters away from center line of road and 1.2 meter below the road surface.

The Optical Fibre Cable is laid through PLB Pipes/Coils buried at a nominal depth of 165 cm. The steps involved in OF Cable construction are as under:

1.3 Excavation of trench upto a nominal depth as specified, according to construction specifications along National/State Highways/other roads and also in city limits as mentioned in the notice inviting tender.

1.4 Laying of PLB pipes/coils coupled by PLB sockets in excavated trenches, on bridges and culverts, drawing of 6 mm polypropylene para pro rope (P.P. rope) through the PLB pipes/coils as per Construction Specifications and sealing of PLB pipe ends at every manhole by PLB end caps of suitable size.

1.5 Providing of mechanical protection by R.C.C. Pipes/GI pipes and/or concreting/chambering according to construction specifications, wherever required.

1.6 Fixing of GI pipes/troughs with clamps at culverts/bridges and/or chambering or concreting of G.I. Pipes/troughs, whenever necessary.

1.7 Back filling and dressing of the excavated trenches according to construction specifications. Digging of pits 1 meter towards jungle side at every manhole chamber to a depth of 60 cm, fixing of route indicator, concreting and back filling of pits. Painting of route indicators with yellow colour and sign writing denoting route indicator number as per construction specifications.

1.8 Earthing (Grounding) of the splice closure and the cable (For Armoured Optical Fiber Cable):

The splice closure and the cable armour shall be earthed at regular intervals (may be at every splice point or as directed by the site-in-charge) to avoid any damage to the cable/splice or the persons working on the cable. Though there are different types of Earthing techniques available, it is proposed to use Spike Earth to obtain necessary earthing potential of <5Ω. It is suggested taking into consideration the limited space availability around the splice pit. Ground Conditions The spikes will be exposed to a range of soil types, including the following: • Clay - soft and compacted; • Granite and gravel; • Rock, shale and sandstone; • Sand; and • Alluvial sediment. Across this range of ground conditions, both the electrical conductivity and corrosive properties of the soils, vary markedly.

Spike Earth: The spike earth arrangement of earth electrode system may consist of one or more numbers of 25.0 mm or 38.0 mm dia. GI pipes, each of 275 cm or 375 cm length. Each pipe is used as earth spike and is so driven into the ground that its top remains at some depth below the ground surface. The spacing between any two spikes should not be less than 375 cm ordinarily. In no case, it should be less than 250 cm. There are different types of Earth rods are available and if required the installer may choose appropriate one for earthing to obtain <5Ω. **For e.g. one such earthing rod of 1853 mm (i.e. 25 + 1752 + 76) long fabricated from 20 mm dia M.S. Rod, the bottom of which to be made cut of the same rod in the shape of a cone – 76 mm long and the forged head made out of the same rod with 30 mm (dia) & 25 mm (height).**

The positions of the spikes (may be one or more, may be 4 Nos. at every corner of the pit) should be marked out around the splice pit and a trench no wider than necessary should be excavated. The trench should be 70 to 100 cm deep or it may be around the RCC splice chamber. The spike should be driven vertically into the trench until the top of the driving head is 30 cm above the bottom of the excavation. Spikes can be driven in quite easily with a sledge hammer for round electrodes. A new arrangement has been developed consisting of a sliding hammer placed on the electrode so that the work may be carried out at a level convenient to the workman without a ladder or auxiliary platform. The blows are delivered to the rod at a point near to the ground. The driving head or a suitable bolt is supplied with each earth spike and this must be used to prevent the top of the spike being damaged during the driving operation, when the spike has been driven fully into the ground, it will probably be found that the driving head will be tight fit on the spike and in these circumstances, it may be left in position.

If the earthing requirement of <5Ω is not achieved with one spike, then, multiple spikes are driven and the spikes should be connected together by a continuous main earth conductor, bare tinned copper protected by PVC pipe to prevent corrosion. The PVC pipe containing the earth conductor should be lightly dressed down and wiped on to the earth conductor at each side of the binder. The binder should be soft copper wire, 1.4 mm. diameters wound round the earth conductor at the points where it is held in the clamp on the spike. The PVC pipe used should be of sufficient diameter to carry the copper conductor and thickness. The binder and the exposed part of the earth conductor should be thoroughly tinned. Earth Enhancing Compound in the form of granular mixture of Calcium Bentonite (clay) & Gypsum may also be used wherever necessary. After completion of earthing arrangements, the pit shall be filled with sand incorporating warning bricks etc. as per standard procedures.

1.9. **Termination &earthing arrangement at the exchanges (Cable terminating ends)(For Armoured Optical Fiber Cable)**:

The armoured optical fibre cable shall be terminated in the termination box (FTB) as per TEC GR No: TEC/GR/TX/FTB-01/03/MAY-2010 with amendments (if any) or the latest. This box is usually ordered and supplied as part of cable supply. This termination box shall be installed inside the exchange building. The fibres of the cable shall be spliced to the pigtails for connection to the optical line systems. In the termination box arrangements have been provided to electrically parallel the armouring of the cable, the metallic components of the cable and the body of the termination box. This common point is extended to outside the body of termination box to facilitate the earthing arrangements. A standard earth of less than 5 ohms is to be prepared and connected to this point of the termination box. It is proposed to use **Spike Earth** to obtain necessary earthing potential of <5Ω. **This earth shall be separate from Equipment Earth and Electrical Earth**. Also, the importance of earthing shall be made known to all the maintenance staff by educating the hazards it poses if not properly earthed. It is preferred that an insulated multi-strand earth wire (generally green in colour)

shall be used to identify the earth link. Apart from the above guidelines, specific instructions mentioned in the user manuals of the respective cable manufactures may also be referred and adhered to.

2. Documentation.

The documentation, consisting of the following shall be prepared exchange to exchange for intra SSA OFC links and for each section in case of inter SSA/National long-distance routes.

2.1 Route Index Diagrams – General: This diagram shall consist of Cable Route Details on Geographical Map drawn to scale the prominent land marks and alignment of cable with reference to road. This shall be prepared on A-3 sheets of 80 GSM.

2.2 Route Index Diagrams – Profile: These diagrams will contain:

2.2.1 Geo Co-ordinates (Latitude & Longitude) of HDD pit and Joint/Loop chamber.
2.2.2 Offset of HDD pit from center of the road. At least two Offset of HDD pit from prominent land marks to facilitated locating the HDD in future.
2.2.3 Make and size of the cable.
2.2.4 Offset of cable from center of the road at every 10 meters for Open Trench work.
2.2.5 Depth profile of Cable at every 10 meters.
2.2.6 Details of protection with type of protection depicted on it.
2.2.7 Location of culvert and bridges with their lengths and scheme of laying of PLB Pipe/PLB Blowing Type/Pre-installed ropes thereon.
2.2.8 Important landmarks to facilitated locating the cable in future.
2.2.9 Location of joints and pulling manholes.

These diagrams shall be prepared on **A-4 sheets** of 80 GSM. On one sheet profile of maximum 400 meters shall be given to ensure clarity.

2.3 Joint location Diagram: This diagram will show:

2.3.1 Geographical location of all the joints.
2.3.2 Depth of Joint Chamber covers from ground level
2.3.3 Type of chamber (Brick/Pre-cast)
2.3.4 Length of O.F. Cable kept inside the joint chamber from either direction.

This shall be prepared on **A-4 sheets** of 80 GSM. All the diagrams 2.1, 2.2&2.3 shall bear the signatures of the contractor, the Engineer-in-charge as a proof of accuracy of the details. The diagrams shall be bound in A-4 size book with cover. The cover sheets shall be of 110 GSM and laminated. The front cover shall have the following details.

2.3.5 Name of the SSA/Project Organization.
2.3.6 Name of the OFC Link
2.3.7 Name of the Contractor
2.3.8 Date of commencement of work.
2.3.9 Date of completion of work for each route/section 6 sets of above-mentioned document shall be submitted.

3. PLB/ROPE Materials used:

3.1 PLB PIPE

3.2 PLB Socket: For coupling PLB Pipes.

3.3 PP Rope:6mm PP rope is drawn through the PLB pipes and safely tied to the end caps at either ends with hooks to facilitate pulling of the OF cables at a later stage.

3.4 PLB End Caps: For pulling the cable through the pipes, it is necessary to have suitable manholes at every 200m length and also at bends and corners suitably located. The pipes are laid for 200m or less at a time for the distance between two manholes. The ends of the PLB pipes/coils are closed with PLB End Caps. A suitable arrangement should be provided in the End Cap to tie PP Rope.

3.5 MATERIALS FOR MECHANICAL PROTECTION: For lesser depths requiring mechanical protection as per specifications and in built-up areas, in towns and cities falling within the municipal limits, suitable mechanical protection is provided to PLB pipes using concreting of size 25 cm x 25 cm reinforced with MS weld mesh or a combination of any of these as per the instructions of the SDE-in-charge of the work as per instructions of the Engineer-in-Charge with DWC pipes.

3.6 M.S. Weld Mesh: The PLB pipes can also be protected by embedding it in concrete of size of 25 cm x 25 cm reinforced with MS weld mesh. The strength of RCC/CC is dependent on proper curing; therefore, it is imperative that water content of CC/RCC mix does not drain out into the surrounding soil. In order to ensure this, the RCC/CC work should be carried out by covering all the sides by yellow PVC sheets of weight not less than 1 kg per 8 sq. m. to avoid seepage of water into the soil.

3.7 RUBBER BUSH: To prevent entry of rodents into PLB pipes, the ends of PLB pipes are sealed at every manhole and joint using rodent resistant hard rubber bush (cap), so that it should be able to prevent the entry of insects, rodents, mud, and rainwater into the PLB pipe/Coil.

3.8 ROUTE INDICATOR: The route indicators are co-located with each manhole. In addition, route indicators are also to be placed where route changes direction like road crossings etc. The route indicators made of precast RCC should have the following dimensions: **Base (250 mm x 150mm);Top(200 mm x 75 mm);Height(1250 mm)**.The word '**BSNL OFC'** should be engraved on the route indicators. (**See figure 'A' for details**). The route indicators are painted yellow and the same are placed at 2 ft. away from the center of the trench towards jungle side. The engraved word '**BSNL OFC'** should be painted in white, on route indicators**.** Numbering of route indicators should be done in white paint. The numbering scheme for route indicators will be Joint No/Route Indicator No. for that joint. For example, 2/6 marking on a route indicator means 6th route indicator after 2nd joint. Enamel paints of reputed brand should be used for painting and sign writing of route as well joint indicators.

3.9 JOINT CHAMBER: -The joint chamber is provided at every joint normally at a distance of 2 km to keep the O.F.C. joint well protected and also to keep extra length of cable which may be required in the event of faults at a later date. **For all new routes, joint/loop chamber is to be placed at a distance of 500M for easy fault restoration**. **OFC joint may be done as per OFC drum length and in intermediate chamber, 20M OFC coiling may be kept**. The joint chambers are made at site using bricks and mortar or are of pre-cast RCC type.

Supply and fixing of Prefabricated RCC Closure as Joint Chamber:

For supplying RCC Ring, firstan RCC ring closure of **size 1.2m Φ(dia) x 0.8m depth** and **rim thickness 50mm made of standard 1:2:4 concrete mix**. It should have **a split (with two halves) RCC cover of thickness 50mm** made of cement concrete mix of 1:2:4 (1: cement, 2: coarse sand, 4 stone aggregate mm nominal size reinforced with MS fabric 75 x 75 mm mesh of **weight not less than 7.75 Kg per sq. meter** having **galvanized iron handles for each half** to facilitate easy lifting, to be used as joint chamber**.** Base should be circular RCC plate 40 mm thick made of cement concrete mix of 1:2:4.**The word 'OFC' should be engraved on each half of the cover**. The joint chamber should be so designed that PLB pipe/PLB Blowing Type/Pre-installed rope ends remain protruding minimum 5 cm inside the chamber. The PLB pipes should be embedded within the chamber in such a way that PLB pipe remains free from the weight of the construction. The joint chamber should be plastered on all internal surfaces and top edges with cement mortar of 1:3 (1: cement, 3: coarse sand) with a floating coat of complete cement as per standard. **The joint chamber is to be filled with clean sand before closing**.

In Golden Quadrilateral & East West Corridor routes (every 10 km):

For fixing Precast RCC chamber, first a pit of size 1.5 m x 1.5 m x 1.65 m depth is required to be dug. Precast RCC chamber consists of three parts (i) round base plate in two half of 140 cm dia and 5 cm thickness (ii) full round RCC joint chamber with diameter of 120 cm and height of 100 cm and thickness of 5 cm (iii) round top cover will be in two halves with diameter of 140 cm and thickness of 5 cm having one handle for each half in centre and word **'BSNL**

OFC' engraved on it**.** After, fixing the precast RCC joint chamber, the joint chamber is filled with clean sand before closing. Lastly, back filling of joint chamber pit with excavated soil is carried out.

In case of Armoured Optical Fiber Cable work, necessary earthing to be done as per specification using Spike Earth rod at every joint Chamber. The maximum earth resistance when measured shall not be more than 5 ohms.

3.10. CABLE PULLING AND JOINTING/SPLICING:

3.10.1 CABLE PULLING:

Manholes marked during PLB pipe Type/Pre-installed ropes laying of approx. size of 3m length x 1.0m width x 1.65m depth shall be excavated for pulling the cables. There may be situations where additional manholes are required to be excavated for some reasons, to facilitate smooth pulling of cable. Excavation of additional manholes will be carried out, without any extra cost. Dewatering of the manholes, if required, will be carried out without any extra cost. Dewatering/ De-gasification of the ducts, if required, will be carried out without any extra cost. The existing 6mm PP rope shall be replaced by with PP rope of 8mm dia between the two consecutive manholes. This is to ensure that the PLB pipes Type/Pre-installed ropes are cleaned for pulling the cable without exerting undue tension on the OF cable. While cleaning, excessive pressure should not be put this may result in breaking of P.P. rope and thus required opening of additional manholes. However, in case the cleaning rope gets stuck up during pulling, the location of clogging of PLB Pipe Type/Pre-installed ropes should be measured and located accurately. The trench at that location should be opened and the PLB Pipe Type/Pre-installed ropes should be cleaned properly or if not possible, it should be changed by a clean new PLB Pipe Type/Pre-installed ropes to facilitate easy cable pulling at a later stage without any breakage. If clogging of PLB Pipe Type/Preinstalled ropes is in the location where the pipes are protected either by RCC Pipe Type/Pre-installed ropes or by concreting and the protection is broken for cleaning/changing the PLB Pipe, the protection thus removed should be brought back to normal by the contractor without any extra cost. However, PLB Pipes Type/Pre-installed ropes, RCC Pipes required for this purpose will be supplied by BSNL.

3.10.2 CABLE BLOWING

3.10.2.1 The tenderer should blow/de blow the cable in the PLB duct by air jet blowing as per Specification. Within 7 days of notice, the contractor shall commence Blowing /de blowing and complete the work at an average speed of 4 KM per day. There shall not be any damage to the OF cable blown/de blown. Cable blowing shall be done using blowing machine which can be kept inside the manhole, as it will avoid formation of loop outside. Compressor of suitable capacity shall be used.

3.10.2.2 Blowing has to be done for the route length, end to end. This implies work may have to be done beyond the section awarded to the contractor, and he is bound to do it. The works involved in blowing are listed below.

3.10.2.3 Transportation of all materials issued by BSNL from stores including cable drums, joint rings and handling of the same (including loading, unloading, handling and local distribution).

3.10.2.4 Digging of manhole and reinstatement after blowing of cable.

3.10.2.5 The opening of End plug, de-roping of rope in PLB pipes.

3.10.2.6 Duct integrity test as per specifications.

3.10.2.7 If any coupler is not connected properly, they should be connected at the coupling points. If required the coupling points are to be dug open for this purpose and the existing duct cut, aligned and coupled properly. Extra piece of pipe if required has to be coupled.

3.10.2.8 Blowing of cable and fixing cable seal plug.

3.10.2.9 Keeping 30metres coil in every joint chamber.

3.10.2.10 Necessary earthing at each Chamber by Spike earth (Earth resistance value <5 Ω)

3.10.2.11 Digging of joint chambers, placing RCC chambers supplied by BSNL, and filling with river sand.

3.10.2.12 Closing the manhole/joint cover and ensuring that the cover sits properly. If there is time gap after laying of pipes and blowing, the contractor should carry out Duct Integrity test of the pipe to the specified pressure and ensure that there is no block or leak, as per specifications. If there are blocks the same shall be rectified. A checklist may be submitted to the site Engineer of the section after every pipe is tested, indicating the block etc. and the action taken to rectify the duct. No separate rates are payable for duct integrity test or rectification of ducts.

Cable should be blown for the entire section in the pre-assigned pipe & colour. No crossing should take place in manholes between different coloured pipes, if multiple pipes are available. Ensure no mud, water or debris goes inside the pipe during the work. Any obstructions found in the pipeline after cable blowing, shall be removed at the expense of the Contractor.

Depending upon the length of the cable and the addl. length required at manholes, chain measurement should be taken from the starting end of the section, and the drums should be properly located.

The drums are of approx. 2 km to 4 km and may sometimes be smaller. It may be preferable to blow the OF cable by placing the drum at the manhole point nearest to midpoint of the coil size and blow in both directions. The standard practice is to keep the clock-wise end of the cable to 'A' side of the route and the anticlockwise end to 'B' side. Also, it is the convention to have the clockwise end on top and anticlockwise at the bottom of the drum. With that in view, the OF Cable is mounted on the jack and wheel and positioned at the manhole point. It is important to keep the drum shaft (axle) in horizontal position by adjusting the jacks. After it is set in position, the cable should be inserted into the blowing machine.

Proper communication is to be established between the lead-in and take out manholes. This can be done by suitable Trans Receivers (Walkie Talkies) or Magneto telephones with drop wire. This has to be arranged by the contractor and communication with each point should be kept throughout the blowing. OF cable at RCC chambers should be lead along the walls & suitable MS clamps should be provided (4 on each side). Additional protection of cable if required shall be provided at manholes.

4 EXCAVATION OF TRENCHES

4.1 Trenching

Location and Alignment of the Trench: In city areas, the trench will normally follow the footpath of the road except where it may have to come to the edge of the carriage way when cutting across road with specific permissions from the concerned authorities maintaining the road (such permissions shall be obtained by the department). Outside the city limits the trench will normally follow the boundary of the roadside land**.** However, where the road side land is full of burrow pits or a forestation or when the cable has to cross culverts/bridges or streams, the trench may come closer to the road edge or in some cases, over the embankment or shoulder of the Road (Permissions for such deviations for cutting the embankment as well as shoulder of the road shall be obtained by the contractor on behalf of BSNL).

The alignment of the trench will be decided by a responsible departmental official, not below the rank of a Junior Telecom Officer. Once the alignment is marked, no deviation from the alignment is permissible except with the approval of Engineer-in-charge. While marking the alignment only the center line will be marked, and the Contractor shall set out all other work to ensure that, the excavated trench is as straight as possible. The Contractor shall provide all necessary assistance and labour, at his own cost for marking the alignment. Contractor shall remove all bushes, undergrowth, stumps, rocks and other obstacles to facilitate marking the center line without any extra charges. It is to be ensured that minimum amount of bushes and shrubs shall be removed to clear the way and the contractor shall give all consideration to the preservation of the trees.

4.2 Line-Up: The line-up of the trench must be such that PLB pipe(s) shall be laid in a straight line except at locations where it has to necessarily take a bend because of change in the alignment or gradient of the trench, subject to the restrictions mentioned elsewhere.

4.3 Method of Excavation: In city limits as well as in built up areas, the contractor shall resort to use of manual labour only to ensure no damage is caused to any underground of surface installations belonging to other public utility services and/or private parties. However, along

the Highways and cross country there shall be no objection to the Contractor resorting to mechanical means of excavation, provided that no underground installations exist in the path of excavation, if any, are damaged. There shall be no objection to resort to horizontal boring to bore a hole of required size and to push through G.I. Pipe (65mm/40mm dia) through horizontal bore at road crossing or rail crossing or small hillocks etc.

All excavation operations shall include excavation and 'getting out'. 'Getting out' shall include throwing the excavated materials at a distance of at least one meter or half the depth of excavation, whichever is more, clear off the edge of excavation. In all other cases 'getting out' shall include depositing the excavated materials as specified.

In Rocky strata excavation shall be carried out by use of electro mechanical means like breakers or by blasting wherever permissible with express permission from the competent authority. If blasting operations are prohibited or not practicable, excavation in hard rock shall be done upon authorization.

Trenching shall as far as possible be kept ahead of the laying of pipes. Contractor shall exercise due care that the soil from trenching intended to be loose for backfilling is not mixed with loose debris. While trenching, the Contractor should not cause damage to any underground installations belonging to other agencies and any damage caused should be made good at his own cost and expense.

The contractor to prevent any accident to pedestrians or vehicles shall provide necessary barricades, night lamps, warning boards and required watchman. While carrying out the blasting operations, the contractor shall ensure adequate safety by cautioning the vehicular and other traffic. The contractor shall employ sufficient manpower for this with caution boards, flags, sign writings etc.

The contractor should provide sufficient width at the trench at all such places, where it is likely to cave in due to soil conditions without any extra payment. A free clearance of minimum 15 cm should be maintained above or below any existing underground installations. No extra payment will be made towards this.

In order to prevent damage to PLB pipes over a period of time, due to the growth of trees, roots, bushes, etc. the contractor shall cut them when encountered in the path of alignment of trench without any additional charges.

In large burrow pits, excavation may be required to be carried out for more than 165 cm in depth to keep gradient of bed less than 15 degrees with horizontal. If not possible as stated above, alignment of trench shall be changed to avoid burrow pit completely.

4.4. Depth and Size of the Trench:

For Normal and Hard soil for all areas, minimum depth shall be 165 cm and Table-1 of Section–IV, Clause 14.1.4.1shall be followed for penalty. For Soft Rock& Hard Rock for all areas in Odisha Circle, minimum depth shall be 135 cm andTable-2 of Section–IV, Clause 14.1.4.1 shall be followed for penalty. For Hilly terrain, minimum depth shall be 90 cm shall be for types of soil and Table-3 of Section–IV, Clause 14.1.4.1 shall be followed for penalty.

However, Engineer-in-charge in exceptional cases due to adverse site conditions encountered, may allow to lay PLB Pipes at a lesser depth with additional protection. In all cases, the slope of the trench shall not be less than 15 degrees with the horizontal surface. The width of the trench shall normally be 45 cm at the top and 30 cm at the bottom. In case, additional pipes (PLB/GI/RCC Pipes) are to be laid in some stretches, the same shall be accommodated in this normal size trench.

When trenches are excavated in slopes, uneven ground, inclined portion, the lower edge shall be treated, as top surface of land and depth of trench will be measured accordingly. In certain locations, such as uneven ground, hilly areas and all other places, due to any reason whatsoever it can be ordered to excavate beyond standard depth of 165 cm to keep the bed of the trench as smooth as possible. Near the culverts, both ends of the culverts shall be excavated more than 165 cm. to keep the gradient less than 15 degree with horizontal. In case where depth beyond

specified depth is necessary it shall be clearly indicated in the technical specification of bid documents and of each work order. No extra payment shall be made for such extra depth**.**

If excavation is not possible to the minimum depth of 165 cm, as detailed above, full facts shall be brought to the notice of the Engineer-in-charge in writing giving details of location and reason for not being able to excavate that particular portion to the minimum depth. The competent authority shall grant approval in writing under genuine circumstances. The decision of the competent authority shall be final and binding on the contractor.

4.5. De-watering: The Contractor shall be responsible for all necessary arrangements to remove or pump out water from trench. The Contractor should survey the soil conditions encountered in the section and make his own assessment about dewatering arrangements that may be necessary. No extra payment shall be admissible for this.

4.6 Wetting: Wherever the soil is hard due to dry weather conditions, if watering is to be done for wetting the soil to make it loose, the same shall be done by the contractor. No extra payment shall be admissible for this.

4.7 Blasting: For **excavation in hard rock**, where blasting operations are considered necessary, the contractor shall obtain approval of the Engineer-in-Charge in writing for resorting to blasting operations.

The contractor shall obtain license from the competent authority for undertaking blasting work as well as for obtaining and storing the explosive as per the Explosive Act, 1884 as amended up to date and the explosive Rules, 1983. The contractor shall purchase the explosives fuses, detonators, etc. only from a licensed dealer. Transportation and storage of explosive at site shall conform to the aforesaid Explosive Act and Explosive Rules. The contractor shall be responsible for the safe custody and proper accounting of the explosive materials. Fuses and detonators shall be stored separately and away from the explosives. The Engineer-in-Charge or his authorized representative shall have the right to check the contractor's store and account of explosives. The contractor shall provide necessary facilities for this.

The contractor shall be responsible for any damage arising out of accident to workmen, public or property due to storage, transportation and use of explosive during blasting operation. Blasting operations shall be carried out under the supervision of a responsible authorized agent of the contractor (referred subsequently as agent only), during specified hours as approved in writing by the Engineer-in-Charge. The agent shall be conversant with the rules of blasting. All procedures and safety precautions for the use of explosives drilling and loading of explosives before and after shot firing and disposal of explosives shall be taken by the contractor as detailed in **IS:4081** safety code for blasting and related drilling operation.

4.8. Trenching Near Culverts/Bridges: The PLB pipes shall be laid in the bed of culvert at the depth not less than 165 cm protected by G.I. pipes/ DWC/ RCC and concreting as decided by Engineer-in-Charge. Both ends of culverts shall be excavated more than 165 cm in depth to keep the gradient of not less than 15 degree with horizontal. The bed of trench should be as smooth as possible. While carrying out the work on bridges and culverts, adequate arrangement for cautioning the traffic by way of caution boards during day time and danger lights at night shall be provided. In case of small bridges and culverts, where there is a likelihood of their subsequent expansion and authorized, the cable should be laid with some curve on both sides of the culvert or the bridge to make some extra length available for re-adjustment of the cable at the time of reconstruction of culvert or the bridge.

4.9. LAYING OF PLB PIPES:

After the trench is excavated to the specified depth, the bottom of the trench has to be cleared of all stones or pieces of rock and leveled up properly. A layer of soft soil/or sand(in case the excavated material contains sharp pieces of rock/stones) of not less than 5 cm is required for leveling the trench to ensure that the cable when laid will follow a straight alignment. Adequate care shall be exercised while laying so that the OF cables are not put to undue tension/pressure after being laid as this may adversely affect the optical characteristics of cables with passage of time.

The contractor shall ensure that trenching and pipe laying activities are continuous, without leaving patches or portions incomplete in between. In case intermediate patches are left, measurement of the completed portions will be taken only after work in such leftover patches are also completed in all respects.

Preparatory to aligning the pipe for jointing, each length of the PLB pipe shall be thoroughly cleaned to remove all sand, dust or any other debris that may clog, disturb or damage the optical fibre cable when it is pulled at a later stage. The ends of each pipe and inside of each PLB socket shall be thoroughly cleaned of any dirt or other foreign materials.

After the trench is cleaned the PLB Pipes shall be laid in the cleaned trench, jointed with "O" ring type PLB Sockets & 6 mm PP rope should be drawn through the PLB Pipes at the time of laying the pipes to facilitate cable pulling at a later stage. At every manhole approximately at every 200m or at bends or turns the PP rope will be tied to the PLB end caps used for sealing the PLB pipes, to avoid entry of rodents/mud etc.

At the end of each day work, the open ends of the pipes' sections shall be tightly closed with end caps to prevent the entry of dirt/mud, water or any foreign matter into PLB pipes until the work is resumed**.**

In City, Town, Urban area falling within Municipal/Corporation limits, the PLB Pipes shall be laid with protection using RCC/G.I. Pipes/Concreting reinforced with weld mesh. Moreover, in cross country routes, if depth is less than 1.2 meters, protection by using RCC/G.I. Pipe/Concreting reinforced with weld mesh shall be provided. Engineer-in-charge shall decide about such stretches and type of protection to be provided in view of the site requirements. Normally 100mm RCC Pipes shall be used for protecting PLB Pipe but if more than one PLB Pipe is to be laid and protected, RCC Pipe of suitable size to accommodate the required number of PLB Pipes shall be used.

The PLB Pipes shall be laid in RCC Full Round spun Pipes/DWC/GI pipes as required at road crossings. The RCC pipes/GI pipes shall extend at least 3 meters on either side of the road at road crossings. At road crossings, extra GI/PLB Pipes may be laid as per the direction of the Engineer-in-charge. On Rail bridges and crossings, the PLB Pipes shall be encased in suitable cast iron as prescribed by the Railway Authorities.

Whenever RCC pipes are used for protection, the gaps between the RCC collars and the RCC pipes shall be sealed using cement mortar 1:3 (1:53 grade cement of reputed brand, 3: fine sand without impurities) to bar entry of rodents. Every third collar of RCC pipes (Normally of 2 meters length) and also both ends of RCC Pipes will be embedded in a concrete block of size 40 cm (L) x 40 cm (W) x 25 cm (H) of 1:2:4 cement concrete mix (1:53 grade cement of reputed brand, 2: coarse sand, 3: stone aggregate of nominal size of 20mm) so that the alignment of RCC pipes remain firm and intact and to avoid entry of rodents.

Wherever GI Pipes are used, special care should be taken to ensure that G.I. Pipes are coupled properly with the sockets so as to avoid damage to PLB Pipe and eventually the OF Cable in the event of pressure coming on the joint and G.I. Pipe joint giving its way. Rubber bushes shall be used at either ends of the GI pipes to protect PLB Pipe. Both the ends of G.I Pipe will be embedded in a concrete block of size 40 cm (L) x 40 cm (W) x 25 cm (H) of 1:2:4 cement concrete mix (1:53) grade cement of reputed brand, 2: coarse sand, 3: stone aggregate of nominal size of 20mm) so that the alignment of G.I. Pipes remain firm and intact and to avoid entry of rodents.

In case of protection by concreting at site, the nominal dimension of concreting shall be 250mm x 250mm section. Cement Concrete Mixture used shall be of 1:2:4 composition i.e. 1. 53 grade Cement of a reputed company, 2.Coarse Sand, 3. Graded Coarse Stone aggregate of 20mm nominal size, reinforced with MS weld mesh. As the RCC is cast at site, it is imperative to ensure that special care is taken to see that proper curing arrangements are made with adequate supply of water. The contractor shall invariably use mechanical mixer at site for providing RCC protection to ensure consistency of the mix.

For carrying out concreting work in trenches, yellow PVC sheets of width not less than 1.0M and of weight not less than 1 kg. per 8 sq. meters shall be spread and nailed on sides of the trench to form trapezoidal section for concreting in the cleaned trench, to avoid seepage of water into the soil. A bed of cement concrete mixture of appropriate width and 75 mm thickness shall be laid on the PVC sheet, before laying PLB pipes. The PLB pipes shall then be laid above this bed of concrete. After laying the PLB pipes, MS weld mesh should be wrapped around, and tied and concrete mix is to be poured to form the cross-sectional dimensions as instructed by the Engineer-in-charge. The strength of RCC is dependent on proper curing, therefore, it is imperative that water content of RCC mix does not drain out into the surrounding soil. Portions where cement concreting have been carried out shall be cured with sufficient amount of water for reasonable time to harden the surface.

After curing, refilling of the balance depth of the trench has to be carried out with excavated soil. The PLB pipes/RCC/GI pipes shall be laid only in trenches accepted by Engineer-in-charge or his representative. The Contractor shall exercise due care to ensure that the PLB pipes are not subjected to any damage strain.

Water present in the trench at the time of laying the PLB Pipes shall be pumped out by the contractor before laying the pipes in the trench to ensure that no mud or water gets into the pipes, thus choking it.

In case of nallahs, which are dry for nine months in a year, the PLB Pipes shall be laid inside the RCC Pipes/ or GI Pipes and concrete laid at a minimum depth of 165 cm, as instructed by the Engineer-in-charge. The mechanical protection shall extend at least 5 meters beyond the bed of nallah on either side.

Notwithstanding anything contained in clauses referred above, the Engineer-in-charge may order, based on special site requirements, that the PLB Pipes may be encased in reinforced cement concrete, as detailed, ibid.

While laying the pipes, a gap of 2m is kept at convenient locations approx. 200m apart and at the bends and turns, which will be used as manholes during OF cable pulling. Ends of the PLB pipes at the manholes shall be sealed using PLB end caps after tying the PP rope to the end caps to avoid choking of the pipes. In a similar manner, manholes shall be kept while approaching bridges, road crossings etc., as instructed by the Engineer-in-charge. The location of the manholes will be decided by the Engineer-in-charge.

4.10. Laying Protection Pipes on Bridges and Culverts:

In case trenching and pipe laying is not possible in the beds of the culverts, the pipes shall be laid over the culverts/bridges after getting due permission from the competent authority. Of late the bridge construction authorities are providing ducts below the footpaths on the bridges for various services. The telecom officers need to maintain good liaison with the concerned authority to get one side of the duct allotted for Telecom Cables. In such ducts, G.I. Pipes can be coupled and laid for pulling the cables. It would be pertinent to mention here that close liaison with bridge construction authorities would be of immense advantage in ensuring provision of ducts on one or both the sides of the bridges as per future requirements. However, for laying cables on existing bridges, where duct arrangement does not exist, one of the following methods may be adopted**.**

Normally in the Bridges/Culverts, where there are no ducts and where the cushion on the top of the Arch is 50 cm to 100 cm or more, G.I. Pipe (Carrying PLB pipe and cable) may be buried on the top of the Arch adjourning the parapet wall, by digging close to the wheel guards. Every precaution shall be taken to see that no damage occurs to the arch of the culvert. After burying the GI pipe, the excavated surface on the arch shall be restored.
Where the thickness of the Arch is less than 50 cm, the pipe must be buried under the wheel guard masonry and the wheel guard rebuilt.

If neither of the two methods is possible, the G.I. Pipes/GI Troughs / DWC pipes must be clamped outside the parapet wall with clamps. If necessary, the pipe may be taken through the parapet wall at the ends where the wall diverges away from the road.

In case where the methods explained in clauses referred above are not possible, the G.I. Pipes/GI Troughs /DWC pipe scan be fixed on the top of the road curb close to the inside face of the parapet wall by means of clamps, using raw plugs and wood screws or small diameter bolts, without damaging the concrete and limiting the external diameter of the bolts to 7.5 mm. The permission for carrying out this work will be obtained from the Road Authorities by the contractor. Methods cited in above clauses should be carried out under close supervision of Road authorities.

The surface to be concreted should be thoroughly cleaned and leveled before concreting. At both ends of the Bridges/Culverts, where the GI Pipes/GI Troughs slope down and get buried, the concreting should be extended to ensure that no portion of the GI Pipes/GI Troughs/DWC pipes are exposed as ordered by the Engineer-in-charge to protect the Pipe/Trough from any possible damage externally caused. Where white wash/colour wash is existing on the Bridges/Culverts, the same should also be carried out on the concreted portion to ensure uniformity.

4.11. Backfilling and Dressing of the Trench:

The earth used for filling shall be free from all roots, grass, shrubs, vegetation, trees, saplings and rubbish. Provided that the PLB Pipes have been properly laid in the trench at the specified depth, the back-filling operation shall follow as closely as practicable. The back-filling operation shall be performed in such a manner so as to provide firm support under and above the pipes and to avoid bend or deformation of the PLB pipes when the PLB pipes get loaded with the back filled earth.

At locations where the back filled materials contain stones/sharp objects which may cause injury to the PLB pipes and where the excavated or rock fragments are intended to refill the trench in whole or in part, the trench should be initially filled, with a layer of ordinary soil or de-rocked loose earth of not less than 10 cm above the pipes.

Backfilling on public, private roads, railway crossings, footpath in city areas shall be performed immediately after laying the PLB pipes. Back filling at such locations shall be thoroughly rammed, so as to ensure original condition and made safe for traffic. All excess soil/material left out on road/footpath/railway crossing shall be removed by contractor. However, along the highways and cross-country, the dug-up material left out should be kept as heap above the trench while refilling.

In city limits, no part of the trench should be kept open for more than 50 meters length at any time and in all places where excavation has been done, no part of the trench should be kept open over night to prevent any mishap or accident in darkness.

5. MICROTUNNELING

5.1 GENERAL WORK DESCRIPTION

Laying of PLB pipe of different types and sizes in general, including some or all of the following activities:

Horizontal boring and laying the PLB pipe through it:

Horizontal boring and PLB pipe laying through it along road and under railway/ road/ canal/ streams crossing in all type of soil at suitable depth including all civil works. Supply of all tested lifting tools and tackles other required equipment and consumables, labour etc. and making all such necessary arrangements are under vendor's scope. Total job has to be completed within time schedule mentioned in work order. The work is specified under two major heads.

5.1.1 Laying along the road.

5.1.2 Road/ Rail/ canal crossing (road crossing means crossing a road of minimum width of 8 meters)**Under each head there will be again three different sizes of boring:**

5.1.2.1 Boring of single pipe.

5.1.2.2 Boring from two pipes to four pipes.

5.1.2.3 Boring more than four pipes.

The PLB pipe will be pulled through the horizontal bore with insertion of nylon rope provided by the department. Balloon (25mm. dia and 45mm length) test both ways is to be carried out by the bidder including the rope insertion if necessary for each pipe line. Duct integration test is to be carried out for each pipe in case of without rope pipes

5.2 SPECIFICATION FOR PLB PIPE LAYING BY MICROTUNNELING

5.2.1 In this system of cable laying following job specification should be maintained:

5.2.1.1 Guided boring / drilling technology is to be used. Mechanical bores are not permitted.

5.2.1.2 Radio or any other detection system should be used for avoiding damage to existing underground utilities i.e. electric supply, water pipe lines, telecom copper cables and optical fiber cables, gas pipelines, sewerage etc.

5.2.1.3 The depth of boring should be such as to clear any underground utilities/obstacles. However, in no case the depth of boring be less than 2m from the road surface.

5.2.1.4 In horizontal and vertical boring, the system should be capable of going up to 10 meters below the ground level.

5.2.1.5 In a shift of 8 hrs at least 500(Five hundred) metre of drilling should be achieved and it supersedes any other condition if available on the same issue.

5.2.1.6 As far as possible cable should be laid in drum lengths and cutting of cable should be avoided while laying along road.

5.2.2 Ground penetrating Radar survey and damage to Telephone cable and other Underground Utility Services:

5.2.2.1 The contractor should collect all available information regarding underground services and take necessary care to protect them. He should also have sufficient arrangements to identify and protect such underground services.

5.2.2.2 The contractor shall carry out ground penetrating Radar survey along the proposed cable route at his own cost to identify the underground services and for setting out the rules as directed by executive officer. The executive officer shall provide the route map to the contractor if available for plotting the existing utilities and setting the route for horizontal drilling. However, for safe drilling the contractor should make requisite inspection and test pits whenever necessary.

5.2.3 LIFTING OF PLB PIPE:

5.2.3.1 The PLB pipe will be handed over to the contractor at stores. The contractor is to lift the store from the store depot to the work site.

5.2.3.2 Guarding at work site is to be done by the contractor.

5.2.3.3 At the time of laying, jack and wheel should be used so that the pipe is not damaged.

5.2.3.4 The contractor is to make arrangement of water and power at the site of work.

5.2.3.5 Permission for laying is to be taken by the contractor from appropriate civic authorities.

5.2.3.6 Digging of test pit and restoration of pits are to be done by the contractor.

5.2.4 Fixing of Route Indicators/Joint Indicators:

Pits shall be dug 1 m towards jungle side at every manhole for fixing of Route Indicator. In addition, Route Indicators are also required to be placed where O.F Cable changes directions like road crossing etc. The pits for fixing the indicator shall be dug for a size of 60 cm X 60 cm and 75 cm (depth). **The indicator shall be secured in upright position by ramming with stone and murrum up-to a depth of 60 cm. And concreting in the ratio of 1:2:4(1: cement, 2: coarse sand, 4: stone aggregate 20 mm nominal size) for the remaining portion of 15 cm.** Necessary curing shall be carried out for the concreted structure with sufficient amount of water for reasonable time to harden the structure. The route indicator shall be painted with primer before painting with oil paint. The material used should bear ISI mark. The size of each written letter should be at least 3.5 cm. The colour of painting is yellow for Route Indicator and colour of sign writing white for Letters. The numbering scheme for route indicators will be Joint No./Route Indicators No. for that joint. For example, 2/6 indicator means 6th route indicator after 2nd joint.

5.2.5 DOCUMENTATIONS:

The documentation, consisting of the following shall be prepared exchange to exchange for intra SSA OFC links and for each section in case of Inter SSA/National Long-distance routes.

5.2.5.1 Route Index Diagrams – General:

This diagram shall consist of pipe Route Details on Geographical Map drawn to scale with prominent landmarks and alignment of pipe with reference to road. This shall be prepared on A-3 sheets of 80GSM.

5.2.5.2 Route Index Diagrams – Profile:

These diagrams will contain: **Geo Co-ordinates (Latitude & Longitude) of HDD pit and Joint/Loop chamber. Offset of HDD pit from center of the road. At least two Offset of HDD pit from prominent land marks to facilitate locating the HDD in future,** Make and size of the pipe.; Offset of pipe from enter of the road at every 10 meters; Depth profile of pipe at every 10 meters; Details of protection with type of protection depicted on it; Location of culvert and bridges with their lengths and scheme of laying of PLB Pipe/coil thereon.; Important landmarks to facilitate locating the pipe in future; Location of pulling manholes.

These diagrams shall be prepared on A-4 sheets of 80 GSM. On one sheet profile of maximum 400 meters shall be given to ensure clarity. All the diagrams (i) & (ii) shall bear the signatures of the contractor, the SDE-Site-in-charge as a proof of accuracy of the details. The diagrams shall be bound in A-4 size book with cover. The cover sheets shall be of 110 GSM and laminated. The front cover shall have the following details: Name of the SSA/Project Organization, Name of the OFC Link, Name of the Contractor, Date of commencement of work & Date of completion of work. For each route/section 6 sets of above-mentioned document shall be submitted.

5.3. SAFETY PRECAUTION:

5.3.1 Safety Precautions when excavating or working in excavations close to electric cables.

The Engineer-in-charge of the work should get full information from Electricity undertaking regarding any electric cables, which are known or suspected to exist near the proposed excavation and unless this is done, excavation should not be carried out in the section concerned. The electricity undertaking should be asked to send a representative and work should be preceded with close consultation with them.
Only wooden handled hand tools should be used until the electric cables have been completely exposed. A cover slab of concrete, brick or stone from above should usually be used to protect power Cables, not laid in conduits. They may or may not be protected on the sides. It is safer, therefore, always to drive the point of the pickaxe downwards then uncovering a cable, so that there is less chance of missing such warning slabs. No workman should be permitted to work alone where there are electric cables involved. At least one more man should be working nearby so that help can be given quickly in case of an accident. If disconnection of power could be arranged in that section it will be better. No electric cables shall be moved or altered without the consent of the Electric Authority and they should be contacted to do the needful. If an electric cable is damaged even slightly, it should be reported to the Electric Authority and any warning bricks disturbed during excavation should be replaced while backfilling the trench. Before driving a spike into the ground, the presence of other underground properties should be checked. Information on plans regarding the location of power cables need not to be assumed as wholly accurate. Full precautions should be taken in the vicinity until the power cable is uncovered. All electric cables should be regarded as being live and consequently dangerous. Any power is generally dangerous, even low voltage proving fatal in several cases.

5.3.2 Electric shock-Action and treatment:

Free the victim from the contact as quickly as possible. He should be jerked away from the live conductors by dry timber, dry rope or dry clothing. Care should be taken not to touch with bare hands as his body may be energized while in contact. Artificial respiration should begin immediately to restore breathing even if life appears to be extinct. Every moment of delay is serious, so, in the meanwhile, a doctor should be called for.

Safety precautions while working in Public Street and along railway lines where a road or footpath is to be opened up in the course of work, special care should be taken to see that proper protection is provided to prevent any accidents from occurring. Excavation work should be done in such a manner that it will not unduly cause inconvenience to pedestrians or occupants of buildings or obstruct road traffic. Suitable bridges over open trenches should be so planned that these are required for the minimum possible time. Where bridges are constructed to accommodate vehicular traffic and is done near or on railway property, it should be with the full consent and knowledge of the competent railway authorities.

5.3.3. Danger from falling material:

Care should be taken to see that apparatus, tools or other excavating implements or excavated materials are not left in a dangerous or insecure position so as to fall or be knocked into the trench thereby injuring any workman who may be working inside the trench.

5.3.4 Care when working in Excavations:

Jumping into a trench is dangerous. If it is deep, workmen should be encouraged to lower themselves. Workers should work at safe distance so as to avoid striking each other accidentally with tools. If the walls of the trench contain glass bits, corroded wire or sharp objects they should be removed carefully. If an obstruction is encountered, it should be carefully uncovered and protected if necessary. Care must be taken to see that excavated material is not left in such a position that it is likely to cause any accident or obstruction to a roadway or waterway. If possible, the excavated material should be put between the workmen and the traffic without encroaching too much on the road.

5.3.5 Danger of cave in:

When working in deep trenches in loose soil, timbering up/shoring the sides will prevent soil subsidence. The excavated material should be kept at sufficient distance from the edge of the trench or pit. Vehicles or heavy equipment must not be permitted to approach too close to the excavation. When making tunneled opening, it should be ensured that the soil is compact enough to prevent cave in even under adverse conditions of traffic. Extra care should be taken while excavating near the foundations of buildings or retaining walls. In such cases, excavation should be done gradually and as far as possible in the presence of the owners of the property.

5.3.6 Protection of Excavations:

Excavations in populated areas, which are not likely to be filled up on the same day should be protected by barriers or other effective means of preventing accidents and the location of all such openings must in any event be indicated by red flags or other suitable warning signs. During the hours from dusk to dawn, adequate number of red warning lamps should be displayed. Supervisory officers should ensure that all excavations are adequately protected in this manner as serious risk and responsibility is involved. Notwithstanding adoption of the above-mentioned precautions, works involving excavations should be so arranged as to keep the extent of opened ground and the time to open it to a minimum.

5.3.7 Precautions while working on roads:

The period between half an hour after sun-set and half an hour before sunrise, and any period of fog or abnormal darkness may also be considered at night for the purpose of these instructions, for the purpose of providing the warning signs. Excavation liable to cause danger to vehicles or the public must at all times be protected with fencing of rope tied to strong uprights or bamboo poles at a suitable height or by some other effective means. Any such temporary erection which is likely to cause obstructions, and which is not readily visible should be marked by posts carrying red flags or boards with a red background by day and by continuously lighted lamps at night.

The flags and the lamps should be placed in conspicuous positions so as to indicate the pedestrians and drivers of vehicles the full expanse i.e. both width and length of the obstruction. The distance between lamps or between floors should not generally exceed 1.25m along the width and 6m along length of the obstruction in non-congested area, but 4 meters along the length in congested areas. If the excavation is extensive, sufficient notice to give adequate warning of the danger should be displayed conspicuously not less than 1.25m above

the ground and close to the excavation. Where any excavation is not clearly visible for a distance of 25m to traffic approaching from any direction or any part of the carriage way of the road in which the excavation exists, a warning notice should be placed on the side or edge of all such roads from which the excavation or as near the distance as is practicable but not less than 10m from the junction of an entering or intersecting road in which the excavation exists. All warnings, in these should have a red background and should be clearly visible and legible. All warning lamps should exhibit a red light, but white lights may be used in addition to facilitate working at night. Wherever required a passage for pedestrians with footbridge should be provided. At excavations, cable drums, tools and all materials likely to offer obstructions should be properly folded round and protected. This applied to jointer's tents as well. Leads, hoses etc. stretched and across the carriageway should be guarded adequately for their own protection and also that of the public.

5.3.8 Traffic Control:

The police authorities are normally responsible for the control of traffic and may require the setting up of traffic controls to reduce the inconvenience occasioned by establishment of a single line of traffic due to restriction in road width or any other form of obstruction caused by the work. As far as possible, such arrangements should be settled in advance. If there are any specific regulations imposed by the local authorities, these should be followed.

5.3.9 Work along Railway Lines:

Normally all works at Railway crossing is to be done under supervision of the railway authorities concerned, but it is to be borne in mind that use of white, red or green flags by the Departmental staff is positively forbidden to be used when working along a railway line as this practice may cause an accident through engine drivers mistaking them for railway signals. When working along a double line of railway, the men should be warned to keep a sharp look on both the "UP" and "DOWN" lines to avoid the possibility of any accident when trains pass or happen to cross one another near the work spot.

5.3.10Procedure and Safety precautions for use of explosives during blasting for Trenching:

In areas where the cable trench cannot be done manually on account of boulders and rocks, it is necessary to blast the rocks by using suitable explosives. The quality of explosive to be used depends on the nature of the rocks and the kind of boulders. A few types of explosive fuses and detonators normally used for making trenches for cable works are detailed as: Gunpowder, Nitrate, Mixture, Gelatin, Safety Fuse, Electric Detonator, and Ordinary Detonator.

5.3.10.1 Procedure:

A detailed survey of the route is to be done to assess the length of the section where trenching is to be done with the help of blasting. A route diagram of the rocky section may be prepared indicating the length of the route where the explosives are to be used. For the purpose of obtaining license, a longer length of route should be given in the application as in many cases, after digging, rocks appear where blasting was not initially anticipated.

Next a license will have to be obtained for use and storing of explosive in that section. If the area falls under a police commissioner, the authority for granting such license is the police commissioner of the concerned area. When the route does not fall in the jurisdiction of a police commissioner, the authority for issuing license is the District Magistrate.

The concerned authority should be applied in prescribed form with a route map. The concerned authority will make an enquiry and issue a license for using/storing explosives for cables trenching work. Such license will be valid for 15 days only. The license should be got renewed if the blasting operation need to be extended. Once the license is granted, it is the responsibility of the holders of the license for the proper use of explosives, its transportation and storing.

5.3.10.2Method of using:

The safest explosive is the gelatin and electric detonator. Gelatin is in the form of a stick. Electric detonator is a type of fuse used for firing the explosive electrically. Holes are made at suitable intervals on rocky terrain or boulders either by air compressor or by manual chipping. The depth of the holes should be 2 to 3 ft. Fill up the holes with small quantity of sand for about 6". First the electric detonator is to be inserted into the gelatin and the gelatin is to be inserted into the holes keeping the + ve and –ve wirings of electric detonators outside the holes. Again, refill the holes with sand. These + ve and –ve insulated wires of detonator are than extended and finally connected to an EXPLODER kept at a distance of not less than 100m. Now the explosive is ready for blasting. But, before connecting wires to exploder for blasting, all necessary precautions for stopping the traffic, use of red flags, exchange of caution signals, etc., should be completed and only then Exploder should be connected and operated.

5.3.10.3Operation of exploder (IDL shuffler type 350 type exploder):

The type 350 blasting machine consists of a bearing block with blasting machine system and the explosion proof light – alloy injection moulded housing. The exploder is held with the left hand. The twist handle is applied to the drive pin, clapped with the right hand turned in the clockwise direction in continuous measurements at the highest speed from the initial position until it reached to a stop. At this stage an indication lamp will glow. When the indication lamp glows, "press button switch" should be pressed. This will extend the electric current to detonator and gelatin will be detonated. The rock will be blasted out of the trench. Number of holes can be blasted in a single stroke by connecting all such detonators in series connection and finally to the exploder. After blasting, again mazdoors are engaged on the work to clear the debris. If the result of the first blasting is not satisfactory, it should be repeated again on the same place.

5.3.11Warning:

There may be two reasons for unsatisfactory results of the blasting:

5.3.11.1 Misfire of gelatin due to leakage for current from detonator.

5.3.11.2 Over loading because of overburden. Never pull the broken wire pieces from the holes in such cases. Attempt should not be made to re-blast the misfired gelatin. The safest way is to make a fresh hole by its side and put fresh gelatin in that hole and blast it.

5.3.12.Precautions:(Restriction of delivery and dispatch of explosives)

No person shall deliver or dispatch any explosives to anyone other than a person who is the holder of a license to possess the explosives or the agent of holder of such a license duly authorized by him in writing on his behalf or is entitled under these rules to possess the explosives without a license. The explosives so delivered or authorized shall in no case exceed the quantity, which the person to whom they are delivered or authorized is authorized to possess with or without a license under these rules.

No other shall receive explosives from any person other than the holder of a license granted under these rules. No person shall receive from or transfer explosives to any person for a temporary storage or safe custody in a licensed premise unless prior approval is obtained from the Chief Controller. A person holding license for possession of explosives granted under these rules shall store the explosives only in premises specified in the license.

5.3.13Protection from Lightning during Storing:

Every magazine shall have attached there to one or more efficient lightning conductors designed and erected in accordance with the specification laid down in Indian Standard Specifications No.2309 as amended from time to time. The connections to various parts of earth resistance of the lightning conductor terminal on the building to the earth shall be tested at least once in every year by a qualified electrical engineer or any other competent person holding a certificate of competency in this behalf from the State Electricity Department. A certificate showing the results of such tests and the date of the last test shall be hung up in conspicuous place in the building.

5.3.13.1 Precautions during thunderstorm:

When a thunderstorm appears to be imminent in the vicinity of a magazine or store house every person engaged in or around such magazine and store house shall be withdrawn to a safe distance from such magazine or storehouse and the magazine and store house shall be kept closed and locked until the thunderstorm has ceased or the threat of it has passed.

5.3.13.2 Maintenance of records:

Every person holding a license granted under these rules for possession, sale or use of explosives shall maintain records in the prescribed forms and shall produce such record on demand to an Inspection Officer.

5.3.13.3 Explosives not be kept in damaged boxes:

The licensee of every magazine or store house shall ensure that, the explosives are always kept in their original outer package. In case, the outer package gets damaged so that the explosives contained therein cannot be stored or transported, such explosives shall be repacked only after the same are examined by controller of explosives.

5.3.13.4 Storage of explosives in excess of the licensed quantity:

The quantity of any kind of explosives kept in any licensed magazine or store house shall not exceed the quantity entered in the license against such kind of explosives. No explosives in excess of the licensed quantity shall be stored in the magazine or store house unless a permit in this behalf is obtained from the licensing authority by a letter or telegram**.**

5.3.13.5 Precautions to be observed at Site:

The electric power at the blasting site shall be discontinued as far as practicable before charging the explosives. No work other than that associated with the charging operations shall be carried out within 10 meters of the holes unless otherwise specified to the contrary by the licensing authority.

When charging is completed, any surplus explosive detonators and fuses shall be removed from the vicinity of the hole and stored at a distance which should prevent accidental detonation in the event of a charge detonating prematurely in any hole. The holes which have been charged with explosives shall not be left unattended till the blasting is completed. Care shall be taken to ensure that fuse or wires connected to the detonation are not damaged during the placing of stemming materials and tamping.

5.3.13.6 Suitable warning procedure to be maintained:

The licensee or a person appointed by the licensee to be in charge of the use of explosives at the site shall lay down a clear warning procedure consisting of warning signs and suitable signals and all persons employed in the area shall be made fully conversant with such signs and signals.

5.3.13.7Precautions to be observed while firing:

The end of the safety fuse (If used in place of a detonator should be freshly cut before being lighted. The exploders shall be regularly tested and maintained in a fit condition for use in firing. An exploder shall not be used for firing a circuit above its rated capacity. The electric circuits shall be tested for continuity before firing. All persons other than the shot-firer and his assistant, if any, shall be withdrawn from the site before testing the continuity. For the purpose of jointing, the ends of all wires and cables should have the insulation removed for a maximum length of 5 cm and should, then be made clear and bright for a minimum length of 2.5 cm. And the ends to be joined should be twisted together so as to have a positive metal contact. Then these should be taped with insulation to avoid leakage when in contact with earth.

In case of blasting with dynamite or any other high explosive, the position of all the bore holes to be drilled shall be marked in circles with while paint. These shall be inspected by the Contractor's agent. Bore holes shall be of a size that the cartridge can easily pass down. After

the drilling operation, the agent shall inspect the holes to ensure that drilling has been done only at the marked locations and no extra hole has been drilled. The agent shall then prepare the necessary charge separately for each bore hole. The bore holes shall be thoroughly cleaned before a cartridge is inserted. Only cylindrical wooden tamping rods shall be used for tamping. Metal rods or rods having pointed ends shall never be used for tamping. One cartridge shall be placed in the bore hole and gently pressed but not rammed down. Other cartridges shall then be added as may be required to make up the necessary charge for the bore hole. The top most cartridge shall be connected to the detonator which shall in turn be connected to the safety fuses of required length. All fuses shall be cut to the length required before being inserted into the holes. Joints in fuses shall be avoided. Where joints are unavoidable, a semi-circular niche shall be cut in one piece inserted into the niche. The two pieces shall then be wrapped together with string. All joints exposed to dampness shall be wrapped with rubber tape.

The maximum of eight bore holes shall be loaded and fired at one occasion. The charges shall be fired successively and not simultaneously. Immediately before firing, warning shall be given, and the agent shall see that all persons have retired to a place of safety. The safety fuses of the charged holes shall be ignited in the presence of the agent, who shall see that all the fuses are properly ignited.

Careful count shall be kept by the agent and others of each blast as it explodes. In case all the charged bore holes have exploded, the agent shall inspect the site soon after the blast but in case of misfire the agent shall inspect the site after half an hour and mark red crosses (X) over the holes which have not exploded. During his interval of half an hour, nobody shall approach the misfired holes. No driller shall work near such bore until either of the following operations have been done by the agent for the misfired boreholes.

The contractor's agent shall very carefully (when the tamping is a damp day) extract the tamping with a wooden scraper and withdraw the primer and detonator. The holes shall be cleaned for 30 cm of tamping and its direction ascertained by placing a stick in the hole. Another hole shall then be drilled 15 cm away and parallel to it. This hole shall be charged and fired. The misfired holes shall also explode along with the new one. Before leaving the site of work, the agent of one shift shall inform another agent relieving him for the next shift, of any case of misfire and each such location shall be jointly inspected and the action to be taken in the matter shall be explained to the relieving agent. The Engineer-in-charge shall also be informed by the agent of all cases of misfire, their causes and steps taken in that connection.

5.4. General Precautions:

For the safety of persons red flags shall be prominently displayed around the area where blasting operations are to be carried out. All the workers at site, except those who actually ignite the fuse, shall withdraw to a safe distance of at least 200 meters from the blasting site. Audio warning by blowing whistle shall be given before igniting the fuse.

Blasting work shall be done under careful supervision and trained personnel shall be employed. Blasting shall not be done within 200 meters of an existing structure, unless specifically permitted by the Engineer-in-charge in writing.

5.4.1 Precautions against misfire:

The safety fuse shall be cut in an oblique direction with a knife. All saw dust shall be cleared from inside of the detonator. This can be done by blowing down the detonator and tapping the open end. No tools shall be inserted into the detonator for this purpose.

If there is water present or if the borehole is damp, the junction of the fuse and detonator shall be made watertight by means of tough grease or any other suitable material. The detonator shall be inserted into the cartridge so that about one-third of the copper tube is left exposed outside the explosive. The safety fuse just above the detonator shall be securely tied in position in the cartridge. Waterproof fuse only shall be used in the damp borehole or when water is present in the borehole.

If a misfire has been found to be due to defective fuse, detonator or dynamite, the entire consignment from which the fuse, detonator or dynamite was taken shall be got inspected by

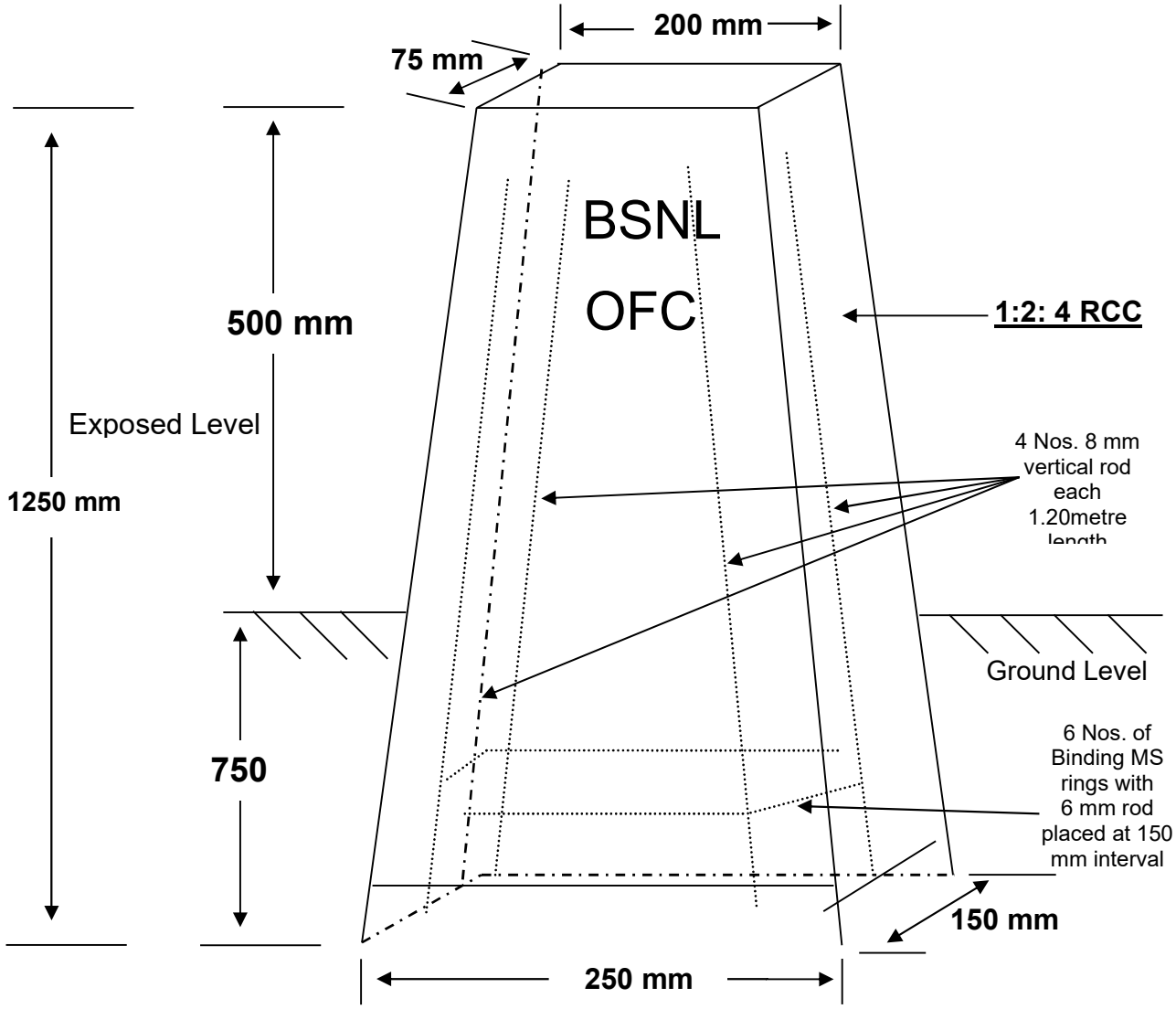
the Engineer-in-charge or his authorized representative before resuming the blasting or returning the consignment.

5.4.2 Precaution against stray currents:

Where electrically operated equipment is used in locations having conductive ground or continuous metal objects, tests shall be made for stray current to ensure that electrical firing can proceed safely.

Figure A

R.C.C ROUTE INDICATOR



----- END OF SECTION II ----

SECTION III

A. SCOPE OF WORK AND JURISDICTION OF CONTRACT

1. SCOPE OF WORK

The scope of the work has been defined in NIT and given again as below-

1.1 The Optical Fibre Cable is to be laid through PLB Pipes buried at a nominal depth of 165 cm. The steps involved in OF Cable construction are as under:

1.1.1 Excavation of trench up to a nominal depth of 165 cm, according to Construction specifications along National/State Highways/other roads and also in city limits as mentioned in the notice inviting tender.

1.1.2 Laying of PLB pipes/coils coupled by PLB sockets in excavated trenches, on bridges and culverts, drawing of 6 mm Polypropylene para pro rope (PP Rope) through the PLB pipe ends at every manhole by PLB end caps of suitable size.

1.1.3 Providing of mechanical protection by R.C.C. Pipes/GI pipes and/or concreting/chambering according to construction specifications, wherever required.

1.1.4 Fixing of GI pipes/troughs with clamps at culverts/bridges and/or chambering or concreting of G.I. Pipes/troughs, wherever necessary.

1.1.5 Back filling and dressing of the excavated trenches according to construction specifications.

1.1.6 Opening of manholes (of size 2 meter x 1 meter x 1.65 meter depth), replacing existing 6mm P.P. ropes by 8mm P.P. rope (from manhole to manhole) for ensuring smooth passage for pulling the cable. Pulling of Optical Fibre Cable with proper tools and accessories as per construction specifications. Sealing of both ends of the manholes by hard rubber bush of suitable size to avoid entry of rodents into the PLB pipes, putting split PLB pipes and split RCC pipes with proper fixtures over cable in the manhole to protect the bare cable in the pulling manhole. Backfilling and dressing of manholes.

1.1.7 **Brick wall Type Joint Chamber:** Digging of pit of size 2meter x 2 meter x 1.8 meter (depth) for construction of jointing chamber of approximately every two kilometers of internal size of1.2 meterx1.2 meter x 1.0 meter made of full brick wall with RCC cover (1:2:4) slab of size 1.7 m x 1.7 m x 1.7 m using 10 mm dia TMT bar of 150 mm C/C both ways. All the 5 internal surfaces should be plastered with 1:6 proportions. The Chamber will be filled with clean sand.

1.1.8 **Pre-Cast RCC Type Joint Chamber**: Pre-Cast RCC Ring of **size 1.2 m Φ(dia) x 0.8m depth and rim thickness 50 mm made** of standard 1:2:4 concrete mix. It should have a split (with two halves) RCC cover of thickness 50mm made of cement concrete mix of 1:2:4 (1: cement, 2: coarse sand, 4 stone aggregate mm nominal size reinforced with MS fabric 75 x 75 mm mesh of weight not less than 7.75 Kg per sq. meter having galvanized iron handles for each half to facilitate easy lifting, to be used as joint chamber. Base should be circular RCC plate 40 mm thick made of cement concrete mix 1:2:4.The word **'OFC'** should be engraved on each half of the cover. The joint chamber should be so designed that PLB pipe/PLB Blowing Type/Pre-installed rope ends remain protruding minimum 5 cm inside the chamber. The PLB pipes should be embedded within the chamber in such a way that PLB pipe remains free from the weight of the construction. The joint chamber should be plastered on all internal surfaces and top edges with cement mortar of 1:3 (1: cement, 3: coarse sand) with a floating coat of complete cement as per standard. **Necessary earthing at each Chamber by Spike earth (Earth value <5 Ω) and connecting with Joint Closer.** The joint chamber is to be filled with clean sand before closing.

1.1.9 Digging of pits 1 meter towards jungle side at every manhole and jointing chamber to a depth of 60 cm, fixing of route indicator/joint indicator, concreting and back filling of pits. Painting of route indicators with yellow colour and joint indicator by red colour and sign writing denoting route/joint indicator number, as per construction specifications.

1.1.10 Documentation.

2. BRIEF DESCRIPTION OF WORK

2.1 The work which shall be carried out as per Construction Specifications issued by BSNL contained in this bid document, involves Conventional Trenching and Reinstatement (CTR); Horizontal Directional Drilling (HDD); Related Allied Works (RAW). The various works can be summarized as below. The construction specification given below covers Conventional trenching and reinstatement and allied works. The specification for Horizontal directional drilling is specified.

2.2 Open trenching at 165 cm depth & width of 45 cm at top and 30 cm at bottom.

2.3. Permanently lubricated PLB pipes (PLB pipes) will be supplied by BSNL in coils of varying lengths from 200 meter onwards. Spools should be de-coiled using jack mechanism and bends removed by proper forming so that the pipes are straight. This will be part of contractor's works and no extra payment will be paid. In addition to the laying of PLB pipes, there may be cases where PVC pipes of straight lengths may have also to be provided.

2.4 Whenever there is any doubt in any of the construction step, BSNL's Engineering Instructions on Underground Optical Fiber Cable Laying Works shall be referred.

3. **COST OF WORK:** The estimated cost of work is as mentioned in the Notice Inviting Tenders which has been worked out based on standard scheduled rates mentioned in this tender document. The actual value of work may vary based on the actual requirement but generally being limited to +/-25% of the contract value.

B. Details of OFC Route / Work to be done

Sl no	Name of the Route	RKM (in Km)	Patch work/New work /EB/4G/BharatNet
1	Bhikapada-Khallikote Pop-Badapalli GP	5.5	Bharat Net
2	Bhagibandha	0.7	4G saturation (CAT-V)
3	Gujulingi	0.3	4G saturation (CAT-V)
4	Puriasahi	0.4	4G saturation (CAT-V)
5	Nuagam	5.5	4G saturation (CAT-V)
6	Chatrapur	5	EB circuit
7	Beguniapada	0.6	IX.2 BTS fiberization
8	Defence Colony Jagganath Pur	1.5	IX.2 BTS fiberization
9	Baunsapalli Ctp	1.2	IX.2 BTS fiberization
10	Batachatrapur	0.5	IX.2 BTS fiberization
11	Purusottampur-3	0.4	IX.2 BTS fiberization
12	Ber_Badaputi	1.2	IX.2 BTS fiberization
13	Khallikote-BIL	0.6	IX.2 BTS fiberization
14	Kodala-Ps-BIL	0.5	IX.2 BTS fiberization
15	Taratarini PSM	2	IX.2 BTS fiberization
16	Collectorate Off Ctp-Ip-Gtl	1	IX.2 BTS fiberization
17	Ber-Chikiti-Patrapur-Jayantipur-Koligam-Icchapur-Ber	3.5	New work
18	Bhejiput Exchage-Nearest Etr Joint	0.5	New work
19	Narayanpur Jn-Ire	3	New work
20	Patrapur-Dhanaghar Chowk	1	Patchwork/ Rehabilitation work
21	Khallikote-Bhejiput	6	Patchwork/ Rehabilitation work
22	Khariaguda-Mandarada	3	Patchwork/ Rehabilitation work
	Total	**43.9**	

----- END OF SECTION III ----

SECTION IV

SPECIAL CONDITIONS OF CONTRACT

1. GENERAL:

1.1 The work shall be accepted only after Acceptance Testing carried out by BSNL team, designated by BSNL, as per prescribed schedule and work/material passing the test successfully.

1.2 BSNL reserves the right to disqualify such bidders who have a record of not meeting contractual obligations against earlier contracts entered into with BSNL.

1.3 BSNL reserves the right to blacklist a bidder for a suitable period in case he fails to honour his bid without sufficient grounds.

1.4 BSNL reserves the right to counter offer price(s) against price(s) quoted by any bidder.

1.5 Any clarification issued by BSNL, in response to query raised by prospective bidders shall form an integral part of bid documents and it may amount to amendment of relevant clauses of the bid documents.

1.6 Tender will be evaluated as a single package of all the items given in the price schedule.

1.7 All work to be executed under the contract shall be executed under the direction and subject to the approval in all respects of the Divisional Engineer/AGM or Site Engineer in-charge of work site who shall be entitled to direct at what point or points and what manner they are to be commenced, and from time to time carried on.

1.8 If the contractor shall desire an extension of time for completion of the work on the grounds of unavoidable hindrance in execution of work or any other ground, he shall apply in writing to the Divisional Engineer/AGM on the same day of occurrence of hindrance on account of which he desires such extension as aforesaid. In this regard the decision of **BA Head,** shall be final.

1.9 If at any time after the commencement of the work, BSNL may feel that execution of whole or part of work, as specified in the tender is not required to be carried out, then BSNL shall give notice in writing of the fact to the contractor who shall have no claim to any payment of compensation whatsoever on account of any profit or advantage which he might have derived in consequence of the full amount of the work not having been carried out neither shall he have any claim for compensation by reason of any alterations having been made in the original specifications, drawings, designs and instructions which shall involve any curtailment of the work as originally contemplated.

1.10 Whenever any claims against the contractor for the payment of a sum of money arises out of or under the contract, BSNL shall be entitled to recover such sum by appropriating in part or whole the security deposit of the contractor, and to sell any Government promissory notes etc., forming the whole or part of such security or running / Final bill pending against any contract with BSNL. In the event of the security being insufficient or if no security has been taken from the contractor, then the balance or the total sum recoverable as the case may be, shall be deducted from any sum then due or which at any time thereafter may become due to the contractor under this or any other contract with BSNL. Should this sum be not sufficient to cover the full amount recoverable the contract or shall pay to BSNL on demand the balance remaining due.

1.11 No official of Gazetted rank or other Gazetted officer employed in Engineering or administrative duties in an Engineering Department or any other Department of the Government of India is allowed to work as a contractor for a period of two years after his retirement from Government service without the previous permission of Government of

India. This contract is liable to be cancelled if either the contractor or any of his employee is found at any time to be such a person who had not obtained the permission of Government of India as aforesaid before submission of the engagement in the contractor's service as the case may be.

1.12 In the event of the contractor being, adjusted insolvent or going voluntarily into liquidation of having received order or other order insolvency act made against him or in the case of company, of the passing of any resolution, or making of any order for winding up whether voluntarily or otherwise, or in the event of the contractor failing to comply with any of the conditions herein specified. The tendering authority, shall have the power to terminate the contract with the recommendation of the BA Head without any notice and an intimation in this regard may be sent to AGM(MM), O/o the CGM, Odisha Circle, Bhubaneswar.

1.13 Without prejudice to any of the rights or remedies under this contract, if the contractor dies, the tendering authority, on behalf of BSNL can terminate the contract without compensation to the contractor, on receipt recommendation of BA Head. However, tendering authority, at his discretion may permit contractor's heirs to perform the duties or engagements of the contractor under the contract, in case of his death on receipt recommendation of BA Head. In this regard the decision of tendering authority, shall be the final.

1.14 In the event of the contractor, winding up his company on account of transfer or merger of his company with any other, the contractor shall make it one of the terms and stipulations of the contract for the transfer of his properties and business, that such other person or company, shall continue to perform the duties or engagement of the contractor under this contract and be subject to his liabilities there under.

1.15 **Interpretation of the contract document:**

1.2.1 The representative of Business Area (BA),and the contractor shall in so far as possible by mutual consultation, try to decide upon the meaning and intent of the contract document. In case of disagreement the matter shall be referred to the Competent Authority whose decision shall be final. Any change in the contract documents shall be set forth in writing by the representative of the parties hereto. It shall be the contractor's responsibility to thoroughly familiarize all of his supervisory personnel with the contents of all the contract documents.

1.3 Notification:

1.3.1 The contractor shall give in writing to the proper person or authority with a copy to the Divisional Engineer/AGM such notification as may be mandatory or necessary in connection with the commencement, suspension, resumption, performance and / or completion of the contracted work. All notice shall be given sufficiently in advance of the proposed operation to permit proper co-relation of activities and the contractor shall keep all proper persons to authorities involved and advised of the progress of operations throughout the performance of the work and/or with such other information and /or supporting figure and data as may from time to time as directed or required.

Shut down on account of weather conditions:

1.3.2 The contractor shall not be entitled to any compensation whatsoever by reason of suspension of the whole or any part of the work made necessary by BSNL or deemed advisable on account of bad weather conditions or other Force majeure conditions.

2. STORES SUPPLIED BY BSNL:

At no point of time the contractor shall be issued stores of value more than the contractor's material security as per clause number 11.5.1.2 of section IV. If at all the work requires more amount of materials to be issued to the contractor, then the security shall suitably be revised before the issue of the store and the contractor will not have any objection to it.

The contractor shall transport (including loading and unloading) all stores issued to him from SDCA / OA HQ Store under the BA, to the site of work at his own cost. The BSNL shall not pay any transportation charges to the contractor.

All materials supplied to the contractor by BSNL shall remain the absolute property of BSNL and shall not be removed from site of the work except for use in the work and shall be at all times open to inspection by the Representative of the Engineer-in-Charge. In case the materials like cable and accessories are taken delivery of by the contractor and stored at the site office/store of the contractor such site office / store will also be treated "as site" for this purpose. Any such material remaining unused at the time of the abandonment, completion or determination of the contract shall be returned to BSNL at a place informed to him by BSNL, failing which the cost of the unused materials shall be deducted from the contractor's material security or any of his pending bills or from any other security.

The contractor shall be responsible for the transportation of store, storage and safe custody of all material supplied to him by BSNL, which in the contractor's custody whether, or not installed in the work. The contractor shall satisfy himself regularly the quantity and quality of the materials supplied to him and he will be responsible for any subsequent deterioration and discrepancy (inclusive of theft) in the quantity/ quality of the materials.

The contractor shall submit a proper account every month of all the materials supplied to him by BSNL and those consumed for items of work any discrepancy of difference between the materials issued to the contractor and those consumed in the work as per the "BSNL's calculation" (which shall be final) will be charged to the contractor or deducted from his bills at 1.5 times of prevailing standard price including freight, handling charges, storage charges etc.

The contractor shall ensure that only the required materials are issued to him. Upon completion of work, the contractor shall return to BSNL at the later designated store in good condition, free of charges, any unused materials that were supplied by BSNL.

3. EASEMENTS, PERMITS, LICENCE AND OTHER FACILITIES:

The contractor shall obtain/provide at his own cost all Licenses, Easements, Permits (RoW Permissions) necessary to do its work including the following.

3.1 "Right of User" Easements and Permits. ROW permission.

3.2 Railway and Highway crossing permits including bridge.

3.3 Canal / Steam crossing permits.

S**uccessful bidder is responsible to obtain the RoW from Road Authority if required**. The successful bidder will be required to prepare route diagram in required format as asked by RoW authority free of cost and shall be responsible for obtaining "Right of User" easements and permits (RoW permission). But BSNL shall make necessary payments directly to the concerned authorities, based on the demand notes issued towards RoW charges.

The contractor shall be fully responsible for angling and obtaining all necessary easements, permits and licenses, for moving all construction equipment, tools, supplied materials and men across Railways and Highways, across public or private road as well as premises of any public utility within the right of user and for bearing all costs that may be incurred in respect of the same.

The contractor is to confine his operation to the provided construction "Right of User" unless it has made other arrangement with the particular property owners and / or tenants such other arrangements shall be entirely at the responsibility of the contractor as to cost and arrangement as also breach and claim and shall be entitled with a copy to the Divisional Engineer/AGM.

The contractor will not be entitled to extra compensation for hardship and increase in cost by the cable trench being routed adjacent to or across other pipeline, Highways, Railways, telephones

or poser poles and wires or guy wires, embankments, cliffs, streams or other obstacles which may physically or otherwise in any manner, restrict or limit the use of the construction "Right of User". Such construction and such contingency shall be deemed to have been providing for in the rates.

At location where the OF cable trench is routed across or along railways or roads the contractor shall without extra cost provide and maintain such detours and road controls as are required by the railways or government or local agencies having jurisdiction.

Due to any reason, if above mentioned permits etc. could not be arranged by the contractor in time then the extension of time limit may be provided as per EoT clause given in tender document.

4. QUALITY OF WORK:

BSNL shall be the final judge of the quality of the work and the satisfaction of BSNL in respect thereof set forth in the contract documents. Laxity or failure to enforce compliance with the contract documents by BSNL and / or its representative shall not manifest a change or intent of waiver, the intention being that, notwithstanding the same, the contractor shall be and remain responsible for complete and proper compliance with the contract documents and the specification therein. The representative of Engineer-in-Charge has the right to prohibit the use of men and any tools, materials and equipment which in his opinion do not produce work or performance meeting the requirement of the contract documents.

5. TAXES AND DUTIES:

Contractor shall pay all rates, levies, fees royalties, taxes and duties payable or arising from out of, by virtue of or in connection with and/or incidental to the contract or any of the obligations of the parties in terms of the contractor documents and / or in respect of the works or operations or any part thereof to be performed by the contractor and the contractor shall indemnify and keep indemnified BSNL from and against the same or any default by the contractor in the payment thereof. However, GSTshall be paid extra by BSNL as applicable.

6. PROTECTION OF LIFE AND PROPERTY AND EXISTING FACILITIES:

The contractor is fully responsible for taking all possible safety precaution during preparation for and actual performance of the works and for keeping the construction site in a reasonable safe condition. The contractor shall protect all life and property from damage or losses resulting from his construction operations and shall minimize the disturbance and inconvenience to the public.

If the excavation of trench alters the contours of the ground around road and highway crossing in such locations dangerous to traffic, the contractor shall at his own cost, take all necessary precautions to protect public and shall comply with all BSNL regulations as to placing of warning boards (Minimum size 3' x 2'), traffic signals, barricades, flags etc. at such location. If the contractor does not put the warning signal as per above directions, then a penalty of Rs. 500/- per day shall be levied on the contractor, till the directions are complied by the contractor. The contractor shall take due precautions to avoid damages to other pipelines, water, mains, sewers, telephones, telegraphs and power conduits, laid wires poles and guy wires, railways, highways, bridges or other underground or above ground structure and/or property crossing or adjacent to the cable trench being excavated.

Attention of the contractor is drawn to the rules regarding laying of cables at road crossing, along Railways Bridges, Highways safety precautions while working in Public Street. The contractor in writing shall obtain the detailed engineering instructions from the Divisional Engineer/AGM of the area.

The contractor shall be solely responsible for location through approved non-destructive means and ensuring the safety of all existing underground pipeline, electrical cables, and or other structures.

The contractor shall be solely liable for all expense for and in respect of repairs and / or damage occasioned by injury of or damage to such underground and above structures or other properties and undertake to indemnify BSNL from and against all actions, cause of actions, damages, claims and demands whatsoever, either in law or in equity and all losses and damages and costs (inclusive between attorney and client), charges and expenses in connection therewith and / or incidental thereto. The contractor shall take all responsibilities and risks in crossing other pipelines and cables and shall be responsible for protecting all such existing pipelines, poles, electric lines, sewers, cables or other facilities from damage by the contractor's operation in connection with the work. The contractor without cost of BSNL shall promptly repair any damage incurred.

The current market value of any commodities lost as a result of any damage to the aforesaid existing facilities shall be paid by the contractor together with such additional sums necessary to liquidate the personal or property damages, resulting therefrom.

7. LABOUR WELFARE MEASURES AND WORKMAN COMPENSATION:

7.1 Obtaining License before commencement of work:

The contractor shall obtain a valid labour license under the Contract Labour (R&A) Act 1970 and the Contract Labour (Regulation and Abolition) Central Rules 1971, as per clause 7 of Section II, and continue to have a valid license until the completion of work. The contractor shall also abide by the provisions of the Child Labour (Prohibition and Regulation) Act 1986. Any failure to fulfill this requirement shall attract the penal provisions of this contract arising out of the resultant non-execution of work. The Contractor shall register himself under "The Building and Other Construction Workers Welfare Cess Act-1996" and "The Building and Other Construction Workers (Regulation of Employment & Conditions of Service) Rules 1998". Deduction of 1% cess is obligatory under central act as labour welfare measure.

7.2 Contractors Labour Regulations:

Working hours:

Normally working hours of an employee should not exceed 9 hours a day. The working day shall be so arranged that inclusive of interval for rest, if any, it shall not spread over more than 12 hours on any day. When a worker is made to work for more than 9 hours on any day or for more than 48 hours in any week, he shall be paid over time for the extra hours put in by him.
Every worker shall be given a weekly holiday normally on a Sunday, in accordance with the provisions of Minimum Wages (Central) Rules 1960, as amended from time to time, irrespective of whether such worker is governed by the Minimum Wages Act or not.

Where the minimum wages prescribed by the Government, under the Minimum Wages Act, are not inclusive of the wages for the weekly day of rest, the worker shall be entitled to rest day wages, at the rate applicable to the next preceding day, provided he has worked under the same contractor for a continuous period of not less than 6 days.

Where a contractor is permitted by the Engineer-in-charge to allow a worker to work on a normal weekly holiday, he shall grant a substituted holiday to him for the whole day, on one of the five days, immediately before or after the normal weekly holiday, and pay wages to such worker for the work performed on the normal weekly holiday at the overtime rate.

7.3 Display of Notice Regarding Wages Etc.

The contractor shall, before he commences his work on contract, display and correctly maintain and continue to display and correctly maintain, in a clear and legible condition in conspicuous places on the work, notices in English and in local Indian languages spoken by the majority of the workers, giving the minimum rates of the wages fixed under Minimum Wages Act, the actual wages being paid, the hours of work for which such wage are earned, wages periods, dates of payments of wages and other relevant information.

7.4 Payment of Wages;

The contractor shall fix wage periods in respect of which wages shall be payable.

No wage period shall exceed one month.

The wages of every person employed as contract labour in an establishment or by a contractor, where less than one thousand such persons are employed, shall be paid before the expiry of seventh day and in other cases before the expiry of tenth day after the last day of the wage period in respect of which the wages are payable.

Where the employment of any worker is terminated by or on behalf of the contractor, the wages earned by him shall be paid before the expiry of the second working day from the date on which his employment is terminated.

All payment of wages shall be made on a working day at the work premises and during the working time and on a date notified in advance and in case the work is completed before the expiry of the wage period, final payment shall be made within 48 hours of the last working day.

Wages due to every worker shall be paid to him direct or to other person authorized by him in this behalf.

All wages shall be paid in current coin or currency or in both. Wages shall be paid without any deductions of any kind except those specified by the Central Government by general or special order in this behalf or permissible under the Payment of Wages Act, 1956.

A notice showing the wages period and the place and time of disbursement of wages shall be displayed at the place of work and a copy sent by the contractor to the Engineer-in-charge under acknowledgement.

It shall be the duty of the contractor to ensure the disbursement of wages in presence of the site Engineer or any other authorized representative of the Engineer-in-charge who will be required to be present at the place and time of the disbursement of wages by the contractor to workmen.

The contractor shall obtain from the site Engineer or any other authorized representative of the Engineer-in-charge, as the case may be, a certificate under his signature at the end of the entries in the "Register of Wages" or the "Wage-cum-Muster Roll", as the case may be, in the following form:

"Certified that the amount shown in the column No…….has been paid to the workman concerned in my presence on ……at……."

7.5. Fines and deductions which may be made from wages:

The wages of a worker shall be paid to him without any deduction at any kind except the following:

7.5.1 Fines.

7.5.2 Deductions for absence from duty i.e. from the place or the places where by the terms of hisemployment he is required to work. The amount of deduction shall be in proportion to the period for which he was absent.

7.5.3 Deductions for damage to or loss of goods expressly entrusted to the employed person for custody, or for loss of money or any other deductions which he is required to account, where such damage or loss is directly attributable to his neglect or default,

7.5.4 Deduction for recovery of advances or for adjustment of overpayment of wages, advances granted shall be entered in a register.

7.5.5 Any other deductions, which the Central Government may from time to time, allow.

7.5.6 No fines should be imposed on any worker save in respect of such acts and omissions on hispart as have been approved of by the Labour Commissioner.

7.5.7 No fine shall be imposed on a worker and no deduction for damage or loss shall be made from his wages until the worker has been given an opportunity of showing cause against such fines or deductions.

7.5.8 Every find shall be deemed to have been imposed on the day of the act or omission in respect of which it was imposed.

7.6 Labour Records

The contractor shall maintain a **Register of Persons employed** on work on contract in Form XIII of the Contract Labour (R&A) Central Rules 1971.

The contractor shall maintain a **Muster Roll** register in respect of all workmen employed by him on the work under Contract in Form XVI of the CL (R&A) Rules 1971.

The contractor shall maintain a **Wage Register** in respect of all workmen employed by him on the work under contract in Form XVII of the CL (R&A) Rules 1971.

Register of accidents: The contractor shall maintain a register of accidents in such form as may be convenient at the work place but the same shall include the following particulars:

7.6.1 Full particulars of the labourers who met with accident.
7.6.2 Rate of wages.
7.6.3 Sex
7.6.4 Age
7.6.5 Nature of accident and cause of accident
7.6.6 Time and date of accident
7.6.7 Date and time when admitted in hospital
7.6.8 Date of discharge from the hospital
7.6.9 Period of treatment and result of treatment
7.6.10 Percentage of loss of earning capacity and disability as assessed by Medical Officer.
7.6.11 Claim required to be paid under Workmen's Compensation Act.
7.6.12 Date of payment of compensation.
7.6.13 Amount paid with details of the person to whom the same was paid.
7.6.14 Authority by whom the compensation was assessed.
7.6.15 Remarks

The contractor shall maintain a Register of Fines in the Form XII of the CL (R&A) Rules 1971. The Contractor shall display in a good condition and in a conspicuous place of work the approved list of acts and omission for which fines can be imposed.

The contractor shall maintain a Register of deductions for damage or loss in form XX of the CL (R&A) Rules 1971.The contractor shall maintain a Register of Advances in Form XXII of the CL (R&A) Rules 1971.The contractor shall maintain a Register of Overtime in Form XXIII of the CL (R&A) Rules 1971. The contractor shall deposit EPF and ESI subscriptions to concerned authorities of the person employed by him as per provision of EPF and ESI Acts and maintain the necessary records as per the requirement under EPF & ESI rules.

7.7 Attendance card-cum wage slip

The contractor shall issue an Attendance card cum wage slip to each workman employed by him. The card shall be valid for each wage period. The contractor shall mark the attendance of each workman on the card twice each day, once at the commencement of the day and again after the rest interval, before he actually starts work. The card shall remain in possession of the worker during the wage period under reference. The contractor shall complete the wage slip portion on the reverse of the card at least a day prior to the disbursement of wages in respect of the wage period under reference. The contractor shall obtain the signature of thumb impression of the worker on the wage slip at the time of disbursement of wages and retain the card with him.

7.8 Employment card

The contractor shall issue an Employment Card in the Form XIV of CL (R&A) Central Rules 1971 to each worker within three days of the employment of the worker.

7.9 Service certificate

On termination of employment for any reason whatsoever the contractor shall issue to the work man whose services have been terminated, a Service Certificate in the Form XV of the CL (R&A) Central Rules 1971.

7.10 Preservation of Labour Records

The Labour records and records of Fines and deductions shall be preserved in original for a period of three years from the date of last entries made in them and shall be made available for inspection by the Engineer-in-charge or Labour Officer or any other officers authorized by the Ministry of Communication in this behalf.

7.11. Power of labour officer to make investigations or enquiry

The labour officer or any person authorized by the Central Government on their behalf shall have power to make enquiries with a view to ascertaining and enforcing due and proper observance of Fair Wage Clauses and provisions of these Regulations. He shall investigate into any complaint regarding the default made by the contractor in regard to such provision.

7.12. Report of Investigating officer and action thereon

The Labour Officer or other persons authorized as aforesaid shall submit a report of result of his investigation or enquiry to the Engineer in-charge indicating the extent, if any, to which the default has been committed with a note that necessary deductions from the contractor's bill be made and the wages and other dues be paid to the labourers concerned. The Engineer in-charge shall arrange payments to the labour concerned within 45 days from the receipt of the report from the Labour Officer or the authorized officer as the case may be.

7.13. Inspection of Books and Slips

The contractor shall allow inspection of all the prescribed labour records to any of his workers or to his agent at a convenient time and place after due notice is received or to the Labour Officer or any other person, authorized by the Central Government on his behalf.

7.14. Submission of Returns

The contractor shall submit periodical returns as may be specified from time to time.

7.15 Amendments

The Central Government may from time to time add to or amend the regulations and on any question as to the application / interpretation or effect of those regulations.

8.0 INSURANCE:

8.1. Without limiting any of his other obligations or liabilities, the contractor shall at his own expense, take and keep comprehensive insurance including third party risk for the plant, machinery, men, materials etc. brought to the site and for all the work during the execution. The contractor shall also take out workmen's compensations insurance as required by law and undertake to indemnify and keep in demnified BSNL from and against all manner of claims and demands and losses and damages and cost (including between attorney and client) charges and expenses that may arise in regard the same or that BSNL may suffer or incur with respect to end / or incidental to the same. The contractor shall have to furnish originals and /or attested copies as required by BSNL to the policies of insurance taken within 15 (fifteen) days of being called upon to do so together with all premium receipts and other papers related thereto which the department may require.

9. COMPLIANCE WITH LAWS AND REGULATION:

9.1 During the performance of the works the contractor shall at his own cost and initiative fully comply with all applicable laws of the land and with any and all applicable by-laws rules, regulations and orders and any other provisions having the force of law made or promulgated or deemed to be made or promulgated by the Government, Governmental agency or BSNL, municipal board, Government of other regulatory or Authorized body or persons and shall provide all certificates of compliance therewith as may be required by such applicable law, By-laws, Rules, Regulations, orders and / or provisions. The contractor shall assume full responsibility for the payment of all contributions and pay roll taxes, as to its employees, servants or agents engaged in the performance of the work specified in the contractor documents. If the contractor shall require any assignee or sub-contractor to share any portion of the work to be performed hereunder may be assigned, sub-leased or sub-contracted to comply with the provisions of the clause and in this connection the contractor agrees as to undertake to save and hold BSNL harmless and indemnified from and against any/all penalties, actions, suits, losses and damages, claims and demands and costs (inclusive between attorney and client) charges and expenses whatsoever arising out or occasioned, indirectly or directly, by failure of the contractor or any assignee or sub-contractor to make full and proper compliance with the said by-laws, Rules, Regulations, Laws and Order and provisions as aforesaid.

9.2 The contractor shall also comply with the rule and regulation of EPF & ESI as per government rule and regulation from time to time at his own cost and expenses. The contractor shall indemnify BSNL from any act or action in future or during the course of work in this regard at his own cost and expense.

BSNL reserves the right to withhold the contractor's amount up to 25% of tender value from the bills of contractor/security deposit in case of non-compliance of these rules and regulations by the contractor.

10. TOOLS AND PLANTS

The contractor shall provide at his own cost all tools, plants appliances, implements, measuring instruments etc. required for proper execution of works. The contractor shall also supply without charge the requisite number of persons with the means and materials necessary for the purpose of setting out works, counting, weighing and assisting the measurements for examination at any time and from time to time. The contractor shall be responsible to make all arrangements, at his own cost for dewatering of trenches / ducts and de-gasification of the ducts before carrying out the work. The contractor shall also be responsible to make arrangements at his own cost, for water required for carrying of works at sites including curing of CC/RCC works. Failing which the same may be provided by the Engineer-in-charge at the expense of the contractor and the expenses shall be deducted from any money due to the contractor under this contract or otherwise.

11. GENERAL (COMMERCIAL) CONDITIONS OF THE CONTRACT

11.1 APPLICATION:

The General conditions shall apply in contracts made by BSNL for the execution of Laying PLB pipe and associated works by Open Trench Method/Trenchless Technology, Optical Fibre Cable Pulling/Blowing and splicing works.

11.2 STANDARDS:

The works to be executed under the contract shall confirm to the standards prescribed in the OF Cable construction practices issued by BSNL using open trenching and trenchless technology methods.

11.3 PRICES:

Prices charged by the Contractor for the works performed under the Contract shall not be higher from the prices quoted by the Contractor in his Bid.

Price once fixed will remain valid for the period of contract. Increase and decrease of taxes/duties will not affect the price except GST, which will be paid at prevailing rate, during this period (proof of payment to the GST authority shall be submitted by the contractor).

11.4.SUBCONTRACTS:

The contractor shall not assign, sub contract or subject the whole or any part of the works covered by the contract, under any circumstances.

11.5 SECURITY:

11.5.1 Material Security:

11.5.1.1 The successful tenderer will have to deposit of Rs. 3 lac for works of approved amount upto Rs.25 lac and of Rs. 5 lac for works of approved amount beyond Rs.25 lac in the form of e-bank guarantee/Bank guarantee/DD, valid for the period of contract plus six months, from a scheduled bank and in the **material security bond form** provided in the bid document's **Section-II**. BSNL reserves the right to increase the amount of material security based on the quantum of work/maximum store to be issued at the time of execution of the work at any point of time.

11.5.1.2 The contractor at any point of time will not be issued stores costing more than material security. If due to any reason more store has to be issued to the contractor, then the material security shall be suitably enhanced. In this regard the decision of the Engineer-in-Charge shall be final and binding.

11.5.1.3 The proceeds of the material security shall be payable to BSNL as a compensation for any loss resulting from the contractor's failure to account for the material issued to him.

11.5.1.4 The Material security shall be released / refunded within three months from the date of commissioning or the final settlement of the material account, whichever is later, on production of **'No Dues Certificate'** from the 'Engineer In-Charge'.

11.5.2 PERFORMANCE SECURITY:

11.5.2.1 For all successful bidder including MSME, the bidder shall submit an amount equivalent to 10%* of the tender approved value as security deposit by way of E-Performance Bank Guarantee (EPBG)/Bank guarantee/DD within 15 working days from the date of issue of LoI/APO in favor of Accounts Officer (A&P), Circle office, Bhubaneswar from any commercial or scheduled bank in India and valid for 36 months from the date of issue of LoI/APO.

**** PBG amount mentioned in GeM bid (i.e default value of 5 % of estimated value) is taken for bid creation purpose only. Actual PBG amount is 10 % of contract value including GST valid for 36 months as per ATC.***

11.5.2.2 On receipt of EPBG/PBG/DD from the bidder and confirmation of the genuineness of the EPBG/PBG/DD from the concerned bank, the EMD if any received earlier may be returned.

11.5.2.3 The proceeds of the performance security shall be payable to BSNL as compensation for any loss resulting from the contractor's failure to perform the obligations under the contract.

11.5.2.4 The performance security deposit shall be refunded after 6 months from expiry of warranty period of last work executed provided there are no recoveries to be made arising out of poor quality of work, incomplete work and/or violation of any terms and conditions of the contract as stipulated in the bid document.

11.5.2.5 If, there is recoveries to be made, contractor shall deposit the money before the release of EPBG/PBG/DD for getting the EPBG/PBG/DD released and in failure to do so, EPBG/PBG/DD will be forfeited and recovery will be realized from the EPBG/PBG/DD amount and the balance amount if any after adjustment of recovery will be refunded to the bidder.

12. ISSUE OF WORK ORDERS AND TIME LIMIT:

Work will be executed by way of issuing work orders. The work order shall be issued for full orfor a part of work which will have to be completed in time as specified in work order.

The work orders shall be issued by the AGM in-charge of OF cable construction works of the BA after Pro-forma work order (contract order) approval from Tender Approving Authority. The AGM in charge of work shall issue the work order after examining the technical and planning details of the works to be executed.

BSNL reserves the right to cancel or modify the scope of work stipulated to be carried out against the work order in the event of change of plan necessitated on account of technical reasons or if in the opinion of work order issuing authority or competent authority, the contractor is not executing the work at the required pace.

13. EXTENSION OF THE TIME LIMIT:

General: In each work order, the work order issuing authority shall specify the time allowed for completion of work consistent with the magnitude and urgency of work. The time allowed for carrying out the work is to be strictly observed by the contractor and shall be reckoned from seventh day from the date of issue of work order.

In as much as "the time being deemed to be the essence of contract", throughout the stipulated period of contract, the work is to be proceeded with all due diligence on the part of the contractor.

Application for Extension of Time and Sanction of Extension of Time (EoT): There may be some hindrances, other than covered under force majeure during execution of work and in such cases the contractor shall apply in writing to the engineer-in-charge for extension of time (EoT), on account of which, he desires such extension, on the same days of occurrence of hindrance. The Divisional Engineer/AGM In-charge shall forward the request to the competent authority with his detailed report, within two days of receipt of request from the contractor. The competent authority, not below the rank of DGM is empowered to grant extension of time for completion of work on certain conditions. He shall exercise such powers, if the following conditions are satisfied.

13.1 The application contains the ground(s), which hindered the contractor in execution of work.

13.2 The Engineer-in-charge is of the opinion that the grounds shown for extension of time are reasonable.

13.3 The competent authority shall consider the request keeping all the facts and circumstances in view and shall grant extension of time, if in his opinion, there are reasonable and sufficient grounds for granting such extension and the reasons for delay are not ascribable to the contractor.

13.4 The competent authority may also grant extension of time for completion of work in cases where reasons for delay are ascribable to the contractor, but such extension of time shall be with LD charges as per clause dealing with penalty for delays in execution of works.

13.5 The competent authority shall grant EOT with time period for completion of work clearly mentioned. The sanction of the competent authority of EOT shall be issued under the signature of the Engineer-in-charge.

13.6 If the competent authority is of the opinion that the grounds shown by the contractors are not reasonable and sufficient and declines to grant the extension of time, the contractor cannot challenge the soundness of the opinion by reference to arbitration. The decision of the competent authority on period of extension of time or refusal for extension of time shall be final and binding on the contractor.

13.7 Grant of Extension of Time without Applications

There are, at times, practical difficulties like non-availability of material, delay in providing permissions/right of way etc. reasons of which are ascribable to BSNL. In such cases, the Engineer-in-charge with the approval of competent authority to sanction EoT may issue extension of time suomotu without waiting for contractor to make an application for EoT. Entry of hindrances shall be made in the Hindrance Register. BSNL will, however, not be liable to the contractor for any losses or damages, costs, charges, or expenses that the contractor may in any way sustain / suffer due to delay in making the above available.

14. MEASUREMENT, INSPECTION, TESTING AND ACCEPTANCE TESTING:

Open Trenching

14.1 Measurement:

14.1.1 The measurement books are to be maintained by the officer-in-charge of the work or his immediate engineering subordinate not below the rank of Junior Telecom Officer. The entry shall be made in ink. No entry shall be erased. If a mistake is made, it should be corrected by crossing out the incorrect words or figures and inserting the corrections, the corrections thus made shall be initialed & dated by the officer concerned.

14.1.2 Responsibility of taking and recording measurements: The measurement of various items of work shall be taken and recorded in the measurement book issued with each work order. The measurement shall be taken and recorded by an officer not below the rank of Junior Telecom Officer, supervising the work. The Junior Telecom Officer/Sub Divisional Engineer/AGM, directly responsible for supervision of work, shall be responsible for accuracy of 100% of measurements (as specified in the Work Order /PO). The Divisional Engineer/AGM shall be responsible for conducting test check of 20% of measurements.

14.1.3 Method of recording of nomenclature of items: Complete nomenclature of items, as given in the agreement need not be reproduced in the measurement book for recording the measurements but corresponding Item Code as provided, shall be used.

14.1.4 Method of measurements: The measurements of the work shall be done activity wise as and when the item of work is ready for measurement. The methods of measurement of various items are enumerated as under:

14.1.4.1 Measurement of depth of trenches:

The cable routes of one work order shall be divided into a number of segments each of maximum 200 meters length bounded by identifiable landmarks at both the ends of the segments. If landmarks are not available, length of segment may be maintained at 200 meters. The measurement of depth shall be recorded at each point of measurement (POM) in the measurement book in meters in the multiples of 5 cm. For example, 97 cm will be recorded as 95 cm and 103 cm as 105 cm. The points of measurements shall be at a distance of 10 meters starting from 0 (zero) Meter. For example, if the length of segment is 75 meters, the POMs shall be at 0 m, 10 m, 20 m, 30 m, 40 m, 50 m, 60 m, 70 m. The last POM shall be at 75^{th}m to be recorded against Residual POM.

The efforts required to excavate trenches is not proportionate especially with reference to depth. Therefore, normally the workers tend to dig shallow trenches. As standard depth of the trench is important for future life and protection of cables, this tendency has to be discouraged. In order to encourage the contractor to achieve best possible depth in the face of site constraints, the following scale of payment shall be applied for digging trenches of lesser depths, subject to condition that relaxation has been granted by the competent authority for lesser depths.However,the requisite protection is to be provided in each case as per the statutory guidelines of BSNL.

TABLE-1: For Normal & Hard soil for all areas in Odisha Circle

Depth between	Reduction in rate pro rata basis
< 165 cm to≥ 150 cm.	5 % of approved rate
< 150 cm to≥ 130 cm.	12.5 % of approved rates

< 130 cm to≥ 105 cm.	25 % of approved rates
Below 105 cm	40 % of approved rates

TABLE-2: For Soft Rock & Hard Rock for all areas in Odisha Circle.

Depth between	Reduction in rate pro rata basis
< 135 cm to≥ 125 cm.	5 % of approved rate
< 125 cm to≥ 105 cm.	15 % of approved rates
Below 105 cm	40 % of approved rates

TABLE-3: For Hilly Terrain in all areas in Odisha Circle

Depth between	Reduction in rate pro rata basis
< 90 cm to≥ 80 cm.	5 % of the approved rate
< 80 cm to≥ 65 cm.	12.5 % of approved rates
< 65 cm to≥ 50 cm.	25 % of approved rates
Below 50 cm	40 % of approved rates

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The payment for sub normal length should be calculated using the following formula:

P= (100-ROR) X **RA** X **D/ (100** X **ND)**

P = Payment for one meter

ROR = Reduction in rate in % as given above

RA = Approved rate of trenching per meter

D = Actual depth in cm.

ND = Nominal depth of trench 165/135/90 cm for which the tender has been floated.

14.1.4.2 Measurement of Lengths and profiles of strata and protection:

The measurements of length of trenches are on running meter basis, irrespective of type of soil encountered while digging. The type of protection provided (item code-wise) in a segment shall be recorded in the measurement book in the sheet provided for this purpose.

14.1.4.3 Measurement of length of cable: The length of cables laid in trenches, through pipes and through ducts shall be measured by use of RODO Meter/Measuring Tape. The length should be cross verified with the making of lengths on the cables. The lengths shall be recorded in sheet provided in the measurement book.

14.1.4.4 Measurement of other items: The measurement/numerical details of other items shall be recorded in the sheets provided for respective items viz.

14.1.4.4.1 Digging of joint pit and preparation of joint chamber along with its type i.e. Brick chamber or Pre-Cast RCC type.

14.1.4.4.2 Fixing, Painting and sign writing of route/joint indicators.

14.1.4.4.3 Termination of Cable in equipment room and no. of joints.

14.1.4.4.4 The maximum earth resistance when measured shall not be more than 5 ohms at every joint chamber and Termination box (Applicable in case of Armoured OFC).

The contractor shall sign all the measurement recorded in the measurement book. This will be considered as an acceptance by the contractor, of measurements recorded in the MB. In case

contractor fails to attend at the measurements or fails to countersign or to record the difference within a week, then in any such events the measurements taken by Engineer-in-charge or by his subordinate as the case may be shall be final and binding on the contractor and the contractor shall have no right to dispute the same.

The Divisional Engineer/AGM before passing the bill for sections covered by each set of measurement may carry out test check by re-opening trench at as many locations as necessary as specified in document 'procedures for underground cable construction' and bills will be passed only when he is personally satisfied of the correctness of entries in the "measurement Book" and also when he is satisfied of other aspects of the work as per the terms of the contract. The contractor shall provide the necessary assistance of labour for re-opening of trench for test check by the Divisional Engineer/AGM. Separate payment shall not be made to the contractor for excavation of such test checks, however such test pits shall not be more than 10% of the cable laying work

Measurement of the work of cable pulling through pipe/duct will be taken equal to the length of the pipe / duct through which the cable has been pulled and not the total length of the cable pulled through pipe/duct.

15. Guidelines for Depth AT of Horizontal Directional Drilling

Brief Description of the work: A high-capacity drilling machine is used to drill the earth in almost horizontal manner and hence the name of the technology. Two manholes of 2 m x 2 m x 2 m pits are taken at the places marked by the site engineer. The distance between these manholes can vary from a minimum of 40 meters to 250 meters.

The drilling is done from Entry pit towards the Exit pit. At the Exit pit the PLB pipes are attached to the drill rods and the pipes are pulled back towards the Entry pit. In general, RCC manholes are constructed at the Entry pit and Exit pit after the HDD work is completed.

HDD work along with manholes:

15.1 The location of manholes shall not only depend on the capacity of the machine but also on the roads. The manholes where the drilling commences is called the Entry Pit. The second manhole is called as Exit pit.

15.2 Normally if the HDD machine is kept near the Entry pit and the drilling is carried out, then the depth at which the Machine enters the earth shall not be less than 1.2 m. As per the T&D standard any pipe laying less than 1.2 m requires additional protection. In order to achieve higher depth at the Entry pit the HDD machine is kept at around 6 to 8 meters away from the Entry pit. A separate pit called as pilot entry pit is dug up at 4 to 6 meters away from the Entry pit.

15.3 Wherever the roads are not in straight line, the span length reduces between the Entry and Exit pit.

15.4 In general, in overlay access works, the manholes are constructed at every 200 meters approximately.

15.5 The PLB pipes are coupled using the PLB couplers.

15.6 The HDD machine drills the earth using 3-meter-long rods. The first rod contains the drill bit and an electronic beacon house. While the underground drilling work is in progress the drill bit position can be monitored from the surface of the earth using a tracker. The depth and the direction of the bit are monitored from the tracker. This is due to beacon house located in the drill bit.

15.7 In the onward drilling from the Entry pit to Exit pit normally a lesser diameter (4-inch diameter) drill bit is used. After the drill bit reaches the Exit pit, the drill bit and the beacon house are removed from the first rod. A higher diameter drill bit called Back reamer is attached to the first rod. The pipes are attached to the reamer and the rods are pulled back. Thus, in the return path the tracker cannot detect the depth and direction of the reamer. Hence depth of the pipes can be monitored during the drilling from Entry pit towards Exit pit only.

15.8 The following tests and suggestions are recommended for the PLB pipes being laid using the HDD technology. Apart from the depth check, the pipes continuity check is also recommended especially for the ACCESS network jobs where multiple pipes are laid.

Standard Tests:

15.9 Online depth and offset check of the PLB pipes. The depth can be monitored from the tracker. The accuracy of the tracker may be verified by site in-charge before the work is commenced. The depth of the pipe should not be less than 2.0 meters.

15.10 The depths at least at two coupling points are to be verified in one km. The depth should not be less than 1.65 metres.

15.11 The offset of the pipe may be recorded both from the center of the road and also from the edge of the road wherever the compound wall exists (City Areas).

15.12 The colours of the pipes are to be recorded in case of multiple PLB pipe laying.

15.13. Protection:

It is recommended that the regular depth protection standards may be applied for HDD works also.

15.14 Depth Relaxation:

The depth relaxation is required at some situations. In general, when HDD is used only for crossing the roads and short spans of 15 to 25 meters length, due to the paucity of the road margin and due to the site conditions, the depth of 2.0 meters may not be achieved. The depth also cannot be achieved at places where very old underground concrete structures, water and drainage pipes exist. It is recommended that the depth relaxation may be given by the competent authority.

S/No.	Depth measured at 10 metres from the trailing edge of the Entry Pit	Recommending Authority	Relaxing Authority
1	> or = 1.65 m to 2.0 m	SDE in charge	DET in charge
2	> or = 1.5 m to 1.65 m	DET in charge	DGM in charge

Note: For depth more than 165 cm, rates for horizontal drilling will remain same. In case of depths being less than 1.65 m up to 1.5 m, the payment of horizontal drilling will be made on pro-rata basis.

It is hereby recommended that no pipe shall be laid by HDD method for depths less than 1.5 m. Hence no protection is required for the Pipes.

16. <u>Inspection and Quality Control:</u>

16.1 **The Quality of Works:** The importance of quality of Optical Fibre Cable Construction works especially laying of multiple PLB pipes/coils using open trenching and trenchless technology method cannot be over-emphasized. The quality and availability of long-distance media, efficiency of the reliable media connectivity to exchanges and success of

new concept of overlay access network depends upon quality of laying of Optical Fibre Cable. Further, the OF cables are vulnerable to damages due to work of other agencies.

16.2 **It is imperative that the contractor(s) is/are fully conversant with the construction practices especially laying multiple pipes by trenchless technology using HDD machine and shall be fully equipped to carry out the work in accordance with the specifications**. The contractors are expected and bound to ensure quality in construction works in accordance with specifications laid down. The contractor shall engage adequate and experienced supervisors to ensure that works are carried out as per specifications and with due diligence and in a professional manner. The contractors shall satisfy himself/themselves that the work conforms to the quality specifications before offering the same to A.T. Wing for Acceptance and Testing.

16.3 An assessment of extent of interest shown by the contractors in executing the works with requisite quality shall be recorded and used in evaluating the Contractors' Performance Rating (CPR).

16.4 In addition to Acceptance Testing being carried out by A.T. Wing and supervision by concerned Officers, all works at all times shall be open to inspection of the department. The contractors shall be bound, if called upon to do so, to offer the works for inspection without any extra payment.

16.5 **Site Order Book**: The site order book is one of the primary records to be maintained by the JTO/SDE supervising the work during the course of execution of works. The noting made by officers as well as contractors, will form as basis for operation of many contractual clauses. The contractor shall remove all the defects pointed out by the department in the Site order book. The contractor or their authorized representatives shall also be at liberty to note their difficulties etc. in these books. The site order books shall invariably be consulted at the time of making final payments to the contractor.

17. Testing and Acceptance Testing:

17.1 The work shall be deemed to have been completed only after the same has been accepted by the A.T. Officer. The contractor shall make test pits at the locations desired by A.T. Officer for conducting test checks without any extra payment. The contractor shall restore the pits after test measurements to its original shape. The contractor shall be responsible to provide test/measurement tools and testers for conducting various tests.

17.2 Scope of Acceptance and Testing: The purpose of acceptance and testing is to verify integrity of measurement and quality of work done. The A.T. Officer shall not be responsible for recording of measurements for the purpose of billing and contractual obligations. However, if the measurements taking by A.T. officer are found to be lesser than the measurements recorded by the officer responsible for recording the measurements, the measurement taken by A.T. Officer shall prevail without prejudice to any punitive action against the contractor as per provisions of the contract and the officer recording the measurements. The contractor shall be obligated to remove defects/deficiencies pointed out by the A.T. Officer without any additional cost to the department.

17.3 Offering the work for acceptance and testing: The Sub Divisional Engineer/AGM responsible for construction, after having satisfied himself of completion of work ready for A.T., shall offer the work to A.T. Officer for conducting Acceptance and Testing. The work shall be offered for A.T. as soon as part of work is complete in all respects. The work against any work order can be offered for A.T. in a number of stages.

17.4 The contractor shall provide labours, if demanded by the A/T officer for digging of test pits and other necessary infrastructure for carrying out the A/T work. No extra payment will be made for the digging of test pit

18. WARRANTY:

18.1 The contractor shall warrant that the material supplied for the work shall be new and free from all defects and faults in material, workmanship and manufacture and shall be of the highest grade and consistent with the established and generally accepted standards for materials of the type ordered and shall perform in full conformity with the specifications and drawings. The contractor shall be responsible for any defects that may develop under the conditions provided by the contract and under proper use, arising from faulty materials, design or workmanship such as corrosion of the equipment, inadequate quantity of materials etc. and shall remedy such defects at his own cost when called upon to do so by BSNL who shall state in writing in what respect the stores are faulty. This warranty shall survive inspection or payment for, and acceptance of goods, but shall expire except in respect of complaints notified prior to such date, **twelvemonths after the Acceptance Testing.**

18.2 If it becomes necessary for the contractor to replace or renew any defective portion/portions of the material under this clause, the provisions of the clause shall apply to the portion / portions of material so replaced or renewed or until the end of the above-mentioned **period of twelve months**, whichever may be later. If any defect is not remedied within a reasonable time, as prescribed by BSNL, BSNL may proceed to do the work at the contractor's risk and costs, but without prejudice to any other rights which BSNL may have against the contractor in respect of such defects.

18.3 **The Cable joint shall be guaranteed for a period of ONE year from the date of closing of joint**. In case of failure of the joint due to poor workmanship i.e. failure of joint without external damage, within the stipulated period of guarantee the contractor shall repair the joint(s) at his own cost within 24 hours of informing him**, failing which BSNL may carry out the repairs and penalty equivalent to five times of the approved rate of the jointing work plus the cost of materials used shall be** recovered from the contractor from his pending bill, PBG or any amount due to him without prejudice to any other action as per terms and conditions of the tender. The cost of joining kit, supplied by BSNL, so used to revive the joint shall be deducted from the running bills of the contractor pending for payment or from PBGif all bills have been settled.

18.4 Replacement under warranty clause shall be made by the contractor free of all charges at site including freight, insurance, cost of works and other incidental charges.

19 <u>AUDIT AND TECHNICAL EXAMINATION:</u>

19.1 BSNL shall have the right to cause an audit and technical examination of the work and the final bills of the contractor including all supporting vouchers, abstract etc. to be made after payment of the final bill and if as a result of such audit and technical examination any sum is found to have been overpaid in respect of any work done by the contractor under the contract or any work claimed by him to have been done by him under the contract and found not to have been executed, the contractor shall be liable to refund the amount of over payment and it shall be lawful for BSNL to recover the same from him in the manner prescribed in clause with the heading payment of bills (same chapter), or in any other manner legally permissible and if it is found that the contractor was paid less than what was due to him under payment shall be duly paid by BSNL to the contractor.

19.2 Provided that BSNL shall be entitled to recover any sum overpaid, nor the contractor shall be entitled to payment of any sum paid short where such payment has been agreed upon between the Divisional Engineer/AGM or his subordinate officer on one hand and

the contractor on the other under any term of the contract permitting payment for work after assessment by the **Competent Authority** or his subordinate officer.

19.3 Any sum of money due and payable to the contractor (including security deposit returnable to him) under this contract may be appropriated by BSNL for the payment of a sum of money arising out or under any other contract made by the contractor with BSNL

----- END OF SECTION IV ----

SECTION - V

SERVICE LEVEL AGREEMENTPROFORMA

The successful tenderer shall have to execute the following agreement.

This agreement made on this _____ day of (month)_____ (year)_____ between M/s _____ _____ hereinafter called "The Contractor" (which expression shall unless excluded by or repugnant to the context, include its successors, heir, executors, administrative representative and assignee) of the one part & Chairman & Managing Director, BSNL hereinafter referred to as BSNL, of other part.

Whereas the contractor has offered to enter into contract with BSNL for the execution of work of OFC laying on behalf of BSNL using open trenching and trenchless technology and other associated works of OF Cable on the terms and conditions herein contained and the rates approved by BSNL (copy of Rates annexed) have been duly accepted and whereas the necessary security deposits have been furnished in accordance with the provisions of the tender document and whereas no interest will be claimed on the security deposits.

Now these presents witness and it is hereby agreed and declared by and between the parties to these presents as follows.

1) The contractor shall, during the period of this contract that is to say from……. to ……. or completion of work for Rs._____(In words) _____ whichever is earlier or until this contract shall be determined by such notice as is hereinafter mentioned, safely carry out, by means of labours employed at his own expenses and by means of tools, implements and equipment etc. to be supplied by him to his labour at his own expenses, all trenching and trenchless pipe laying, cable pulling, cable splicing, joint chamber preparation, fixing, painting and sign writing or route indicators and other associated works as described in tender documents (annexed to the agreement), when the Tender Issuing Authority or any other persons authorized by the competent authority of BSNL in that behalf require. It is understood by the contractor that the quantity of work mentioned on the schedule is likely to change as per actual requirements as demanded by exigencies of service.

2) The NIT (notice inviting tender), Bid documents (Qualifying and Financial), letter of intent, approved rates, annexed hereto and such other additional particulars, instructions, drawings, work orders as may be found requisite to be given during execution of the work shall be deemed and taken to be an integral part of the contract and shall also be deemed to be included in the expression "The Agreement" or "The Contract" wherever herein used.

3) The contractor shall also supply the requisite number of workmen with means & materials as well as tools, appliances, machines, implements, vehicles for transportation, cartage etc. required for the proper execution of work within the time prescribed in the work orders.

4) The contractor hereby declares that nobody connected with or in the employment of BSNL is / shall ever be admitted as partner in the contract.
5) The Contractor shall obtain the RoW Permission from the concerned authority wherever required within the stipulated time as decided by the competent authority of BSNL.

However,BSNL will pay only for the amount of demand note issued towards RoW charges directly to the concerned authorities.

6) The contractor shall assist the concerned supervising officer of BSNL in obtaining and uploading the required data on TRANSNET during the execution of the work.

7) The contractor shall abide by the terms and conditions, rules, guidelines, construction practices, safety precautions etc. stipulated in the tender document including any correspondence between the contractor and BSNL having bearing on execution of work and payments of work to be done under the contract.

In witness whereof the parties present have hereunto set their respective hands and seals this …… day of …… in the year ……

Above written:

Signed, Sealed &Delivered by
the above-named Contractor in
the presence of

Witness:

1.

2.

Signed &Delivered on behalf
of BSNL by the

Witness:

1.

2.

----- END OF SECTION V ---

SECTION - VI
PAYMENT TERMS

1. Procedure for Preparation and settlement of bills:
All items of work involved in the work order shall be completed in all respects before preparing the bills for the work against the work order. The procedure for preparation of final bills is enumerated as under:

1.1 Procedure for preparation, processing and payment of bill: As the work of a route is being executed through work orders for section/segment of the route, there is no provision for running bill. The payment will be released after acceptance and testing of the work ordered against the said work order. The contractor shall prepare the final bill after completion of the entire work entrusted against work order & acceptance and testing of all the works and submit to the SDE in-charge of the work within 30 days of acceptance and testing for payment. The final bill shall be prepared for all the measurements of all items involved in execution of complete work order. The contractor shall prepare the final bill containing the following details:

1.1.1 The bill for all the quantities as per Measurements at the approved rates.
1.1.2 Adjustment of performance security deposit and statutory taxes already recovered.
1.1.3 Store reconciliation statement furnishing account of stores received against the **Work Order** and returned to the designated Store go-down as surplus with requisite verifications from store in-charge/SDE in-charge of work.
1.1.4 Letters of grant of E.O.T(s), if work could not be completed within stipulated time.
1.1.5 Six sets of bound documentation.
1.1.6 Copy of the Wage Register, Attendance Register, Monthly EPF & ESI Deposit Challan have to be submitted by the contractor along with the bills.

1.2. The SDE in-charge of work shall scrutinize the final bill against the works entrusted and accord necessary certificates stating that the work has been executed satisfactorily in accordance with specifications and terms and conditions of the contract. The SDE in charge of work shall verify the quantities of items of work with reference to measurements recorded in the measurement book (and also A/T reports in case of any deviations noted by A/T office). The SDE in-charge of work shall submit the final bills, along with other documents mentioned above, with the documents as mentioned hereunder to the Divisional Engineer/AGM, in-charge of work.

1.2.1 Bill prepared by the contractor
1.2.2Material reconciliation statement
1.2.3 Measurement Book
1.2.4 A/T Certificates
1.2.5 The site order Book
1.2.6 The hindrances register
1.2.7 Details of recoveries/penalties for delays, damages to Departmental/Third party properties as per provisions of the contract. In case no recovery is to be made, NIL report needs to be submitted
1.2.8 Details of empty cable drums cost which needs to be recovered from the bill
1.2.9 Service Completion Certificate
1.2.10 Route Index Diagram
1.2.11 Graphical printout of depth profile in case of HDD

1.3. The Divisional Engineer/AGM shall exercise the prescribed checks on the bills and accord necessary certificates on the bills. The Divisional Engineer/AGM shall retain the third copy of the bill along with photocopies of other documents not available in his estimate file and send first and second copies of the bill, measurement book and other documents

submitted by SDE along with the bills as above to the Planning/MM Cell of the concerned BA for processing and final submission to the Office of Paying Authority duly approved by the BA Head.

1.4. Office of Paying Authority shall process the bills in the estimate file of the concerned work and scrutinize the bills vis-à-vis work order issued and sanctioned provisions in the estimate etc. The office shall also scrutinize the bill to recover all the liabilities of the contractor and statutory taxes. The bill shall be passed, after necessary scrutiny by Works Section, by the officer competent to pass the final bill.

1.5. Procedure for payment for substandard works:

1.4.1 The contractors are required to execute all works satisfactorily and in accordance with the specifications. If certain items of work are executed with unsound, imperfect or unskilled workmanship or with materials of any inferior description or that any materials or articles provided by him for execution of work are unsound or of a quality inferior to that contracted for or otherwise, not in accordance with the contract (referred to as substandard work hereinafter), the Divisional Engineer/AGM in-charge shall make a demand in writing specifying the work, materials or articles about which there is a complaint.

1.4.2 Timely action by Site In-charge: Timely reporting and action, to a great extent, can prevent occurrence of substandard work, which will be difficult or impossible to rectify later on. It is incumbent on the part of site in-charge to point out the defects in work in time during progress of the work. The Junior Telecom Officer/Sub Divisional Engineer/AGM responsible for execution and supervision of work shall without any loss of time submit a report of occurrence of any substandard work to the Divisional Engineer/AGM in-charge besides making an entry in the site order book. A notice in respect of defective work shall be given to the contractor by Divisional Engineer/AGM in-charge in writing during the progress of work asking the contractor to rectify/replace/remove the substandard item of work and also define time period within which such rectification/removal/replacement has to be done. After expiry of the notice period, if the contractor fails to rectify/replace/remove the substandard items, the defects shall be got rectified/replaced/removed departmentally or through some other agency at the risk and cost of the contractor.

1.4.3 Non-reporting of the substandard work in time on the part of site in-charge(s) shall not in any way entitle the contractor to claim that the defects were not pointed out during execution and as such the contractor cannot be absolved of the responsibility for substandard work and associated liabilities.

1.4.4 Authority and Procedure to accept substandard work and payment thereof: There may be certain items of work pointed out as substandard which may be difficult to rectify and in the opinion of the concerned Engineer-in-charge of work, the items in question will not materially deteriorate the quality of service provided by the construction. The concerned BA Head shall appoint committee to work out the reduced rates payable to the contractor for such substandard work. The committee shall constitute one Divisional Engineer/AGM other than the one who is directly in-charge of Cable Construction involving substandard items of work, as Chairman and one SDE and an Accounts Officer as members. The committee shall take into account the approximate cost of material/work pointed out as substandard and recommend the rates payable for substandard work which shall not exceed 60% of the approved rates of the item in question.

1.4.5 Record of substandard work: The items adjudged as substandard shall be entered into the measurement book with red ink.

1.4.6 No claim for delayed payment due to dispute etc.: No claim for interest or damage will be entertained or be payable by BSNL in respect of any amount or balance which may be lying with BSNL owing to any dispute, difference or misunderstanding between the parties or in respect of any delay or omission on the part of BSNL in making intermediate or final payments or in any other respect whatsoever.

2. DISPOSAL OF EMPTY CABLE DRUMS:

2.1 The contractor shall be responsible to dispose-off the empty cable drums after laying of the cable. The competent authority taking into account the prevailing market rates has fixed the cost of Optical Fibre cable drums. The cost of empty cable drums shall be deducted from the bill for the work on which the cable along with the drum has been issued or any other amount due to the contractor or from security deposit.

2.2 Rates fixed for Optical Fibre cable drums are given in Tender Document (Qualifying Bid). The rates are fixed and there is no percentage above or below applicable on these rates**.**

2.3 The contractor shall be responsible for the accounting of the cable drums issued and shall mention the number of cable drums in the bill so that the amount if deducted from the bills due.

2.4 The contractor shall not be allowed to dump the empty cable drums in Govt., public place, which may cause inconvenience to Govt. /Public. If the contractor does not dispose-off the empty cable drums within 3 days of becoming empty, BSNL is at liberty to dispose-off the drum in any manner deemed fit and also recover the amount fixed in this contract for empty cable drums along with the transportation charges to be decided by the **Competent Authority** from the bill / security deposit / any other amount due to the contractor.

----- END OF SECTION VI ----

SECTION - VII
PENALTY CLAUSE

1 Delays in the contractor's performance:

1.1 The time allowed for completion of the work as per work order shall be strictly adhered by the contractor and shall be deemed to be the most important aspect of the contract on the part of the contractor and shall be reckoned from Seventh day from issue of work order by BSNL. The work shall, throughout the stipulated period of contract, be proceeded with all due diligence to achieve the desired progress uniformly, and the contractor shall pay as penalty for delay in execution of the work @ 0.5 % for each week of delay or part thereof, for a period up to ten weeks and thereafter @ 0.7 % for each week of delay or part thereof for another ten weeks subject to a maximum of 12 % of the cost of the work allotted as per the work order.

1.2 On any date the penalty payable as above, reaches 12 (twelve) percent of the cost of the work, the contractor should proceed with the work further only on getting a written instruction from the Divisional Engineer/AGM in charge of work that, he is allowed to proceed further with the work. It will be in the discretion of the Divisional Engineer/AGM to allow the contractor to continue with the work on the basis of any written agreement reached between the contractor and the Divisional Engineer/AGM. One of the conditions of such agreement may be a stipulation for the contractor to agree for realization of penalty for delay at a higher rate as may be agreed between Divisional Engineer/AGM and contractor.

1.3 The days on which work is not done due to reasons beyond the control of contractor, such as natural calamities, law & order situation etc., will not be accounted for in imposing the penalty.

1.4 Penalty for delay in completion of the work shall be recoverable from the bills of the contractor and/or by adjustment from the security deposit or from the bills of any other contract. However, adjustment from PBG will be made only when the contract has been terminated or at the time of final settlement of bills on completion of work.

1.5 In case of slow progress of the work in a section which has been awarded to a particular contractor, and the public interest does not permit extension of time limit for completion of the work, the Divisional Engineer/AGM in charge of work will have the full right to order that the scope of the contractor may be restricted to such fraction of the work and get the balance executed at the risk and cost of the contractor. The details are given in Rescission of the contract clause of the bid document. All such payments shall be recovered from the contractor's pending bills or security deposit.

1.6 The competent authority of BSNL reserves the right to cancel the contract and forfeit the PBG if the contractor fails to commence the work within 7 days after issue of the work order.

2. Penalty for causing inconvenience to the Public:

2.1 To ensure progress during the execution of work and to cause minimum inconvenience to the public, the contractor shall not dig a trench of more than 200 meters at a stretch in a route at a time. He shall cause to lay cable and close such trenches expeditiously. Under any circumstances a stretch of trench of maximum 200 meters shall not be kept open for more than 2 days in case of cable laying by digging paved surfaces. In the event of contractor failing to comply with these conditions, a penalty of recovery up to Rs.300/-

per day, if the trench is kept open beyond the time limit allowed, may be imposed by BSNL. This penalty will be in addition to that payable for delay or slow work.

2.2 The contractor shall not be allowed to dump the empty cable drums/waste materials in Govt./public place, which may cause inconvenience to Govt./Public. If the contractor does not dispose-off the empty cable drums/waste materials within 3 days of becoming empty, BSNL is at liberty to dispose-off the drum in any manner deemed fit, and also to recover the amount fixed in this contract for empty cable drums/waste materials from the bill/security deposit/ along with the costs incurred by BSNL in disposing off such materials. BSNL may also levy a penalty up to Rupees One thousand only for each such default.

3. Penalty for cutting / damaging the old cable:

3.1 During excavation of trench utmost care is to be taken by the contractor, so that the existing underground cables are not damaged or cut. In case of any cut or damage, the contractor shall carry out such repairs for restoration of the damaged cable free of charge. The cost of jointing kit and additional cable required for repair shall also be borne by the contractor.

Size of existing UG cable Cut /damaged	Account of penalty per cut/damage
Upto100 pairs cable	Rs 3000.00 (Three thousand)
Above 100 pairs cable & up to 400 pairs	Rs 5000.00 (Five thousand)
Above 400 pairs	Rs 10000.00 (Ten thousand)

For OF Cable Penalty for cutting/damaging will be as below:

Type of cable Cut /damaged	Charges per fault for penalty and compensation
OF Cable of any size	Rs 25000.00 (Twenty Five thousand)

3.2 If contractor fails to repair the damage, the cost of repair (including cost of Labour and material) and penalty will be charged as per the schedule given above which will be charged from the contractor or the amount will be deducted from his Bill.

3.3 The Engineer-in-Charge will be the sole authority to take the decision if repair has been done satisfactorily within reasonable time and he will decide if any amount will be recovered or otherwise.

4. Penalty to damage stores/materials supplied by BSNL while laying:

The contractor while taking delivery of materials supplied by BSNL at the designated place shall thoroughly inspect all items before taking them over. In case of execution of the work, if any material is found damaged/working unsatisfactorily, then a penalty equivalent to the cost of material + 10% as penalty shall be recovered from the contractor's payments/securities.

However, contractor will not be penalized for any defect in workmanship of the materials, which shall be taken up separately with the supplier of the stores.

5. Penalty to damage the underground installations of other agencies:

While trenching, the contractor should not cause damage to any underground installations belonging to other agencies and any damage caused should be made good at his own cost and expenses.

6. Penalty for achieving Lower Depth:

In no case the statutory depth of boring should be (i) less than 2 m from the road / ground surface and (ii) less than 1.65 m at the pits. In **order** to encourage the contractor to achieve required depth in the face of site constraints, the following payment schedule with penalty are to be adhered to for boring at lesser depths subject to condition that relaxation has been granted by the competent authority for lesser depths.

6.1 For ground surfaces (other than pits):

Actual Micro-tunneling (HDD) Depth between	Penalty
<2 m to > 1.50 m	10% of approved rate
< 1.50 m	No payment. To be treated as non-standard work.

6.2 For surfaces at pits:

Actual Micro-tunneling (HDD) Depth between	Penalty
< 1.65 m to > 1.50 m	10% of approved rate
< 1.50 m	No payment. To be treated as non-standard work.

7. Rescission / Termination of Contract:

7.1 Circumstances for rescission of contract: Under the following conditions the competent authority of BSNL may rescind the contract:

7.1.1 If the contractor fails to start the work within stipulated time as specified in the work order and after issue of Show Cause Notice.

7.1.2 If the contractor commits breach of any item of terms and conditions of the contract.

7.1.3 If the contractor suspends or abandons the execution of work and the engineer in-charge of the work comes to conclusion that work could not be completed by due date for completion or the contractor had already failed to complete the work by that date.

7.1.4 If the contractor had been given by the officer-in-charge of work a notice in writing to rectify/replace any defective work and he/she fails to comply with the requirement within the specified period.

7.2 Upon rescission of the contract, the EPBG/PBG/DD of the contractor shall be liable to be forfeited and shall be absolutely at the disposal of BSNL as under:

7.2.1 Measurement of works executed since the date of last measurement and up to the date of rescission of contract shall be taken in the presence of the contractor or his authorized representative who shall sign the same in the MB. If the contractor or his authorized representative do not turn up for joint measurement, the measurement shall be taken by the officer authorized for this purpose after expiry of due date given for joint measurement. The measurement taken by the officer so authorized shall be final and no further request for joint measurement shall be entertained.

7.2.2 The unused material (Supplied by BSNL) available at site, shall be transported back by BSNL to the Telecom Store at the risk and cost of the contractor. If any such

material is found damaged/lost then the penalty shall also be recovered from the contractor as per conditions in tender documents/bid.

7.2.3 If a contractor left the work incomplete in his zone due to whatever reason may be, then the remaining unexecuted work (balance work of the zone) shall first be offered to the L2 bidder of the same zone at the L1 rate. If the L2 bidder declines the offer, the work will then be offered to the L1 contractors from all adjacent zones at the L1 rate of the zone under consideration. If multiple contractors from adjacent zones accept the offer, the contractor will be selected on the basis of their performance in their respective zones. In such an event no compensation shall be payable by BSNL to the default contractor towards any inconvenience/loss that he may be subjected to as a result of such an action by BSNL. In this regard the decision of concerned BA head shall be final and binding. In all these cases, expenses which may be incurred in excess of the sum which would have been paid to the original contractor if the whole work had been executed by him shall be borne and paid by the original contractor and shall be deducted from any money due to him by BSNL under the contract or any other account whatsoever anywhere in BSNL or from a security deposit.

7.2.4 The certificate of the Divisional Engineer/AGM in-charge of work as to the value of work done shall be final and conclusive against the contractor, provided always that action shall only be taken after giving notice in writing to the contractor.

7.3 Termination for Insolvency:

7.3.1 BSNL may at any time terminate the Contract by giving written notice to the Contractor, without compensation to contractor, if the contractor becomes bankrupt or otherwise insolvent as declared by the competent court provided that such termination will not prejudice or affect any right or action or remedy which has accrued or will accrue thereafter to BSNL.

7.3.2 Optional Termination by BSNL (other than due default of the contractor):
BSNL may, at any time, at its option cancel and terminate this contract by written notice to the contractor, in which event the contractor shall be entitled to payment for the work done up to the time of such cancellation.

7.3.3 In the event of the termination of the contract, the contractor shall forthwith clear the site of all the contractor's materials, machinery and equipment and hand over possession of the work / operations concerned to BSNL or as BSNL may direct.

7.3.4 BSNL may, at its option, cancel or omit the execution of one or more items of work under this contract and/or part of such items without any compensation whatsoever to the contractor.

7.3.5 Issuance of Notice:

The Divisional Engineer/AGM in-charge of work shall issue show cause notice giving details of lapses, violation of terms and conditions of the contract, wrongful delays or suspension of work or slow progress to the contractor directing the contractor to take corrective action. A definite time schedule for corrective action shall be mentioned in the show cause notice. If the contractor fails to take corrective action within the stipulated time frame, the Divisional Engineer/AGM in-charge shall submit a draft of final notice along with a detailed report to the competent authority who had accepted the contract.

7.3.6 The **final notice for rescission of contract to** the contractor shall expressly state the precise date and time from which the rescission would become effective. The following safeguards shall be taken while issuing the final notice.

7.3.6.1 During the period of service of notice and its effectiveness, the contractor should not be allowed to remove from the site any material/equipment belonging to BSNL.

7.3.6.2 The contractor shall give in writing the tools and plants he would like to take away/remove from the site. Such of the materials as belong to him and which may not be required for future execution of balance work may be allowed by the Divisional Engineer/AGM in-charge of work to be removed with proper records.

7.3.6.3 No new construction beneficial to the contractor shall be allowed.

7.3.6.4 Adequate BSNL security arrangement in replacement of the contractor watch and ward shall be made forthwith. Expenses on this account are recoverable from the security deposit or any amount due to the contractor.

8 INDEMNITIES:

8.1 The contractor shall at all times hold BSNL harmless and indemnify from against all action, suits, proceedings, works, cost, damages, charges, claims, and demands of every nature and descriptions, brought or procured against BSNL, its officers and employees and forthwith upon demand and without protest or demur to pay to BSNL any and all losses and damages and cost (inclusive between attorney and client) and all costs incurred in endorsing this or any other indemnity or security which BSNL may now or at any time have relative to the work or the contractors obligation or in protecting or endorsing its right in any suit on other legal proceeding, charges and expense and liabilities resulting from or incidental or in connection with injury, damages of the contractor or damage to property resulting from or arising out of or in any way connected with or incidental to the operations caused by the contract documents. In addition, the contractor shall reimburse BSNL or pay to BSNL forthwith on demand without protest or demur all cost, charges and expenses and losses and damages otherwise incurred by it in consequences of any claim, damages and actions which may be brought against BSNL arising out of or incidental to or in connection with the operation covered by the contractor.

8.2 The contractor shall at his own cost at BSNL's request defend any suit or other proceeding asserting a claim covered by this indemnity, but shall not settle, compound or compromise any suit or other finding without first consulting BSNL.

9 FORCE MAJEURE:

9.1 If any time, during the continuance of this contract, the performance in whole or in part by either party or any obligation under this contract shall be prevented or delayed by reason of any war, or hostility, acts of the public enemy, civil commotion, sabotage, fires, floods, explosions, epidemics, quarantine restrictions, strikes, lockouts or act of God (Here in after referred to as events) provided notice of happenings of any such eventuality is given by either party to the other within 21 days from the date of occurrence thereof, neither party shall by reason of such event be entitled to terminate this contract nor shall either party have any such claim for damages against the other in respect of such non-performance and work under the contract shall be resumed as soon as practicable after such event may come to an end or cease to exist, and the decision of BSNL as to whether the work have been so resumed or not shall be final and conclusive, provided further that if the performance, in whole or part of any obligation under this contract is prevented or delayed by reason of any such event for a period exceeding 60 days party may, at his option terminate the contract.

9.2 Provided also that if the contract is terminated under this clause, BSNL shall be at liberty to take over from the contractor at a price to be fixed by BSNL, which shall be final, all unused, undamaged and acceptable materials, bought out components and stores in the

course of execution of the contract, in possession of the contractor at the time of such termination of such portions thereof as BSNL may deem fit excepting such materials, bought out components and stores as the contractor may with the concurrence of BSNL select to retain.

10. RESOLUTION OF DISPUTES

10.1 Disputes and Excepted Matters:

10.1.1 All disputes and differences between the parties hereto, as to the construction or operation of this contract, or the respective rights and liabilities of the parties on any matter in question; or any other account whatsoever, but excluding the Excepted Matters (detailed below); arising out of or in connection with the contract, that cannot be resolved amicably between the Contractor and Purchaser/BSNL, shall be hereinafter called the "Dispute". The aggrieved party shall give a 'Notice of Dispute' specifying all points of Dispute/ issue, the amount of any quantified claims citing relevant Contractual clause, and, to the extent possible, an estimate of the monetary value of all other claims, along with supporting document(s) to the other party, and requesting for invoking the following dispute resolution mechanisms.

10.1.2 After receipt of a 'Notice of Dispute', as above, the parties shall in good faith, make all reasonable efforts, to arrive at a mutually acceptable resolution to the disputes raised in the "Notice of Dispute" in formal meetings(s) between authorized representatives of the parties.

10.1.3 Parties agree that any effort by either party for arriving at the mutually acceptable resolution of the disputes is to be kept confidential by both Parties. Parties also agree to not rely upon any views expressed, admissions or suggestions made, or willingness to enter into a settlement by either party as evidence in any forum / arbitration / court proceeding.

10.1.4 Parties agree that neither party shall be entitled to any claim or compensation for any consequential, indirect or special losses/damages, including loss of profit, loss of production, loss of use, loss of goodwill, loss of reputation, remote damages, loss of business opportunities, loss of employment opportunities, loss of interest including any pre-reference or pendente-lite interest, idling costs of men and machinery, prolongation costs etc., on account of any dispute/ claim raised under the contract. Parties agree that claim for any such amount shall not be considered and shall be void.

10.1.5 In case the parties fail to amicably resolve the disputes between them within 60 days of receipt of "Notice of Dispute" sent under clause 25.1.1 above, the following provisions, as mentioned below, shall follow:

The Dispute shall be resolved without recourse to courts through dispute resolution mechanisms detailed subsequently, in the sequence as mentioned below, and the next mechanism shall not be invoked unless the earlier mechanism has been invoked or has failed to resolve it within the deadline mentioned therein.

1. **Senior-Level Negotiation**
2. Mediation
3. Litigation

10.2 Excepted Matters

Matters for which specific provision has been made in any clause of the Contract, and where the decision of BSNL has been stated to be final and binding, shall be deemed as Excepted Matters and shall not be subject to **Senior-Level Negotiation**, mediation, or arbitration. These shall include, but shall not be limited to, the following:

1. Any controversies or claims brought by a third party for bodily injury, death, property damage or any indirect or consequential loss arising out of or in any way related to the performance of this Contract ("Third Party Claim"), including, but not limited to, a Party's right to seek contribution or indemnity from the other Party in respect of a Third-Party Claim.
2. Issues related to the pre-award tender process or conditions.
3. Issues related to ambiguity in contract terms shall not be taken up after a contract has been signed. All such issues should be highlighted before the signing of the contract by the contractor.
4. Any claim, difference or dispute relating to, connected with or arising

   i. out of BSNL's decision to initiate proceeding for suspension or banning, or decision to suspend or to ban business dealings with the Bidder / Contractor;
   ii. Any claim, difference or dispute relating to, connected with or arising out of BSNL's decision under the provisions of Integrity Pact executed between BSNL and the Bidder / Contractor;
   iii. Any dispute pertaining to insolvency and bankruptcy, property laws.

5. Provisions incorporated in the contract, which are beyond the purview of BSNL or are in pursuance of policies of Government, including but not limited to

   i. Provisions of restrictions regarding local content and Purchase Preference to Local suppliers in terms of Make in India policy of the Government.
   ii. Provisions regarding restrictions on Entities from Countries having land-borders with India in terms of the Government's policies in this regard
   iii. Purchase preference policies regarding MSEs and Start-ups

10.3 Senior-Level Negotiation

After exhausting efforts to resolve the Dispute with the Purchaser/BSNL executing the contract on behalf of the Procuring Entity, the contractor shall give a 'Notice of **Senior-Level Negotiation**' specifying the matters which are in question, or subject of the dispute or differences indicating the relevant contractual clause, as also the amount of claim item-wise to CMD BSNL or CGM of the BSNL circle as the case may be, or any other authority mentioned in the contract (hereinafter called the " **Negotiating Authority**") for invoking resolution of the dispute through **Senior-Level Negotiation**. During such **Senior-Level Negotiation**, the **Negotiating Authority** shall give adequate opportunity to the contractor to present his case. Within 60 days after receiving the representation, the **Negotiating Authority** shall **record and communicate in writing the outcome of such negotiations on all matters referred to him. It is clarified that Senior-Level Negotiation is an internal bilateral settlement mechanism intended to facilitate resolution of disputes at an elevated management level and does not constitute a judicial or quasi-judicial adjudication.** The parties shall not initiate, during the **Senior-Level Negotiation** proceedings, any conciliation or arbitral or judicial proceedings in

respect of a dispute that is the subject matter of the **Senior-Level Negotiation** proceedings. If not satisfied by the outcome in **Senior-Level Negotiation**, or if the **Negotiating Authority** fails to notify his decision within the abovementioned time-frame, the contractor/ purchaser may proceed to invoke the process of Mediation as follows.

10.4 Mediation of disputes

1. For cases with dispute amount more than 10 Crores, Mediation shall be conducted through the mediation scheme framed by the Department of Telecommunications, Government of India vide OM No. 1-1/4/2024-Law
   (I) dated 22.11.2024, comprising, where considered appropriate, a retired judge, a retired senior officer, and/or a technical expert ("Mediation Scheme").
2. For cases with dispute amount less than 10 Crores, the mediation process will be adopted as per institutional mediation provisions defined under the Mediation Act, 2023.
3. The cost of the mediation shall be shared equally by the parties unless otherwise agreed.
4. If the parties reach an agreement on a dispute settlement, they shall draw up a written settlement agreement duly signed by the parties and Mediator. When the parties sign the settlement agreement, it shall be final and binding on the parties. The dispute shall be treated as resolved on the date of such agreement.
   It is clarified that mediation is a voluntary and confidential process in which the mediator acts solely as a facilitator and shall not have authority to impose any decision or determination on the parties.
5. The parties shall not initiate, during the mediation proceedings, any arbitral or judicial proceedings in respect of a dispute that is the subject matter of the mediation proceedings.
6. Termination of Mediation: Disputes shall remain alive if the mediation is terminated as follows:
   (a) By written declaration of the mediator, after consultation with the parties, to the effect that further efforts at mediation are no longer justified, on the date of such declaration; or
   (b) By a written declaration of any party to the mediator, before the signing of settlement agreement by both sides, to the effect that the mediation proceedings are terminated, on the date of such declaration; or
   (c) If the parties fail to reach an agreement on a settlement of the dispute, within stipulated time from the appointment of mediator.

   On termination of Mediation, if the dispute is still alive, the aggrieved party shall be free to invoke litigation.

10.5 Dispute Settlement in case of the disputes between two Central Public Sector Enterprises or with any government body, the following shall apply:

In case of any dispute between BSNL and another Central Public Sector Enterprise (CPSE), or between BSNL and a Government Department/Organization (excluding disputes involving Railways, Income Tax, or Customs & Excise), the dispute shall be referred for resolution through the Administrative Mechanism for Resolution of CPSEs Disputes (AMRCD) in accordance with DPE OM No. 05/0003/2019-FTS-10937 dated 14.25.2022. The decision of AMRCD shall be final and binding on both parties, and no arbitration or court proceedings shall be initiated until the said mechanism is exhausted.

10.6 Cases covered under the Micro, Small and Medium Enterprises Development

(MSMED) Act, 2006.

In disputes involving registered MSMEs, reference to the Micro and Small Enterprises Facilitation Council (MSEFC) is mandatory under Section 18 of the said Act. Such disputes are to be first taken up for conciliation and, in the event of failure thereof, arbitration is automatically triggered as per the statutory process.

11 SET OFF:

11.1 Any sum of money due and payable to the contractor (including security deposit refundable to him) under this contract may be appropriated by BSNL or any other person or persons contracting through BSNL and set off the same against any claim of BSNL or such other person or persons for payment of a sum of money arising out of this contract made by the Contractor with BSNL or such other person or persons contracting through BSNL.

----- END OF SECTION VII ----

SECTION VIII
SCHEDULE OF RATES / BILL OF QUANTITY

1	Tender No.& Date	GEM/2026/B/7453892
2	Name of the Contractor (in Block Letters)	
3	Name of the Company (in Block Letters)	
4	Postal Address	
5	Phone / Fax Number, Mobile No. & email address	
6	Name of the work	PLB Pipe/Duct Laying by Open Trench Method/ Trenchless Technology (HDD), Optical Fiber Cable Pulling/Blowing, Splicing,earthing and associated works
7	Name & Number of the Zone	BPR02 (Chatrapur)
8	Zone RKM	43.9 km
9	Estimated Cost (including GST):	Rs.13079745.40
10	EMD (2% of Estimated Cost excluding GST):	Rs. 261595
11	Details of EMD Deposited	a) Rs. …… b) Receipt No. of e-payment (RTGS/NEFT/IMPS) …..……. & date …… c) Name of the bank …… d) Branch ……

 Read understood and Signature of the Bidder with Seal

WORK SCHEDULE AND SCHEDULE RATES FOR OFC CONSTRUCTION WORK FOR 43.9 KMs (ZONE - BPR02)					
S.No.	Nature of work	Unit	Rate	Quantity	Total Amount
1	**Excavation of Trenches/HDD**				
1.1	Excavating trenches up to a depth of 1.65 meters and back filling the excavated trenches after laying the HDPE pipe:				
	i) NON ROCKY SOIL	per meter	₹ 158.00	14900	23,54,200.00
	ii) ROCKY SOIL	per meter	₹ 320.00	1060	3,39,200.00
1.2	Horizontal Boring(Manual) at a depth of 1.65 m for crossing roads / railway track inserting 65/40 mm dia GI pipes and pushing HDPE pipes/PLB Pipe inside and drawing PP rope:	per meter	₹ 300.00	0	0.00
1.3	Horizontal Boring(Horizontal Directional Drilling) at a depth of 1.65 m for crossing roads / railway track and pulling HDPE pipes/PLB Pipe inside and drawing PP rope: Horizontal Boring(HDD) at a depth of 1.65m upto 2 pipes	per meter	₹ 250.00	27940	69,85,000.00
2	**Laying of PLB /HDPE Pipe/GI Pipe/RCC Pipe/DWC (BSNL will supply PLB/HDPE Pipes/Coil PP Rope,Socket ,End Cap)**				
2.1	HDPE /PLB Pipe (40 mm) laying coupled with HDPE/PLB Sockets and drawing of PP Nylon Rope through it.(per Mtr)				
	i) 1st Pipe	per meter	₹ 9.00	15590	1,40,310.00
	ii) Each additional Pipe	per meter	₹ 5.00	0	0.00
2.2	Laying and Fixing of 65/40 mm GI Pipes/RCC/DWC pipes in trenches with HDPE pipes/PLB Pipe Inside and drawing PP nylon Rope	per meter	₹ 35.00	100	3,500.00
2.3	Laying and Fixing of GI/RCC/DWC Pipes on Bridges and Culverts with HDPE pipes/PLB Pipe inside and drawing PP Nylon Rope with iron Clamps (including material cost and clamps to be put at a distance of 1M to be supplied by Vendor).The BSNL will supply G.I.Pipes/RCC/DWC Coupler.	per meter	₹ 85.00	26	2,210.00
3	**RCC Protection Work**				0.00
3.1	Providing RCC protection in the ratio of 1:2:4 for encashing PLB /GI/DWC pipes in the road ,bridges,Road crossings and culverts and for low depth protection or on wheel guard of bridge as specified in the construction practices of OFC construction including material at Site .	Per cu mtr	₹ 6,000.00	34	2,01,000.00
4	**Cable Laying**				
4.1	Opening of manholes , pulling/ blowing of OF Cable, sealing of HDPE/PLB Pipe in manholes by ruber bushes/couplers and providing HDPE/PLB/RCC Split pipes in manholes and back filling of manholes per meter:(Mtr)	Per meter	₹ 13.00	43900	5,70,700.00
5	**Cable Splicing Work**				
	Splicing Termination of Optical Fibre Cable.Splicing of all fibres of OF cable laid at every joint within specified splicing loss rang(≤0.2db) and making termination at the ends. All materials required for splicing and labourers except Joint clousers to be supplied by vendor/Bidder				
5.1	Splicing/Termination of Optical Fibre Cable per joint(Joint) 24F	Per joint (all fibres)	₹ 2,000.00	56	1,12,000.00
5.2	48F	Per joint (all fibres)	₹ 2,500.00	0	0.00

5.3	96F/144/288F(Ribbon)	Per joint (all fibres)	₹ 4,000.00	2	8,000.00
6	**Joint Chamber**				
6.1	Supply and fixing of pre cast RCC joint chamber with size 1.2M inner dia x 0.8M depth and ring Thickness of 40mm made of cement concrete mix of 1:2:4. It should have split RCC cover (With two Haves) of thickness 40mm made of cement concrete mix of 1:2:4. Base should be circular . Scope includes required excavation,backfilling with clean sand and all above operations including supply of materials.	per chamber	₹ 3,500.00	54	1,89,000.00
7	**Route Indicator**				
7.1	Digging of Pits 1 m towards jungle side on each manholes/joint chamber for transporting & fixing of route/joint indicator,fixing and concreting of routes/joint indiactor,painting and sign writing of route/joint indicators.(Indicator)	per route indicator	₹ 600.00	178	1,06,800.00
8	**Route Index Diagram**				
8.1	Documentation and providing details of Lat -Long of the route/Videography of the route on which cable is laid	2 sets	₹ 400.00	22	8,800.00
9	**Cutting concrete /Asphalt surface**				
9.1	Cost of cutting concrete /Asphalt surface	per meter	₹ 90.00	209	18,810.00
10	**Earthing of joint closure and Termination Box(For Armoured Cable)**				
10.1	Earthing of joint closure (For armoured cable): supply and fixing of spike earth arrangement consists of one number of 25mm or more dia GI pipe of 275cm or more length,connecting the joint closure earthing point with earth rod by soft copper wire ,minimum 2.5 mm diameters and fixing with nut and bolt properly.Eath value should be less tha 0.5 ohm.(all materilas to be supplied by vendor)	per joint chamber	₹ 3,000.00	15	45,000.00
10.2	Earthing of termination box: Earthing with GI pipe 4.5M Long,40 mm dia including accessories and providing masonary enclosure with cover platehaving locking arrangementand watering pipe etc.Earth value should be less than 0.5 ohm	Set	₹ 5,000.00	0	0.00
10.3	Earthing of termination box: Providing and laying earth connectionfrom earth electrode with 4mm dai copper wire in 15 mm dia GI pipe from earth electrode including connection with copper thimble excavation and refilling as required.Earth value should be less than 0.5 ohm	Meter	₹ 360.00	0	0.00
11	**Drawing OF cable on Over Head**				
11.1	24F and above with support(accessories will be provided by BSNL)	per meter	₹ 20.00	0	0.00
11.2	24F and above with out support	per meter	₹ 15.00	0	0.00
11.3	4/6/12F with support	per meter	₹ 12.00	0	0.00
11.4	4/6/12F without support	per meter	₹ 8.00	0	0.00
	Total Estimated cost (before GST)				**1,10,84,530.00**
				Say	**1,10,84,530.00**

	Total Estimated cost with GST			**With GST**	**1,30,79,745.40**

----- END OF SECTION VIII ----

 Read understood and Signature of the Bidder with Seal

SECTION IX

Qualifying Criteria and Experience

1. Work Experience

1.1 The tenderers who possess experience of having successfully completed the **Optical Fiber Cable Construction works** in BSNL/MTNL/TCIL/ITI/Rites/HCL/ PGCIL/ RailTel/any other Govt. (Central/ State) undertaking having mandate to carry out OFC construction works and also have received the payment for such completed works to the extent as mentioned below, are eligible to participate in the tender:

a) For an amount equal to at least 90% of estimated cost of work put to tender during last 5 Financial years plus the subsequent period ending the month preceding the one in which NIT is issued.

OR

b) For an amount equal to at least 120% of estimated cost of work put to tender during last 7 Financial years plus the subsequent period ending the month preceding the one in which NIT is issued.

1.2 Other mandatory requirement:

1.2.1 The tenderer whose near relative(s) is/are employed in BSNL, is not eligible to Participate in the tender, as per limitations mentioned in Annexure-C

1.2.2 Contractor should have valid EPF registration certificate.

1.2.3 Contractor should have valid ESI registration certificate.

1.2.4 Contractor should have valid GSTIN registration certificate.

1.2.5 Contractor should have valid PAN No.

1.2.6 Financial Turn over: The total financial turnover for the last three years, ending on 31st March of the previous financial years (i.e., 2022-23, 2023-24, and 2024-25), must be at least 150% of the estimated tender cost (inclusive of GST). Self-attested copies of the Profit & Loss statements, duly audited by a Chartered Accountant (CA), along with a Financial Turnover Certificate issued by the CA, must be submitted. Additionally, the certificates issued by the Chartered Accountants should include a verified UDIN for the corresponding years.

1.2.7 Solvency certificate for 25% of the Tender cost or Rs. 5 Lac whichever is higher. ***The solvency certificate shall not be older than the date of issue of NIT***.

Note1: The Experience should be of Optical Fiber Cable construction work only and does not include any other work like Pole less/Up-gradation/Maintenance works of OFC/External Plant/ UG Cable (Copper) etc.

Note 2: The experience certificate for carrying out the works and amounts received should be in the prescribed pro-forma and the certificate should be under the signature of officer responsible for maintaining the contractor's ledger e.g. DGM/ AO (Cash)/DDO or equivalent officer of the concerned PSU i.e. BSNL/MTNL/TCIL/ITI/Rites/HCL/ PGCIL/ RailTel/any other Govt. (Central/ State) undertaking as the case may be. BSNL may verify and ask the bank statement of the certified amount which should be reflected in the bank statement of the bidder.

Note 3: **The experience criteria as mentioned above (vide 1.1) shall supersede any other experience criteria mentioned in GeM portal by default.**

---- END OF SECTION IX ----

SECTION X

TENDERER'S PROFILE and BID FORM

General:

Paste color passport size photograph of the tenderer/authorized signatory holding power of Attorney and having Digital Signature Certificate. Photograph should be self-attested.

1. Name of the tenderer / firm …….

2. Name of the person submitting the tender whose photograph is affixed and who possesses the Digital Signature Certificate(DSC).

(a) Shri/Smt …….

(b) DSC Issuing Agency…….

(**In case of Proprietary / Partnership firms, the tender has to be digitally signed by Proprietor /Partner(s) only, as the case may be)**

3. Address of the firm

…….....……

…….....

…….....

4. Correspondence Address

……

……..…….....……

……

5. Tel. No. (with STD code) (O) …….……(Fax) ……...
(R)….....……

6. Mobile No. ……. email-ID: …….
(Note: These contact details shall be used to communicate with the tenderer / contractor and any communication sent thereon shall be taken as proper communications under this contract)

7. Registration & incorporation particulars of the firm(Tick as applicable):

(i)Proprietorship (ii) Partnership (iii) LLP (iv) Private Limited (v) Public Limited
(Please attach attested copies of documents of registration/incorporation of your firm with the competent authority as required by business law)

8. Name of Proprietor/ Partners/ Directors

.....…….....……

…….....

9. Tenderer's Bank Details:

a. Name of Bank ..... Bank Account Number…….

b. Name of Branch..... City.....

c. Branch Code.....IFSC Code.....

10. Permanent Income Tax Account Number (PAN) …….

Income Tax Circle…….….

11. EPF registration number …….....

12. ESI registration number……

12. GSTIN registration number ……

13. Whether Micro, Small or Medium Enterprises (MSMEs)?

(Yes/No):……

If Yes,

Validity: from…… to ……

Monetary Limit ……...

14. Infrastructural capabilities of the bidder:

I. Tools and Machinery infrastructure

a) Capacity of trenching per day (in meters) …….…….

b) Capacity of PLB pipe laying per day(in meters) …….……

c) Capacity of pulling cable through duct / pipe per day (in meters) …….

d) Availability of HDD machine

Owned…….……...Capacity…….

Leased……Capacity…….

e) Particulars of vehicles available with the tenderer:

Type of Vehicles(s)Registration number

……. …….……

…….…….

……. …….

II. Man power

f) No of trained supervisors……

g) No. of Trained splicers …….

h) Capacity of engaging mazdoors per day …….

Place:Signature of tenderer / Authorized signatory…….

Date:Name of the tenderer……

Seal of the tenderer

BID FORM

NIT No. ……

To

The AGM (MM),
O/o the CGM ,3rd Floor, MM Cell,
Bhubaneswar-751009

Dear Sir,

Having examined the terms & conditions of contract and the specifications including Addendum/Corrigendum(if any), the receipt of which is hereby duly acknowledged, we, the undersigned, hereby submit our offer to execute the work of Optical Fiber Cable construction stipulated against this NIT in conformity with the said drawings, terms & conditions of contract and specifications.

We undertake, if our Bid is accepted, we will execute the work in accordance with specifications, time limits & terms and conditions stipulated in the tender document.

If our Bid is accepted, we shall submit the securities as per the conditions mentioned in the contract.

We agree to abide by this Bid for a period of 150 days from the date fixed for Qualifying Bid opening and it shall remain binding upon us and may be accepted at any time before the expiry of that period.

Until a formal Agreement is prepared and executed, this Bid together with your written acceptance thereof in your notification of award shall constitute a binding contract between us.

Bid submitted by us is in full compliance of the requirements of e-tendering.

Dated this …… day of ……. (the year)

Dated this …… day of ……. (the year)

Signature of Authorized Signatory ……

In capacity of ……

Duly authorized to sign the bid for and on behalf of …….

Witness …….

Address ……

Signature ……

----- END OF SECTION X ----

SECTION XI

ANNEXURE-I

Declaration w.r.t. Rule 144 (xi) to GFR 2017

Certificate to be submitted by Bidders (On Company's Letter Head)

Reference 1: BSNL Gem bid No ..... issued

on ……..

Reference 2: Department of Expenditure Office Memorandums (OMs) No. 7/10/2021-PPD (1) Dated 23.02.2023.

I, in capacity of authorized signatory of M/s……(----Name of the company--)

having Regd. office

at……

……...being a participant bidder in BSNL T.E cited at reference 1 above, do hereby declare that I have read and understood all the clauses regarding Restrictions under Rule 144(xi) of the General Financial Rules (GFRs) 2017 on grounds of Defense of India and National Security issued vide OM cited at reference 2 above, on procurement from bidders from a country or countries which shares a land border with India. It is declared that we fulfill all the requirements in this regard and are eligible to be considered for the Tender Enquiry under reference 1 above.

Further, we undertake that we will also abide by all the requirements of cited OMs during the entire contract period.

(Name of the authorized signatory) Signature Designation in Company

Seal / Stamp of Company

Counter signed by Company Secretary of the Company with seal / stamp.

ANNEXURE-II

NAME AND ADDRESS OF THE BIDDER FIRM /LETETR HEAD

Ref No:_____ Dated_____

Make in India Certificate

Pursuant to the Guidelines of Public Procurement (Preference to Make in India), letter no P-45021/2/2017-PP(BE II)-Part(4) Vol II dated 19.07.2024 of Govt. of India, Min of Commerce and Industries, Dept of Promotion of Industries & Internal Trade, New Delhi we hereby certify that we, M/s _____, are the local supplier meeting the requirement of minimum local content _____ (in %) as defined in above letter in respect of the material intended to be procured by BSNL,Odisha Circle, Bhubaneswar vide their GeM bid No._____ dated_____.

Details of location at which local value addition will be made is as follows:

_____ _____(Place).

We also understand, any false declarations in this matter will be an instance for breach of the 'Code of Integrity' under Rule.175 (1) (i)(h) of the General Financial Rules for which a bidder or its successors can be debarred from participating in any procurement process upto two years as per Rule 151 (iii) of the General Financial Rules along with such other actions as may be permissible under law.

Signature_____

Name_____

Position in the firm_____(Proprietor/Partner/Director)

(Vendor's Seal)

Signature of the Bidder with Seal

----- END OF SECTION XI ----